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Development and evaluation of predictive machine learning methods for forecasting demand for battery electric vehicles

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Abstract

Battery electric vehicle (BEV) adoption in the EU accelerated sharply after 2020, creating a need for forecasting methods that can adapt to structural market change. This thesis develops and evaluates a machine learning framework for forecasting BEV market share across all 27 EU member states from 2011 to 2024, with particular attention to how the temporal composition of training data affects predictive performance.

The study uses a three-trial experimental design that compares models trained on pre-COVID data (2011–2019), extended data including the transition period (2011–2022), and post-COVID data only (2020–2023). Before modelling, PCA and K-Means clustering were used to group countries into two stable clusters, labeled *Leaders* and *Followers*, which were then modeled separately. Nine model classes were tested, including Linear Regression, SVR, ARIMAX, LSTM, Random Forest, XGBoost, and three stacking ensemble variants combining ARIMAX and LSTM with ElasticNet, Gradient Boost, or Weighted Average meta-learners.

The results show that models trained only on pre-2020 data performed poorly on the post-2020 period, indicating a clear structural break in BEV adoption dynamics. Adding transitional data improved performance for several models but weakened sequential models such as LSTM. In contrast, training only on post-2020 data produced the best overall results. The *Proposed (ElasticNet)* stacking model achieved the strongest performance in Trial 3, with a MAE of 2.97% for *Leaders* and 3.68% for *Followers*, outperforming all baseline models. The cluster structure remained stable across trials, suggesting that economic capacity and charging infrastructure are key differentiators of BEV uptake.

These findings show that recent, structurally coherent data are more valuable than longer historical series in fast-changing technology markets. The framework also performed well in a forward-looking 2025 case study, supporting its practical value for near-term BEV market planning and policy analysis.

Keywords: Battery Electric Vehicles, Forecasting, Machine Learning, Stacking Ensemble, European Union

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List of Abbreviations

AIC	Akaike Information Criterion
ARIMA	AutoRegression Integrated Moving Average
ARIMAX	AutoRegression Integrated Moving Average with eXogenous inputs
BEV	Battery Electric Vehicle
CAGR	Compound Annual Growth Rate
CPI	Consumer Price Index
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
EDA	Exploratory Data Analysis
EU	European Union
EV	Electric Vehicle
GDP	Gross Domestic Product
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MedAE	Median Absolute Error
ML	Machine Learning
NAR	Nonlinear Autoregressive
OLS	Ordinary Least Squares
PCA	Principal Component Analysis
RBF	Radial Basis Function
RNN	Recurrent Neural Network
SVR	Support Vector Regressor
VIF	Variance Inflation Factor
WCSS	Within-Cluster Sum of Squares

1 Introduction

The electrification of road transport has emerged as one of the most consequential structural shifts in the European automotive sector over the past decade. Driven by tightening emissions regulations, expanding charging infrastructure, declining battery costs, and a fundamental shift in consumer attitudes toward sustainable mobility, BEV adoption across the 27 member states of the EU has accelerated markedly since 2020, following a period of gradual and uneven diffusion during the 2010s. The EU's commitment to achieving climate neutrality by 2050 and the prohibition of new internal combustion engine vehicle sales from 2035 under the revised CO_2 emissions regulation [Eur19] have placed BEV adoption at the centre of European transport and energy policy, making accurate near-term forecasting of market trajectories a practical necessity for infrastructure planners, energy system operators, policymakers, and industry stakeholders.

The COVID-19 pandemic marked a structural turning point in this transition. Despite the broader economic disruption, global BEV sales grew by 43% in 2020 as overall car sales fell by 16% [LH21], driven by a combination of government stimulus packages, tightening EU CO_2 regulations, and heightened consumer environmental awareness. The post-2020 period has been characterized by an acceleration of BEV adoption that fundamentally differs in pace and structure from the gradual diffusion of the 2010s, creating a market environment that renders historical pre-2020 data increasingly unrepresentative of current dynamics. This structural break poses a direct challenge to the conventional practice of calibrating forecasting models on the longest available historical series. This practice implicitly assumes the structural relationships between BEV adoption and its drivers remain stable over time.

Despite the practical importance of BEV market forecasting and the significance of this structural break, the existing literature has not systematically investigated how the temporal composition of training data, specifically the inclusion or exclusion of the post-2020 period affects the predictive accuracy of different model classes across heterogeneous European markets. Most prior forecasting studies focus on single national markets, particularly China, and assume structural stability without testing whether their models generalise effectively across market regime changes [KSP24] and [DC23]. This gap motivates the central research questions of this thesis.

1.1 Research Questions

This thesis addresses four research questions that collectively investigate the relationship between training data composition, market structure, and forecasting accuracy in the European BEV market:

Research Question 1: To what degree can a model trained on pre-2020 market data accurately forecast BEV adoption rates in the post-2020 period, and do EU member states behave structurally differently in the post-COVID era compared to the pre-COVID era?

Research Question 2: How does the integration of transitional market data from 2020 to 2022 in the training data influence the predictive accuracy and bias of adoption forecasting models?

Research Question 3: Does a regime-specific model trained exclusively on post-2020 data demonstrate higher predictive accuracy than models trained on longer historical windows that include pre-COVID dynamics?

Research Question 4: Is there a common structural grouping among EU member states in terms of BEV adoption profiles, and can separate cluster-specific models improve forecasting accuracy compared to a single pooled model across all 27 markets?

1.2 Scope and Contributions

This thesis makes the following contributions to the BEV forecasting literature:

1. **Comprehensive multi-country panel dataset.** A harmonized country-year panel covering all 27 EU member countries from 2011 to 2024 is constructed, integrating macroeconomic, infrastructural, and technological variables from multiple official sources including Eurostat [Off25a][Off25b], the International Monetary Fund [Int25], and the European Alternative Fuels Observatory [Eur25a]. This dataset spans both the early diffusion stage and the post-2020 acceleration phase of the European BEV market within a single harmonized framework.

2. **Three-trial experimental design.** A novel experimental framework is developed that systematically varies the temporal composition of the training data across three trials, enabling the first systematic comparison of how pre-COVID, transitional, and post-COVID training windows affect the forecasting accuracy of multiple model classes across heterogeneous European markets.

3. **Novel ARIMAX-LSTM stacking architecture.** A two-stage stacking ensemble is proposed that combines ARIMAX and LSTM as complementary base learners. This captures a linear temporal structure and non-linear sequential dynamics respectively, through three meta-learner variants. The architecture is specifically designed to leverage the complementary error structures of the two base learners in the presence of structural market breaks.

4. **Cluster-based forecasting framework.** PCA and K-Means clustering are applied to partition the 27 EU member states into structurally homogeneous market segments, enabling separate modelling of *Leaders* and *Followers* clusters and demonstrating that the fundamental division of EU markets is robust across all three temporal training configurations.

5. **Forward-looking case study.** All nine model classes are retrained on the 2020–2024 configuration and applied to forecast 2025 BEV market share across all 27 EU member states, providing a forward-looking validation of the full proposed framework under real-world conditions beyond the original test horizon.

The remainder of this thesis is organized as follows. Chapter 2 presents a comprehensive literature review covering the theoretical foundations of data preprocessing, feature selection and dimensionality reduction, clustering methods, forecasting models, and evaluation metrics, alongside a review of existing BEV forecasting studies and a synthesis of the research gaps that motivate this work. Chapter 3 describes the data sources, preprocessing procedures, experimental design, and model configurations applied in this study. Chapter 4 presents the results of the PCA and clustering analyses across all three trials, the comparative model performance evaluation, hypothesis validation, post assessment of theoretical expectations against empirical findings, and the 2025 case study. Chapter 5 provides the final outlook, discussing the policy implications, limitations, and directions for future research, and concludes with closing remarks on the broader significance of the findings for European BEV market planning.

2 Literature Review

2.1 Introduction to BEV Adoption in Europe

The transition towards BEVs was majorly driven by tightening emissions regulations, expanding charging infrastructure, declining battery costs, and growing environmental awareness among consumers. BEV adoption across the 27 member states of the EU has accelerated markedly since 2020, following a period of gradual and uneven diffusion during the 2010s. Understanding the drivers of this transition and developing accurate forecasting tools to anticipate near-term market trajectories has become a central concern for policymakers, industry stakeholders, and researchers alike.

The existing literature on BEV adoption identifies a broad set of determinants that influence the pace and pattern of market uptake. [Sie+14] provide one of the most comprehensive early analyses of the relative importance of financial and socioeconomic factors, finding that charging infrastructure availability and financial incentives including purchase subsidies and tax exemptions are among the strongest predictors of BEV adoption rates across countries. Their cross-national analysis of 30 countries highlights that while purchase price parity with conventional vehicles remains a barrier, the presence of supportive policy environments and visible public charging networks plays a disproportionate role in shaping consumer behavior in the early diffusion phase [Sie+14]. These findings establish a foundational understanding of the supply and demand-side conditions that underpin BEV market growth and inform the selection of predictor variables in this study.

Despite this established understanding of adoption drivers, accurately forecasting the pace and trajectory of BEV market growth remains a methodologically contested area. [KSP24] provide a systematic review of forecasting methods applied to EV ownership, identifying a wide range of approaches spanning time series models, machine learning methods, diffusion models, and hybrid architectures. Their review highlights a recurring tension in the literature between the interpretability of traditional econometric models and the predictive flexibility of data-driven approaches, with no single method demonstrating consistent superiority across different market contexts and time horizons. Similarly, [DC23] review EV forecasting models with particular attention to diffusion and substitution effects, noting that models calibrated on historical adoption data frequently fail to generalise to periods of accelerated market growth, a limitation that becomes particularly acute in the context of the post-2020 European BEV market. This disagreement in the literature about the most appropriate forecasting framework motivates the comparative trial design adopted in this study, which systematically evaluates the sensitivity of multiple model classes to the temporal composition of training data.

The challenge of forecasting structural acceleration in BEV markets is further underscored by studies focused on the Chinese EV market, which experienced a similarly rapid transition during the 2010s. [WGW19] propose a hybrid forecasting approach combining wavelet decomposition with a backpropagation neural network to capture both the long-term trend and short-term fluctuations in Chinese new energy vehicle sales, demonstrating that hybrid architectures outperform individual models when the target series exhibits non-stationary and multi-scale dynamics. [Zha+16] similarly compare Grey System Theory with a NAR neural network for EV growth forecasting, finding that data-driven neural approaches better capture the nonlinear acceleration patterns of emerging EV markets compared to traditional Grey models. These findings from the Chinese context are directly relevant to the European case, where the post-2020 acceleration phase introduced similarly non-stationary dynamics into what had previously been a relatively gradual diffusion process.

The application of multivariate time series models to EV forecasting is examined by [Zha+17], who compare univariate and multivariate approaches for forecasting EV sales in China, demonstrating that the inclusion of exogenous variables such as fuel prices, GDP, and charging infrastructure substantially improves forecast accuracy compared to univariate time series models that rely solely on historical sales data. This finding provides direct support for the multivariate modelling approach adopted in this study, where ten predictor variables spanning macroeconomic, infrastructural, and technological dimensions are incorporated alongside the autoregressive structure of the forecasting models. The role of Linear Regression and SVR in time series prediction is further examined by [Lin+07], who demonstrate that SVR with an appropriate kernel function can capture non-linear temporal patterns that elude standard Linear Regression, though its performance is sensitive to the choice of hyperparameters and the stationarity of the underlying series.

What remains missing from the existing literature is a systematic investigation of how the temporal composition of training data specifically the inclusion or exclusion of the structural break introduced by the COVID-19 pandemic affects the forecasting accuracy of different model classes across heterogeneous EU member state markets. The majority of existing EV forecasting studies treat training data composition as fixed and do not explicitly account for the possibility that the market regime governing adoption dynamics may have changed fundamentally over the study period. This study addresses this gap directly through its three-trial experimental design, which progressively isolates the post-COVID market regime to assess its impact on predictive accuracy across a diverse set of model architectures applied to all 27 EU member states.

2.2 Data Preprocessing

Data preprocessing is a foundational step in any empirical modelling study, as the quality and structure of the input data directly determine the reliability of subsequent analyses and forecasts. In the context of BEV market data spanning multiple countries and an extended time horizon, preprocessing challenges arise from three primary sources: missing values due to data unavailability in the early years of the study period, skewed distributions driven by the rapid and uneven growth of BEV adoption across EU member states, and scale heterogeneity across variables measured in fundamentally different units. This section reviews the theoretical foundations of the preprocessing methods applied in this study, covering missing value imputation, logarithmic transformation, and the Yeo-Johnson power transformation.

2.2.1 Missing Value Imputation

Missing data is a challenge in empirical research and can introduce systematic bias into model estimates if not handled appropriately [LR02]. The nature of the missing data mechanism whether values are missing completely at random, missing at random, or missing not at random determines the most appropriate imputation strategy. [LR02] provided the foundational theoretical framework for missing data analysis, establishing that naive approaches such as complete case analysis or mean substitution can produce biased and inefficient estimates when missingness is not completely random. In the context of this study, missing values arise primarily from data unavailability in the early years of the study period, reflecting the infant state of BEV markets in several EU member states prior to 2013, rather than random measurement error. This systematic pattern of missingness concentrated at the boundaries of the time series rather than distributed randomly across observations necessitates imputation methods that respect the temporal structure and growth dynamics of the underlying variables.

Log-Linear Interpolation

Log-linear interpolation is a well-established method for filling missing values in time series that exhibit exponential or multiplicative growth patterns [LAC17]. The method proceeds by applying a natural logarithm transformation to the observed values, performing standard linear interpolation in log space, and then back-transforming the interpolated values to the original scale using the exponential function. For a time series y_t with a missing value at time t, the log-linear interpolated value is given by:

$$\hat{y}_t = \exp\left(\log(y_{t_1}) + \frac{t - t_1}{t_2 - t_1}\left(\log(y_{t_2}) - \log(y_{t_1})\right)\right) \tag{2.1}$$

where t_1 and t_2 are the nearest observed time points before and after the missing value respectively. This formulation ensures that interpolated values follow a geometric rather than arithmetic progression between observed points, which is appropriate for variables such as *BEV market share* and *Recharding Points* that grow multiplicatively rather than additively over time [LAC17]. The log-linear approach also preserves non-negativity a critical property for count and proportion variables and avoids the unrealistic jumps between observed values that can result from simple linear interpolation when growth is exponential.

[LAC17] provide a comprehensive review of interpolation methods for time series, evaluating their performance across a range of criteria including accuracy, smoothness, and preservation of distributional properties. Their analysis confirms that log-linear interpolation consistently outperforms simple linear interpolation for variables exhibiting exponential growth, particularly when missing values occur at the boundaries of the observed series precisely the pattern encountered in this study.

CAGR-Based Extrapolation

For terminal missing values where observations are absent at the end of the time series rather than within it, interpolation methods that require boundary observations on both sides of the gap are not applicable. In such cases, extrapolation based on the CAGR provides a principled approach to projecting the series forward from the last observed value [LAC17]. The CAGR over a period of n years is defined as:

$$CAGR = \left(\frac{y_T}{y_{T-n}}\right)^{\frac{1}{n}} - 1 \tag{2.2}$$

where y_T is the most recent observed value and y_{T-n} is the observed value n years prior. The extrapolated value for the terminal missing year is then given by:

$$\hat{y}_{T+1} = y_T \times (1 + CAGR) \tag{2.3}$$

This approach assumes that the growth rate observed over the most recent n years continues into the unobserved period, which is a reasonable assumption for near-term extrapolation when the series exhibits a stable trend. In this study, the CAGR was computed from the last three observed years for each variable, balancing the trade-off between recent trends and stability of the growth rate estimate.

Hybrid Approach

For variables with missing values at both internal time points and at the terminal year, a hybrid approach combining log-linear interpolation and CAGR-based extrapolation was applied sequentially. Log-linear interpolation was first applied to fill internal gaps, leveraging the growth trend between two observed boundary points. CAGR-based extrapolation was then applied to project the terminal missing value forward from the last observed year. This sequential combination ensures that interpolated values reflect local growth trends while extrapolated terminal values remain consistent with the most recent trajectory of each variable [LAC17]. The theoretical justification for this hybrid strategy rests on the principle that different missing data patterns require different imputation mechanisms, and that a single method applied uniformly across all variables and all types of missingness is unlikely to be optimal [LR02].

[LAC17] further confirm that sequential combinations of interpolation and extrapolation methods produce more reliable reconstructions than any single method applied in isolation, particularly when missing values occur at both internal and terminal positions within the same series.

2.2.2 Transformations

Logarithmic Transformation

Skewness in the distribution of predictor and target variables is a challenge in regression and forecasting contexts, as highly skewed distributions can violate the normality assumptions of many statistical models, inflate the influence of extreme values on parameter estimates, and reduce the signal-to-noise ratio in the feature space [MCD14]. Variables with an absolute skewness value greater than 1 are generally considered highly skewed and candidates for transformation [MCD14].

The natural logarithm transformation is the most widely used approach for addressing right skewness in variables that exhibit multiplicative growth patterns [DGE12]. For a variable $y > 0$, the log transformation is defined as:

$$y^* = \log(y) \tag{2.4}$$

In practice, when variables may take zero values as is the case for BEV market share in countries with no recorded registrations in early years and for recharging points in countries with no charging infrastructure the *log1p* transformation is applied instead:

$$y^* = \log(1 + y) \tag{2.5}$$

This modification ensures that zero values are mapped to zero in the transformed space rather than producing undefined results, preserving the full sample without requiring arbitrary imputation of zero values prior to transformation [DGE12]. The log-transformed variable follows a more symmetric distribution, reducing the influence of extreme high values on model estimation and improving the linearity of relationships between predictors and the target variable. After model estimation, predictions are back-transformed to the original scale using the exponential function:

$$\hat{y} = \exp(\hat{y}^*) \tag{2.6}$$

Yeo-Johnson Transformation

While the logarithmic transformation is effective for right-skewed positive variables, it is mathematically undefined for negative values and inappropriate for left-skewed distributions. [Yeo00] propose a family of power transformations that generalise the Box-Cox transformation to handle variables of any sign and any direction of skewness. The Yeo-Johnson transformation is defined as:

$$\psi(y, \lambda) = \begin{cases} \dfrac{(y+1)^{\lambda} - 1}{\lambda} & \text{if } \lambda \neq 0,\ y \geq 0 \\ \log(y+1) & \text{if } \lambda = 0,\ y \geq 0 \\ \dfrac{-((-y+1)^{2-\lambda} - 1)}{2-\lambda} & \text{if } \lambda \neq 2,\ y < 0 \\ -\log(-y+1) & \text{if } \lambda = 2,\ y < 0 \end{cases} \tag{2.7}$$

where λ is the transformation parameter estimated by maximizing the log-likelihood of the transformed data under a normality assumption [Yeo00]. When $\lambda = 1$ the transformation reduces to the identity, when $\lambda = 0$ it approaches the natural logarithm for positive values, and intermediate values of λ produce transformations that continuously interpolate between these extremes. This flexibility allows the Yeo-Johnson transformation to adapt to both positive and negative skewness by identifying the optimal scaling parameter for each variable independently.

In this study, the Yeo-Johnson transformation was applied to the *Purchase Price (EUR)*, which exhibited moderate left skewness of -0.67. The log transformation was not suitable in this case both because the variable exhibited left rather than right skewness, and because the logarithm cannot be applied when missing values at the boundaries of the series preclude the computation of a reliable growth-based extrapolation.

2.3 Feature Selection and Dimensionality Reduction

In high-dimensional datasets where multiple predictor variables are measured simultaneously, two fundamental challenges arise prior to model development: the identification of features that carry meaningful information about the target variable, and the detection and resolution of multicollinearity among predictors. Multicollinearity is the presence of strong linear relationships among predictor variables and can destabilize regression coefficient estimates, inflate their variance, and reduce the interpretability and generalization capability of fitted models [DGE12].

This section reviews the theoretical foundations of the three sequential methods applied in this study for feature assessment and dimensionality reduction: Pearson correlation analysis, VIF analysis, and PCA.

2.3.1 Pearson Correlation Analysis

The Pearson correlation coefficient is the most widely used measure of linear association between two continuous variables [DGE12]. For two variables X and Y observed over n observations, the Pearson correlation coefficient r is defined as:

$$r = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2 \sum_{i=1}^{n}(Y_i - \bar{Y})^2}} \tag{2.8}$$

where $\bar{X}$ and $\bar{Y}$ are the sample means of X and Y respectively. The coefficient r takes values in the range $[-1, 1]$, where $r = 1$ indicates a perfect positive linear relationship, $r = -1$ indicates a perfect negative linear relationship, and $r = 0$ indicates the absence of any linear association. In the context of feature selection, the Pearson correlation between each predictor and the target variable provides an initial assessment of the strength and direction of linear relationships, enabling the identification of potentially informative features prior to more complex analysis.

However, the Pearson correlation has several important limitations that must be acknowledged. First, it measures only linear association and will fail to detect non-linear relationships between predictors and the target variable, even when such relationships are strong and consistent [DGE12]. Second, it is sensitive to outliers, which can distort the estimated correlation coefficient substantially when the sample contains extreme values. Third and most critical for the purposes of this study, a high correlation between a predictor and the target variable does not preclude the presence of severe multicollinearity between that predictor and other features in the dataset. A variable may be both highly correlated with the target and highly collinear with other predictors, making feature selection based on pairwise correlations with the target variable alone an incomplete strategy for preparing data for multivariate modelling.

2.3.2 VIF

VIF is the standard tool for detecting and quantifying multicollinearity in a set of predictor variables [DGE12]. For a predictor X_j, the VIF is defined as:

$$VIF_j = \frac{1}{1 - R_j^2} \tag{2.9}$$

where R_j^2 is the coefficient of determination obtained by regressing X_j on all other predictor variables in the dataset. A VIF of 1 indicates the complete absence of multicollinearity; in other words, the predictor is orthogonal to all others. As R_j^2 approaches 1, indicating that the predictor can be almost perfectly explained by the other variables in the dataset, the VIF approaches infinity. Values exceeding 10 are generally considered indicative of severe multicollinearity that asks for remedial actions, as the associated regression coefficient estimates become highly unstable and their standard errors become inflated to the point of rendering statistical inference unreliable [DGE12].

It is important to note that a low VIF alone does not guarantee that a predictor is informative about the target variable; it only indicates the absence of linear dependence on other predictors. Conversely, a high VIF does not necessarily mean a predictor should be discarded, as it may carry important information about the target that is not captured by other variables despite being collinear with them. This tension between predictive relevance and multicollinearity where the most informative predictors are often also the most collinear motivates the use of dimensionality reduction rather than variable selection as the primary strategy for resolving multicollinearity in this study.

2.3.3 PCA

PCA is a linear dimensionality reduction technique that transforms a set of possibly correlated variables into a smaller set of uncorrelated variables called principal components, ordered by the proportion of total variance they explain [Pea01] [Jol02]. PCA simultaneously resolves multicollinearity by producing orthogonal components with VIF values of exactly 1 and reduces the dimensionality of the feature space, addressing both challenges identified in the correlation and VIF analyses within a single unified framework.

Mathematical Formulation

Let $\mathbf{X}$ be an $n \times p$ data matrix of n observations and p standardised predictor variables, with covariance matrix $\mathbf{\Sigma}$. PCA seeks a linear transformation $\mathbf{Z} = \mathbf{XW}$ where $\mathbf{W}$ is a $p \times p$ matrix of weights (loadings) such that the columns of $\mathbf{Z}$, the principal components are mutually orthogonal and ordered by decreasing variance [Jol02]. The loading matrix $\mathbf{W}$ is obtained through the eigenvalue decomposition of the covariance matrix:

$$\mathbf{\Sigma}\mathbf{w}_j = \lambda_j \mathbf{w}_j, \quad j = 1, 2, \ldots, p \tag{2.10}$$

where λ_j are the eigenvalues and $\mathbf{w}_j$ are the corresponding eigenvectors. The eigenvalues are ordered such that $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p \geq 0$, and the j-th principal component is given by:

$$\mathbf{z}_j = \mathbf{X}\mathbf{w}_j \tag{2.11}$$

The proportion of total variance explained by the j-th principal component is:

$$\text{Explained Variance}_j = \frac{\lambda_j}{\sum_{k=1}^{p} \lambda_k} \tag{2.12}$$

[Pea01] first introduced the concept of fitting orthogonal lines and planes to data points in multidimensional space, establishing the geometric foundations of PCA. The modern algebraic formulation was developed by [Hot33] and comprehensively formalized by [Jol02], whose monograph remains the definitive reference for the theoretical properties and practical applications of PCA across a wide range of scientific domains.

PCA rests on several important assumptions that must be acknowledged. First, PCA is a linear method and will not capture non-linear relationships among variables [Jol02]. If the underlying structure of the data is non-linear, PCA may fail to identify the most meaningful low-dimensional representation. Second, PCA is sensitive to outliers, as extreme values can disproportionately influence the estimated eigenvectors and distort the component structure. Third, the sign of each principal component is mathematically arbitrary, the algorithm can produce a component or its negative with equal validity, as both capture the same variance structure. This sign indeterminacy must be accounted for when interpreting loading factors and component scores across separate PCA runs, as the same component may appear with opposite signs in different applications of the algorithm to different training datasets [Jol02]. In the context of this study, this property is relevant when comparing PCA solutions across the three trials, where sign differences in loading factors reflect mathematical convention rather than substantive differences in the underlying market structure.

Loading Factors

The loading factors of a principal component are the elements of the corresponding eigenvector $\mathbf{w}_j$, representing the correlation between each original variable and the principal component [Jol02]. Formally, the loading of variable X_k on principal component j is:

$$l_{kj} = w_{kj}\sqrt{\lambda_j} \tag{2.13}$$

Loading factors provide the primary tool for interpreting the meaning of principal components variables with large absolute loadings on a component contribute most strongly to its definition, and the pattern of loadings across variables reveals the underlying dimension of variation that the component captures [Jol02]. A comprehensive tutorial on the interpretation of PCA loading factors and their relationship to the original variables is provided by [Shl14], who demonstrates through worked examples how loading patterns can be used to assign meaningful labels to principal components in applied research contexts.

standardisation

A critical prerequisite for PCA is the standardisation of all input variables to zero mean and unit variance prior to decomposition [Jol02]. Without standardisation, variables measured on larger scales will dominate the principal components regardless of their actual informativeness, simply because their larger numerical values produce larger variances. For each variable X_j, standardisation is performed as:

$$X_j^* = \frac{X_j - \bar{X}_j}{s_j} \tag{2.14}$$

where $\bar{X}_j$ is the sample mean and s_j is the sample standard deviation of variable j. This ensures that all variables contribute equally to the decomposition regardless of their original units of measurement, which is essential in this study where predictors range from percentage points of *BEV market share* to *GDP* values in the tens of billions of euros.

Component Selection Using Elbow Method

Selecting the number of principal components to retain is a fundamental decision in the application of PCA, as retaining too few components loses important information while retaining too many reintroduces collinearity and noise [Jol02]. The elbow method, also known as the scree plot method is one of the most widely used approaches for component selection [Cat66]. It involves plotting the eigenvalues of the principal components in descending order and identifying the point at which the curve bends sharply the elbow after which the marginal contribution of additional components to total explained variance diminishes substantially. Components above the elbow are retained as they capture the most meaningful variance structures in the data, while components below the elbow are discarded as they capture primarily noise.

A complementary criterion is the Kaiser rule, which retains all components with eigenvalues greater than 1 [Kai60]. This threshold corresponds to the condition that a retained component explains more variance than a single standardized input variable, providing a theoretically motivated lower bound for component retention. In this study both criteria are applied jointly and components above the elbow and with eigenvalues exceeding 1 are retained, ensuring that the selected components capture substantively meaningful variance.

Prior Application in EV Market Research

PCA has been applied in several prior studies of EV market dynamics to address multicollinearity among adoption drivers and reduce the dimensionality of feature spaces containing correlated economic, infrastructural and technological variables. [MSJ24] apply a methodological sequence of correlation analysis, multicollinearity assessment, PCA and clustering to segment EU's EV markets, demonstrating that PCA-derived components provide a more stable and interpretable basis for market segmentation than the original correlated features. Their findings validate the suitability of this methodological sequence for the European BEV market context and provide direct support for the approach adopted in this study. A key advantage of PCA in this context is that by reducing the feature space to a small number of orthogonal components, it resolves the tension identified in the VIF analysis between retaining informative predictors and eliminating multicollinearity, both objectives are achieved simultaneously within the PCA framework.

2.4 Clustering Methods

Clustering is an unsupervised machine learning technique that partitions a dataset into groups of similar observations without the use of predefined class labels [Mac67]. In the context of this study, clustering is applied to the principal component scores of the 27 EU member states to identify distinct market segments based on their BEV adoption profiles. By grouping countries with similar economic, infrastructural and technological characteristics, clustering enables separate forecasting models to be trained for each segment, capturing the heterogeneity of BEV adoption dynamics across the EU without requiring a single model to simultaneously account for the vastly different trajectories of leading and lagging markets. This section reviews the theoretical foundations of the clustering methods considered in this study, covering the elbow method & silhouette score for optimal cluster selection, K-Means clustering and DBSCAN as an alternative density-based approach.

2.4.1 Determining the Optimal Number of Clusters

Elbow Method

The elbow method is a widely used heuristic for selecting the optimal number of clusters in K-Means by examining how the WCSS decreases as k increases [KM13]. As k increases, each cluster becomes smaller and more homogeneous, causing WCSS to decrease monotonically. The rate of decrease is typically rapid for small values of k when clusters are being meaningfully differentiated, and slows as additional clusters begin to split already homogeneous groups. The optimal k is identified at the point where the WCSS curve bends sharply forming an elbow like curve beyond which the marginal reduction in WCSS no longer justifies the additional complexity of an extra cluster. Formally, the elbow is the value of k that maximises the second derivative of the WCSS curve with respect to k:

$$k^* = \arg\max_k \left(\frac{d^2\text{WCSS}}{dk^2} \right) \tag{2.15}$$

[KM13] provide a comprehensive analysis of techniques for selecting the appropriate number of clusters in K-Means, evaluating the elbow method alongside alternative approaches including the gap statistic, the Calinski-Harabasz index, and the Davies-Bouldin index. Their analysis confirms that while no single method is universally optimal, the elbow method provides a robust and computationally efficient criterion that performs reliably across a wide range of dataset characteristics, making it a standard first-choice approach in applied clustering research [HAY21] [Kuc+19].

Silhouette Score

The silhouette score is a validation metric that measures the quality of a clustering solution by assessing how similar each observation is to its own cluster relative to other clusters [Rou87]. For observation $\mathbf{x}_i$ assigned to cluster C_j, the silhouette coefficient $s(i)$ is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{2.16}$$

where $a(i)$ is the mean distance between $\mathbf{x}_i$ and all other observations in the same cluster; in other words, measuring intra-cluster cohesion:

$$a(i) = \frac{1}{|C_j| - 1} \sum_{\mathbf{x}_l \in C_j, l \neq i} \|\mathbf{x}_i - \mathbf{x}_l\| \tag{2.17}$$

and $b(i)$ is the mean distance between $\mathbf{x}_i$ and all observations in the nearest neighboring cluster, which means $b(i)$ is measuring inter-cluster separation:

$$b(i) = \min_{l \neq j} \frac{1}{|C_l|} \sum_{\mathbf{x}_m \in C_l} \|\mathbf{x}_i - \mathbf{x}_m\| \tag{2.18}$$

The silhouette coefficient $s(i)$ takes values in $[-1, 1]$ [Rou87]. A value close to 1 indicates that the observation is well matched to its own cluster and poorly matched to neighboring clusters. A value close to 0 indicates that the observation lies near the boundary between two clusters. A negative value indicates that the observation may have been assigned to the wrong cluster [Rou87]. The overall silhouette score for a clustering solution is the mean of the individual silhouette coefficients across all observations, and the optimal number of clusters is the value of k that maximizes this mean score.

[Rou87] introduced the silhouette method as a graphical and numerical aid for interpreting and validating cluster analysis, demonstrating its utility across a range of real datasets and clustering algorithms. The silhouette score is particularly valuable as a complement to the elbow method because it provides an absolute measure of cluster quality rather than a relative measure of improvement allowing the analyst to assess whether the identified clusters are genuinely well-separated or merely the best available partition of a dataset with no clear cluster structure.

2.4.2 K-Means Clustering

Mathematical Formulation

K-Means is a centroid-based partitioning algorithm that divides n observations into k non-overlapping clusters by minimizing the WCSS, also known as inertia [Mac67]. Formally, given a set of observations $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$ where each observation is a p-dimensional vector, K-Means seeks to find a partition $\{C_1, C_2, \ldots, C_k\}$ and a set of cluster centroids $\{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \ldots, \boldsymbol{\mu}_k\}$ that minimize the objective function (WCSS/inertia):

$$J = \sum_{j=1}^{k} \sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2 \tag{2.19}$$

where $\|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2$ is the squared Euclidean distance between observation $\mathbf{x}_i$ and the centroid of cluster C_j, and the centroid is defined as:

$$\boldsymbol{\mu}_j = \frac{1}{|C_j|} \sum_{\mathbf{x}_i \in C_j} \mathbf{x}_i \tag{2.20}$$

The algorithm proceeds iteratively through two alternating steps until convergence [Mac67]:

1. **Assignment step:** Each observation is assigned to the cluster whose centroid is nearest in Euclidean distance:

$$C_j = \left\{\mathbf{x}_i : \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2 \leq \|\mathbf{x}_i - \boldsymbol{\mu}_l\|^2 \ \forall\, l \neq j\right\} \tag{2.21}$$

2. **Update step:** The centroid of each cluster is recomputed as the mean of all observations currently assigned to it.

The algorithm converges when cluster assignments no longer change between iterations, guaranteeing a local minimum of the objective function J. However, K-Means is not guaranteed to find the global minimum, as the solution depends on the initialization of centroids. To mitigate sensitivity to initialization, multiple random restarts are typically performed and the solution with the lowest WCSS is retained [Mac67].

Assumptions and Limitations

K-Means rests on several assumptions that define both its strengths and its limitations. First, the algorithm assumes that clusters are approximately spherical and of similar size, as the Euclidean distance metric used in the objective function is isotropic and does not account for elongated or irregularly shaped cluster structures [Mac67]. Second, cluster labels assigned by K-Means are arbitrary and not guaranteed to be consistent across separate runs of the algorithm, as the numbering of clusters depends on the random initialization of centroids rather than any intrinsic ordering of the groups. This label indeterminacy is an important consideration when comparing cluster assignments across different training periods or datasets, as what appears to be a transition between clusters may simply reflect a relabeling of the same underlying groups. Third, the analyst must specify the number of clusters k before running the algorithm because it is not estimated from the data, requiring an independent criterion for determining the optimal number of clusters.

A Priori Assessment for BEV Market Segmentation

K-Means is well suited to the BEV market segmentation task in this study for several reasons. The principal component scores used as input features are continuous, standardized, and orthogonal, satisfying the spherical cluster assumption more closely than the original correlated features would. The relatively small number of observations of the 27 EU member states means that the computational efficiency advantage of K-Means over more complex methods is less relevant, but its interpretability and the ease of characterizing cluster structure through centroids make it an appropriate choice for a study where understanding the economic and infrastructural meaning of each cluster is as important as the clustering accuracy itself. [MSJ24] confirm the suitability of K-Means for EU's EV market segmentation, finding that it produces stable and interpretable market groups that align with the observed heterogeneity in BEV adoption rates across member states.

2.4.3 DBSCAN

Mathematical Formulation

DBSCAN is an unsupervised density-based clustering algorithm that identifies clusters as regions of high point density separated by regions of low density, and classifies observations in sparse regions as noise points rather than assigning them to any cluster [Est+96]. DBSCAN requires two parameters: ε, the neighborhood radius, and $MinPts$, the minimum number of points required to form a dense region. The algorithm defines three types of points:

- **Core point:** A point $\mathbf{x}_i$ is a core point if at least $MinPts$ points lie within its ε-neighbourhood:

$$|N_\varepsilon(\mathbf{x}_i)| \geq MinPts \tag{2.22}$$

  where $N_\varepsilon(\mathbf{x}_i) = \{\mathbf{x}_j : \|\mathbf{x}_i - \mathbf{x}_j\| \leq \varepsilon\}$

- **Border point:** A point that lies within the ε-neighborhood of a core point but does not itself satisfy the $MinPts$ criterion.

- **Noise point:** A point that is neither a core point nor a border point, lying in a sparse region of the feature space.

Clusters are formed by connecting all core points that are within each other's ε-neighborhoods, along with their associated border points [Est+96]. Unlike K-Means, DBSCAN does not require the number of clusters to be specified in advance, can identify clusters of arbitrary shape, and explicitly handles outliers by classifying them as noise rather than forcing them into the nearest cluster.

Limitations and Rejection

Despite its theoretical advantages over K-Means for datasets with irregular cluster shapes and outliers, DBSCAN has important limitations that render it unsuitable for the BEV market segmentation task in this study. First, DBSCAN requires the analyst to specify the parameters ε and $MinPts$, and its results are highly sensitive to these choices; a small change in ε can cause observations to shift between core, border and noise classifications, substantially altering the resulting cluster structure [Est+96]. Second, DBSCAN assumes that all clusters have approximately uniform density, and struggles to identify clusters of varying density, a common characteristic of real-world datasets including BEV market data where adoption rates vary dramatically across member states. Third, with limited number of observations, the dataset does not provide sufficient density for DBSCAN to form meaningful density-based clusters, as the algorithm requires a minimum of $MinPts$ points in each neighborhood to classify core points, which is difficult to satisfy with such a small sample.

These limitations are consistent with the findings of [MSJ24], who similarly found that density-based clustering was not well suited to EU's EV market data and adopted K-Means as the preferred clustering approach instead. The empirical application of DBSCAN in this study confirmed these theoretical concerns; the algorithm failed to produce meaningful cluster structures across all three trials, further validating the selection of K-Means as the primary clustering method.

2.5 Forecasting Models and Their Preliminary Assessment

The selection of forecasting models in this study reflects a deliberate strategy of evaluating a diverse set of methodological approaches from classical machine learning methods to deep learning architectures against a common experimental framework.

This diversity allows for a systematic comparison of model classes that differ fundamentally in their underlying assumptions, mathematical structures, and expected behavior under structural breaks in the target series. The following sections present the mathematical formulation, theoretical assumptions, bias-variance characteristics, and a-priori performance expectations for each model considered in this study. Where relevant, prior applications of each method to EV market forecasting are discussed to contextualize the theoretical expectations within the empirical literature.

2.5.1 Linear Regression

Mathematical Formulation

Linear Regression is the foundational model of statistical forecasting, establishing a linear relationship between a set of predictor variables and a continuous target variable [DGE12].

For n observations and p predictor variables, the Linear Regression model is defined as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{2.23}$$

where $\mathbf{y} \in \mathbb{R}^n$ is the vector of target observations, $\mathbf{X} \in \mathbb{R}^{n\times(p+1)}$ is the design matrix containing the predictor variables and an intercept column of ones, $\boldsymbol{\beta} \in \mathbb{R}^{p+1}$ is the vector of regression coefficients, and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is the vector of error terms. The OLS estimator minimizes the sum of squared residuals:

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\text{argmin}} \, \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{y} \tag{2.24}$$

provided that $\mathbf{X}^\top\mathbf{X}$ is invertible, which requires the absence of perfect multicollinearity among the predictors. The fitted values and predictions are given by:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} \tag{2.25}$$

Assumptions

The validity of OLS estimation and the associated statistical inference rests on the Gauss-Markov assumptions [DGE12]:

1. **Linearity:** The relationship between the predictors and the target variable is linear in the parameters
2. **Strict exogeneity:** The error terms have zero conditional expectation given the predictors: $\mathbb{E}[\boldsymbol{\varepsilon}|\mathbf{X}] = \mathbf{0}$
3. **Homoscedasticity:** The error terms have constant variance: $\text{Var}(\varepsilon_i|\mathbf{X}) = \sigma^2 \; ; \forall\, i$
4. **No autocorrelation:** The error terms are uncorrelated: $\text{Cov}(\varepsilon_i, \varepsilon_j|\mathbf{X}) = 0 \; ; \forall\, i \neq j$
5. **No perfect multicollinearity:** The columns of $\mathbf{X}$ are linearly independent

Under these assumptions, the Gauss-Markov theorem establishes that the OLS estimator is the Best Linear Unbiased Estimator (BLUE) as it has the lowest variance among all linear unbiased estimators [DGE12].

Bias-Variance Trade-off

Linear Regression is traditionally categorized as a model with potential for a high bias due to its rigid parametric assumptions, but often exhibits low variance in its parameter estimates compared to more complex models [DGE12]. The strong parametric assumption of linearity means that if the true relationship between predictors and the target variable is non-linear, the model will systematically underfit the data producing predictions that are consistently biased in a predictable direction. However, the simplicity of the model means that its predictions are stable across different training samples, resulting in low variance. In the context of BEV market forecasting, where the relationship between adoption rates and their drivers is likely to be non-linear and subject to structural breaks, the high bias of Linear Regression is expected to limit its predictive accuracy, particularly when the model is trained on pre-COVID data and applied to the post-COVID acceleration phase.

A Priori Assessment

Linear Regression is included in this study as one of the baseline models against which all other approaches are benchmarked. Given its strong linearity assumption and its inability to capture non-linear dynamics, threshold effects, or structural breaks in the target series, linear regression is not expected to be a competitive forecasting model for BEV market share in the post-COVID period. Its inclusion serves to establish a performance floor for any model that fails to outperform Linear Regression provides no meaningful forecasting advantage over the simplest possible approach. [Lin+07] demonstrate that Linear Regression provides a useful baseline for time series prediction tasks but is consistently outperformed by more flexible methods such as SVR when the underlying series exhibits non-linear patterns, a finding that is expected to be replicated in this study given the non-linear nature of BEV adoption dynamics.

2.5.2 SVR

Mathematical Formulation

SVR extends the principles of Support Vector Machines to regression tasks, introduced by [CV95] as part of the broader framework of statistical learning theory. SVR seeks a function $f(\mathbf{x})$ that deviates from the observed target values by at most ε for all training observations, while remaining as flat as possible. For a linear SVR, the function takes the form:

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b \tag{2.26}$$

where $\mathbf{w}$ is the weight vector and b is the bias term. The flatness of f is controlled by minimizing $\frac{1}{2}\|\mathbf{w}\|^2$, and the ε-insensitive loss function is defined as:

$$L_\varepsilon(y, f(\mathbf{x})) = \max\left(0, |y - f(\mathbf{x})| - \varepsilon\right) \tag{2.27}$$

This loss function assigns zero penalty to predictions within an ε-tube around the true value and penalizes deviations outside the tube linearly. The full SVR optimization problem with slack variables ξ_i and ξ_i^* for violations above and below the tube respectively is:

$$\underset{\mathbf{w}, b, \xi, \xi^*}{\text{minimize}} \ \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \tag{2.28}$$

subject to:

$$\begin{cases} y_i - \mathbf{w}^\top \mathbf{x}_i - b \leq \varepsilon + \xi_i \\ \mathbf{w}^\top \mathbf{x}_i + b - y_i \leq \varepsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \tag{2.29}$$

where $C > 0$ is the regularization parameter that controls the trade-off between the flatness of f and the tolerance of training errors exceeding ε.

Kernel Trick and RBF Kernel

For non-Linear Regression tasks, SVR employs the kernel trick to implicitly map the input features into a higher-dimensional feature space where a linear function can capture non-linear relationships in the original input space [CV95]. The kernel function $K(\mathbf{x}_i, \mathbf{x}_j)$ computes the inner product of two observations in the transformed feature space without explicitly computing the transformation. The RBF kernel is defined as:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma \left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2\right) \tag{2.30}$$

where $\gamma > 0$ controls the width of the kernel and determines the reach of each training observation's influence. The RBF kernel is the most widely used kernel in SVR applications as it is capable of approximating any continuous function given sufficient data and appropriate hyperparameter selection [CV95].

Assumptions

SVR makes relatively few distributional assumptions compared to Linear Regression [CV95]. It does not assume normality of the error terms, homoscedasticity, or linearity of the relationship between predictors and the target variable. The primary assumption is that the training and test data are drawn from the same underlying distribution, an assumption that is directly challenged in this study by the structural break between the pre-COVID training period and the post-COVID test period.

Bias-Variance Trade-off

The bias-variance characteristics of SVR are governed by the regularization parameter C and the kernel width γ [CV95]. A large value of C reduces bias by allowing the model to fit the training data more closely at the cost of increased variance, while a small C increases bias but reduces variance. Similarly, a large γ produces a narrow kernel that captures fine-grained local patterns but generalises poorly, which means there is a high variance while a small γ produces a smooth function with higher bias but lower variance.

A Priori Assessment

SVR with an RBF kernel is theoretically well suited to capturing non-linear relationships between BEV adoption rates and their drivers. [Lin+07] demonstrate that SVR consistently outperforms Linear Regression for non-linear time series prediction tasks, supporting its inclusion as a competitive baseline. However SVR's reliance on the assumption of distributional similarity between training and test data makes it potentially vulnerable to the structural break introduced by the COVID-19 pandemic. When trained on pre-COVID data, the RBF kernel will learn a similarity structure based on pre-2020 market dynamics that may no longer be applicable in the post-COVID acceleration phase, potentially leading to substantial prediction errors in Trial 1. SVR is therefore expected to be competitive in Trial 3 where training and test data are drawn from the same post-COVID regime.

2.5.3 ARIMAX

From ARIMA to ARIMAX

The ARIMA model is the classical framework for univariate time series forecasting, introduced by [Box+08] in their foundational monograph on time series analysis. An ARIMA(p, d, q) model describes a stationary time series as a linear combination of its own past values and past forecast errors, with d rounds of differencing applied to achieve stationarity. The ARIMA model is defined as:

$$\phi(B)(1-B)^d y_t = \theta(B)\varepsilon_t \tag{2.31}$$

where B is the backshift operator such that $By_t = y_{t-1}$, $\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \ldots - \phi_p B^p$ is the autoregressive polynomial, $\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \ldots + \theta_q B^q$ is the moving average polynomial, and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ is white noise [Box+08].

While ARIMA is a powerful tool for univariate time series forecasting, its fundamental limitation is that it models only the temporal autocorrelation of a single series and cannot incorporate the influence of external drivers on the target variable. In the context of this study, applying ARIMA to a panel dataset of 27 EU member states would require aggregating country-level observations into a single European average series, producing a synthetic trajectory that obscures the cross-country heterogeneity that is central to this analysis. Furthermore, the univariate nature of ARIMA means it cannot leverage the rich set of macroeconomic, infrastructural, and technological predictors that have been identified in the literature as key drivers of BEV adoption [Sie+14] and [Zha+17]. For these reasons, ARIMA was evaluated and discarded in favor of its multivariate extension, ARIMAX.

Mathematical Formulation of ARIMAX

The ARIMAX(p, d, q) model extends ARIMA by incorporating a vector of exogenous predictor variables $\mathbf{x}_t$ alongside the autoregressive and moving average components [Box+08]:

$$\phi(B)(1-B)^d y_t = \boldsymbol{\beta}^\top \mathbf{x}_t + \theta(B)\varepsilon_t \tag{2.32}$$

where $\boldsymbol{\beta}$ is the vector of coefficients for the exogenous predictors. This formulation allows the model to capture both the temporal autocorrelation structure of the BEV adoption series and the contemporaneous influence of external drivers, addressing the fundamental limitation of pure ARIMA for this application. In this study, the principal component scores derived from the PCA of the ten predictor variables are used as the exogenous inputs $\mathbf{x}_t$, ensuring that the exogenous component is free of multicollinearity.

Order Selection Using AIC

The autoregressive order p and moving average order q of the ARIMAX model are selected using the AIC, introduced by [Aka74] as an information-theoretic framework for model selection. The AIC is defined as:

$$AIC = 2k - 2\ln(\hat{L}) \tag{2.33}$$

where k is the number of estimated parameters and $\hat{L}$ is the maximized value of the likelihood function of the model. The AIC penalizes model complexity reflected in k to guard against overfitting, while rewarding goodness of fit reflected in $\hat{L}$ to ensure adequate description of the data. The model order that minimizes the AIC provides the best balance between fit and parsimony [Aka74]. In this study, the AIC is computed over a grid of values $p, q \in \{0, 1, 2\}$ for each country, with the order combination achieving the minimum AIC selected for that country's model.

Stationarity and Integration

A fundamental requirement of ARIMA and ARIMAX is that the modeled series be stationary; in other words, having constant mean, variance, and autocovariance over time [Box+08]. BEV adoption series typically exhibit strong upward trends and are therefore non-stationary in levels. The integration parameter d addresses this by applying d rounds of differencing to the series prior to modelling, where the first difference is:

$$\Delta y_t = y_t - y_{t-1} = (1-B)y_t \tag{2.34}$$

A fixed integration order of $d = 1$ was applied in this study, as a single round of differencing is typically sufficient to achieve stationarity in time series exhibiting linear trends [Box+08].

Assumptions

The ARIMAX model rests on several key assumptions [Box+08]. The error terms ε_t are assumed to be independently and identically distributed white noise with zero mean and constant variance. The exogenous predictors are assumed to be weakly exogenous and not influenced by the current or past values of the target variable. The differenced series is assumed to be stationary after d rounds of differencing. Violations of these assumptions are particularly the white noise assumption in the presence of structural breaks and can lead to biased coefficient estimates and unreliable forecasts.

Bias-Variance Trade-off

ARIMAX is a parametric linear model with a relatively small number of estimated parameters, giving it a bias-variance profile similar to Linear Regression but has a relatively high bias due to the linearity assumption but low variance due to the parsimony of the model [Box+08]. The autoregressive component adds temporal structure that Linear Regression lacks, potentially reducing bias in time series contexts where past values are informative about future ones. However, the linearity assumption of both the autoregressive and exogenous components means that ARIMAX cannot capture non-linear dynamics or threshold effects in BEV adoption.

A Priori Assessment

ARIMAX is expected to perform well in Trial 1 for countries where BEV adoption followed a relatively stable and predictable trend during the pre-COVID period, as its autoregressive structure is well suited to capturing smooth temporal trajectories. However, the structural break introduced by the COVID-19 pandemic which fundamentally altered the growth dynamics of BEV adoption across all EU member states is expected to challenge the stationarity assumption and degrade ARIMAX performance when trained on pre-COVID data and tested on the post-COVID period. In Trial 3, where the training data is drawn exclusively from the post-COVID regime, ARIMAX is expected to recover competitive performance as the stationarity assumption is more likely to hold within the more homogeneous post-2020 period.

2.5.4 LSTM

Background of RNNs and the Vanishing Gradient Problem

RNNs are a class of neural network architectures designed for sequential data, maintaining a hidden state $\mathbf{h}_t$ that summarizes information from previous time steps and is updated at each step according to:

$$\mathbf{h}_t = \sigma\left(\mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t + \mathbf{b}\right) \tag{2.35}$$

where $\mathbf{W}_h$ and $\mathbf{W}_x$ are weight matrices, $\mathbf{b}$ is a bias vector, and σ is a nonlinear activation function. RNNs are trained using BPTT, an extension of standard backpropagation [RHW86] that computes gradients by unrolling the network across time steps and propagating errors backwards through the sequence. However, standard RNNs suffer from the vanishing gradient problem as gradients are propagated backwards through many time steps, they are multiplied repeatedly by the weight matrix $\mathbf{W}_h$, causing them to shrink exponentially and preventing the network from learning long-range temporal dependencies [HS97].

Mathematical Formulation

[HS97] introduced LSTM architecture to address the vanishing gradient problem through a gating mechanism that explicitly controls the flow of information across time steps. The LSTM unit maintains two state vectors: a hidden state $\mathbf{h}_t$ and a cell state $\mathbf{c}_t$, updated at each time step through three multiplicative gates:

Forget gate: determines what proportion of the previous cell state to retain:

$$\mathbf{f}_t = \sigma\left(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f\right) \tag{2.36}$$

Input gate: determines what new information to write to the cell state:

$$\mathbf{i}_t = \sigma\left(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i\right) \tag{2.37}$$

$$\tilde{\mathbf{c}}_t = \tanh\left(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c\right) \tag{2.38}$$

Cell state update:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \tag{2.39}$$

Output gate: determines what information from the cell state to expose as the hidden state:

$$\mathbf{o}_t = \sigma\left(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o\right) \tag{2.40}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \tag{2.41}$$

where $\odot$ denotes the Hadamard (element-wise) product, σ is the sigmoid activation function, $\tanh$ is the hyperbolic tangent activation function, and $\mathbf{W}_f, \mathbf{W}_i, \mathbf{W}_c, \mathbf{W}_o$ and $\mathbf{b}_f, \mathbf{b}_i, \mathbf{b}_c, \mathbf{b}_o$ are the learned weight matrices and bias vectors for each gate respectively [HS97].

Assumptions

Unlike the parametric models discussed previously, LSTM makes minimal distributional assumptions about the data [HS97]. It does not assume linearity, normality of errors, or stationarity of the target series. The primary assumption is that the temporal dependencies and non-linear relationships learned from the training data are representative of the patterns that will be encountered during testing, an assumption that is directly challenged by the structural break between the pre-COVID and post-COVID periods in this study.

Bias-Variance Trade-off

LSTM networks are low-bias, high-variance models [HS97]. Their flexible non-linear architecture allows them to approximate complex temporal patterns with low systematic bias, but this flexibility comes at the cost of high sensitivity to the training data, particularly when the training sample is small, as is the case in this study where the annual panel dataset provides only a limited number of annual observations per country. Dropout regularization and early stopping are applied in this study to mitigate overfitting and control the variance of LSTM predictions.

A Priori Assessment

LSTM is theoretically well suited to capturing the non-linear sequential dynamics of BEV adoption, including gradual acceleration phases and threshold effects that may not be well described by linear models. However, its high variance makes it particularly sensitive to the temporal composition of the training data when conflicting pre-COVID and post-COVID growth patterns are present in the same training window, as in Trial 2, the gating mechanisms may struggle to reconcile the fundamentally different sequential dynamics of the two periods, leading to unstable and potentially degraded performance. In Trial 3, where the training data contains only post-COVID observations, LSTM is expected to benefit from the more homogeneous temporal dynamics of the acceleration phase, though the small training sample size of four years per country may limit its ability to fully exploit its architectural capacity. [WGW19] demonstrate that neural network architectures combining sequential learning with feature decomposition outperform individual models for non-stationary EV sales series, supporting the theoretical motivation for including LSTM as a base learner in the proposed stacking architecture.

2.5.5 Random Forest

Mathematical Formulation

Random Forest is an ensemble learning method introduced by [Bre01] that constructs a large number of decision trees during training and outputs their aggregated prediction. Each tree in the forest is trained on a bootstrap sample of the training data, a random sample of n observations drawn with replacement and at each node split, only a random subset of m features out of the total p features is considered as candidates for splitting. The prediction of the Random Forest for a new observation $\mathbf{x}$ is the average of the predictions of all T trees:

$$\hat{f}_{RF}(\mathbf{x}) = \frac{1}{T}\sum_{t=1}^{T} \hat{f}_t(\mathbf{x}) \tag{2.42}$$

where $\hat{f}_t(\mathbf{x})$ is the prediction of the t-th decision tree. Each tree is grown to maximum depth without pruning, producing individual predictors of low-bias but high-variance. The aggregation step reduces the overall variance of the ensemble without substantially increasing bias, exploiting the variance reduction property of averaging [Bre01].

The key innovation of Random Forest relative to simple bagging where trees are averaged but all features are considered at each split is the subsampling of random characteristics at each node, which decorates individual trees

[Bre01]. For T trees with pairwise correlation ρ and individual variance σ^2, the variance of the averaged prediction is:

$$\text{Var}\left(\hat{f}_{RF}\right) = \rho\sigma^2 + \frac{1-\rho}{T}\sigma^2 \tag{2.43}$$

As $T \to \infty$, the second term vanishes and the ensemble variance converges to $\rho\sigma^2$. By reducing the correlation ρ between trees through random feature subsampling, Random Forest achieves lower ensemble variance than bagging for the same number of trees [Bre01].

Assumptions

Random Forest makes minimal parametric assumptions about the data [Bre01]. It does not assume linearity, normality, homoscedasticity, or stationarity, making it broadly applicable across a wide range of data types and distributions. It is also invariant to monotone transformations of the features and robust to irrelevant features, as the random feature subsampling at each node naturally dilutes the influence of uninformative predictors over the ensemble.

Bias-Variance Trade-off

Random Forest achieves a favorable bias-variance trade-off through the combination of deep unpruned trees which have low bias but high variance individually and averaging across many decorrelated trees, which reduces variance without substantially increasing bias [Bre01]. The number of trees T and the feature subsampling parameter m are the primary hyperparameters governing this trade-off. In this study, $T = 100$ trees were used.

A Priori Assessment

Random Forest is expected to perform competitively in scenarios where the relationship between BEV adoption rates and their drivers is non-linear and where interactions between features are important. Its robustness to irrelevant features and its resistance to overfitting through averaging make it a strong general-purpose baseline. However, its reliance on bootstrap sampling from the training data means that it is fundamentally limited by the distributional properties of the training set if the training data does not contain observations representative of the post-COVID acceleration phase, Random Forest cannot extrapolate beyond the range of its training distribution. This limitation is expected to appear in Trial 1, where the model is trained exclusively on pre-COVID observations and tested on a period characterized by fundamentally different adoption dynamics.

2.5.6 XGBoost

Mathematical Formulation

XGBoost is a scalable and regularized gradient boosting framework introduced by [CG16]. Gradient boosting builds an ensemble of decision trees sequentially, with each tree trained to correct the residual errors of the cumulative ensemble from the previous iteration. The prediction of a gradient boosting model with T trees is:

$$\hat{y}_i = \sum_{t=1}^{T} f_t(\mathbf{x}_i), \quad f_t \in \mathcal{F} \tag{2.44}$$

where $\mathcal{F}$ is the space of regression trees. At each iteration t, a new tree f_t is added to minimize the regularized objective:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)\right) + \Omega(f_t) \tag{2.45}$$

where l is a differentiable loss function and $\Omega(f_t)$ is the regularization term defined as:

$$\Omega(f_t) = \gamma T_t + \frac{1}{2}\lambda \left\|\mathbf{w}_t\right\|^2 \tag{2.46}$$

where T_t is the number of leaves in tree t, $\mathbf{w}_t$ is the vector of leaf weights, γ penalizes tree complexity, and λ penalizes large leaf weights [CG16]. Using a second-order Taylor expansion of the loss function, the optimal weight for leaf j is:

$$w_j^* = -\frac{\sum_{i\in I_j} g_i}{\sum_{i\in I_j} h_i + \lambda} \tag{2.47}$$

where $g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$ and $h_i = \partial^2_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$ are the first and second order gradients of the loss function [CG16].

Assumptions

XGBoost makes minimal distributional assumptions and is applicable to a wide range of regression problems[CG16]. Like Random Forest, it does not assume linearity, normality, or stationarity. The regularization terms γ and λ provide explicit control over model complexity, making XGBoost more resistant to overfitting than standard gradient boosting. Early stopping is used only if the validation score does not improve for a specified number of rounds, the training is terminated. This provides additional protection against overfitting in this study.

Bias-Variance Trade-off

XGBoost achieves a lower bias than Random Forest by sequentially fitting residuals. Each tree is specifically trained to correct the errors of the previous ensemble, allowing the model to progressively reduce systematic bias [CG16]. However, this sequential fitting also makes XGBoost more prone to overfitting than Random Forest if the regularization parameters are not carefully tuned, as each new tree can potentially memorize noise in the training residuals.

A Priori Assessment

XGBoost is expected to be a strong competitor in this study given its demonstrated empirical performance across a wide range of structured regression tasks. However, like Random Forest, its inability to extrapolate beyond the range of its training distribution is expected to limit its performance in Trial 1. The early stopping mechanism employed in this study uses an internal validation set of 10% of the training data that provides a degree of protection against overfitting, but cannot compensate for the fundamental distributional shift between the pre-COVID training period and the post-COVID test period.

2.5.7 Stacking Ensemble

Mathematical Formulation

Stacking, also known as stacked generalization is an ensemble learning framework in which the predictions of multiple base learners are used as input features for a second-level meta-learner, which learns the optimal combination of the base learner outputs [Wol92]. For M base learners $\{f_1, f_2, \ldots, f_M\}$ trained on a training dataset $\mathcal{D}_{train}$, the stacking procedure generates a meta-feature matrix:

$$\tilde{\mathbf{X}} = \begin{bmatrix} f_1(\mathbf{x}_1) & f_2(\mathbf{x}_1) & \cdots & f_M(\mathbf{x}_1) \\ f_1(\mathbf{x}_2) & f_2(\mathbf{x}_2) & \cdots & f_M(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(\mathbf{x}_n) & f_2(\mathbf{x}_n) & \cdots & f_M(\mathbf{x}_n) \end{bmatrix} \tag{2.48}$$

where each column contains the predictions of one base learner across all observations. The meta-learner g is then trained on $\tilde{\mathbf{X}}$ with the true target values $\mathbf{y}$ as the response:

$$\hat{y} = g(\tilde{\mathbf{x}}) = g(f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_M(\mathbf{x})) \tag{2.49}$$

In this study, $M = 2$ base learners are used — ARIMAX and LSTM, with their predictions on both the training and test periods forming the meta-feature matrix. The motivation for selecting ARIMAX and LSTM as base learners is their complementary modelling strengths: ARIMAX captures linear temporal autocorrelation and the influence of exogenous drivers, while LSTM captures non-linear sequential dependencies. Three variants of the meta-learner are evaluated, as described below.

ElasticNet Meta-Learner

ElasticNet regression was introduced by [ZH05] as a regularized regression method that combines the L1 penalty of the Lasso with the L2 penalty of Ridge regression:

$$\hat{\boldsymbol{\beta}}_{EN} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left\| \mathbf{y} - \tilde{\mathbf{X}}\boldsymbol{\beta} \right\|^2 + \lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\beta}\|_2^2 \tag{2.50}$$

where λ_1 and λ_2 control the strength of the L1 and L2 penalties respectively. The L1 penalty encourages sparsity which drives some coefficients to exactly zero while the L2 penalty stabilizes the solution when predictors are correlated [ZH05]. In the meta-learning context, ElasticNet assigns sparse and stable weights to the base learner predictions, effectively selecting the most informative combination of ARIMAX and LSTM outputs while preventing either learner from dominating the ensemble through an unstable coefficient. A positivity constraint was applied in this study to ensure that both base learners contribute positively to the final prediction, reflecting the assumption that both ARIMAX and LSTM provide meaningful information about BEV adoption dynamics.

Gradient Boost Meta-Learner

A Gradient Boosting Regressor was used as an alternative meta-learner to capture potential non-linear interactions between the ARIMAX and LSTM predictions [Fri01]. Gradient Boosting builds an additive ensemble of weak learners that typically gives shallow decision trees by sequentially fitting each new learner to the residual errors of the cumulative ensemble from the previous iteration. The ensemble prediction after T iterations is:

$$\hat{y} = \sum_{t=1}^{T} \nu \cdot h_t(\tilde{\mathbf{x}}) \tag{2.51}$$

where h_t is the t-th weak learner fitted to the residuals of the previous ensemble, $\nu \in (0, 1]$ is the learning rate that controls the contribution of each tree, and $\tilde{\mathbf{x}} = [\hat{y}_{ARIMAX}, \hat{y}_{LSTM}]$ is the two-dimensional meta-feature vector of base learner predictions. At each iteration t, the new learner h_t is fitted to the negative gradient of the loss function with respect to the current ensemble predictions:

$$h_t = \underset{h}{\operatorname{argmin}} \sum_{i=1}^{n} \left[-\frac{\partial L(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}} - h(\tilde{\mathbf{x}}_i) \right]^2 \tag{2.52}$$

For the squared error loss function $L(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$, the negative gradient reduces to the ordinary residual $y_i - \hat{y}_i^{(t-1)}$, and the algorithm reduces to iterative residual fitting [Fri01]. The ensemble is updated as:

$$\hat{y}^{(t)} = \hat{y}^{(t-1)} + \nu \cdot h_t(\tilde{\mathbf{x}}) \tag{2.53}$$

In the meta-learning context, the Gradient Boosting regressor operates on the two-dimensional meta-feature space $\tilde{\mathbf{x}} = [\hat{y}_{ARIMAX}, \hat{y}_{LSTM}]$, learning a non-linear combination function that maps the base learner predictions to the final ensemble output. This allows the meta-learner to capture interactions between the ARIMAX and LSTM predictions that a linear combination such as ElasticNet or the weighted average. As a results the approach provides additional flexibility in scenarios where the optimal combination of the two base learners varies non-linearly across the range of predicted values [Fri01].

Weighted Average Meta-Learner

The weighted average combination method was introduced by [BG69] as one of the earliest and most theoretically grounded approaches to forecast combination. The combined forecast is defined as:

$$\hat{y} = w \cdot \hat{y}_{ARIMAX} + (1 - w) \cdot \hat{y}_{LSTM} \tag{2.54}$$

where $w \in [0, 1]$ is the weight assigned to the ARIMAX prediction and $(1 - w)$ is the complementary weight assigned to the LSTM prediction. The optimal weight is determined by minimizing the Mean Squared Error of the combined forecast on the training data:

$$w^* = \underset{w \in [0,1]}{\text{argmin}} \frac{1}{n} \sum_{i=1}^{n} (y_i - w\hat{y}_{ARIMAX,i} - (1 - w)\hat{y}_{LSTM,i})^2 \tag{2.55}$$

[BG69] demonstrate theoretically that a weighted combination of forecasts will often generally outperform the individual component forecasts, provided that the component forecasters make uncorrelated errors, a condition that is approximately satisfied in this study by the complementary modelling approaches of ARIMAX and LSTM.

Assumptions

The stacking framework assumes that the base learners capture complementary aspects of the target variable's dynamics, such that their combined predictions contain more information than either individual prediction alone [Wol92]. It also assumes that the meta-learner can be trained on a sufficient number of observations to learn a reliable combination rule. In this study, the small number of country-year observations per cluster represents a potential limitation for the more complex meta-learners particularly the Gradient Boosting meta-learner which may overfit the training-period base learner predictions.

Bias-Variance Trade-off

The stacking ensemble inherits the bias-variance characteristics of its base learners but is designed to reduce the overall prediction error by leveraging their complementary strengths [Wol92]. The ElasticNet meta-learner introduces regularization that prevents the combination from being dominated by the noisier of the two base learners, while the weighted average provides a simple and interpretable combination that is robust to the small sample sizes encountered in this study.

Theoretical Justification for Base Learner Selection

The selection of ARIMAX and LSTM as base learners for the proposed stacking architecture is grounded in both theoretical and empirical considerations. From a theoretical perspective, the two models capture fundamentally different and complementary aspects of the BEV adoption series. ARIMAX is a parametric linear model that explicitly models temporal autocorrelation and the linear influence of exogenous macroeconomic, infrastructural, and technological drivers, making it well suited to capturing the stable and predictable component of BEV adoption dynamics [Box+08]. LSTM, by contrast, is a non-parametric deep learning architecture that learns non-linear sequential dependencies through its gating mechanisms, capturing threshold effects, acceleration phases, and structural transitions that are beyond the representational capacity of linear models [HS97].

The theoretical basis for combining these two models rests on the ensemble learning principle established by [BG69], which demonstrates that a weighted combination of forecasters will outperform its individual components when those components make weakly correlated errors. Because ARIMAX and LSTM differ fundamentally in their modelling assumptions and learning mechanisms, the errors they produce are unlikely to be strongly correlated. ARIMAX will systematically misestimate where the true relationship is non-linear, while LSTM is prone to overfit where the true relationship is linear and data are insufficient to support deep learning. Their combination therefore provides error cancellation that neither model can achieve alone [Wol92].

The choice of meta-learner further reinforces the theoretical properties of the ensemble. The ElasticNet meta-learner introduces L1 and L2 regularization that stabilizes the combination coefficients under the small sample sizes encountered in this study, preventing either base learner from dominating the ensemble through unstable coefficients [ZH05]. The Gradient Boosting meta-learner captures potential non-linear interactions between the ARIMAX and LSTM predictions that a linear combination cannot represent, providing an additional layer of flexibility [Fri01]. The weighted average meta-learner offers the most interpretable combination, directly quantifying the relative contribution of each base learner and providing theoretical guarantees of potential improvement over the individual forecasters under the conditions established by [BG69].

From a bias-variance perspective, the proposed architecture is designed to achieve a balanced trade-off across the three meta-learner variants. The ensemble bias is approximately:

$$\text{Bias}(\hat{y}_{ensemble}) \approx w \cdot \text{Bias}(\hat{y}_{ARIMAX}) + (1 - w) \cdot \text{Bias}(\hat{y}_{LSTM}) \tag{2.56}$$

Since ARIMAX has higher bias but lower variance and LSTM has lower bias but higher variance, their combination produces an intermediate bias-variance profile that is expected to outperform both individual models in terms of overall prediction error, particularly in the post-COVID period where both linear temporal structure and non-linear acceleration dynamics are simultaneously present in the target series.

A Priori Assessment

The proposed stacking architecture is expected to outperform all individual baseline models in Trial 3, where the post-COVID training data provides a coherent and structurally consistent basis for both base learners. The combination of ARIMAX's linear temporal structure and LSTM's non-linear sequential learning is expected to produce predictions that are more accurate and more robust to country-level heterogeneity than either model alone. In Trial 1, the proposed model is expected to be limited by the same structural break that constrains all models trained on pre-COVID data, but the ensemble combination is expected to provide some degree of error cancellation, as the errors of ARIMAX and LSTM are not perfectly correlated resulting in modestly better performance than the individual base learners in some clusters.

Based on the assumptions and theoretical considerations discussed above, Table 2.1 summarizes the justification for selecting the respective models.

Model	**Justification for Inclusion in BEV Forecasting**
Linear Regression	Serves as a simple and interpretable baseline model to establish a minimum performance benchmark against which more complex models can be evaluated.
SVR	Captures non-linear relationships and threshold effects expected in BEV adoption dynamics, particularly during rapid growth phases.
ARIMAX	Models temporal autocorrelation and incorporates macroeconomic and technological drivers influencing BEV adoption.
LSTM	Captures non-linear temporal dynamics, acceleration phases, and structural breaks in BEV adoption patterns.
Random Forest	Captures feature interactions and heterogeneous country-level effects in BEV adoption drivers.
XGBoost	Provides high predictive accuracy and reduces bias through sequential residual learning in structured tabular datasets.
Stacking Ensemble	Combines linear temporal structure from ARIMAX and non-linear sequential learning from LSTM to improve predictive robustness.

Table 2.1: Justification for Model Selection

2.5.8 Validation Practices in EV Forecasting Literature

A recurring weakness identified across the EV forecasting literature is the inconsistency and insufficiency of model validation practices. [DC23] and [KSP24] both highlight that many existing studies either neglect formal out-of-sample validation entirely or perform only in-sample assessments, where the model is evaluated on the same data used for training. Such in-sample evaluations systematically overestimate predictive accuracy by failing to assess the model's ability to generalise to unseen data.

Few studies in the EV forecasting literature implement rolling-origin or cross-validation methods that are standard practice in time series forecasting, and sensitivity analyses and scenario testing that is essential for evaluating model robustness under policy or market uncertainty, however, remain inconsistent across model types. Furthermore, the majority of studies rely exclusively on a single accuracy metric such as MAPE or RMSE, without examining the distribution of prediction errors across heterogeneous market contexts. As a result, cross-study comparison of forecasting performance remains difficult, and the reproducibility of many reported results is uncertain. This study addresses these validation gaps through a strict out-of-sample evaluation framework across three temporally distinct trials; although cross-validation was applied within the ElasticNet meta-learner and a 20% validation split was used inside LSTM, full rolling-origin or time-series cross-validation was not feasible due to the limited number of annual observations. Performance were assessed using two complementary metrics — *MAE* and MedAE that are reported separately for two market clusters across all 27 EU member states.

2.6 Evaluation Metrics

The selection of appropriate evaluation metrics is a fundamental component of any forecasting study, as the choice of metric directly determines how model performance is measured, compared, and interpreted. Different metrics emphasize different aspects of prediction error; some penalize large errors disproportionately, others are robust to outliers, and some are scale-dependent while others are normalized. In the context of this study, where BEV adoption rates vary enormously across EU member states from near-zero values in smaller markets to over 50% in front runner countries; the choice of metric must account for the heterogeneity of the target variable across the panel. This section reviews the theoretical foundations of the two evaluation metrics used in this study, *MAE* and MedAError, and establishes the information-theoretically sound framework for evaluating and comparing forecasting performance across models, clusters, and trials.

2.6.1 *MAE*

Mathematical Formulation

MAE is one of the most widely used metrics for evaluating the accuracy of regression and forecasting models [WM05]. For n observations with true values y_i and predicted values $\hat{y}_i$, the *MAE* is defined as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{2.57}$$

The *MAE* measures the average magnitude of prediction errors across all observations, giving equal weight to every error regardless of its size. It is expressed in the same units as the target variable. *BEV%* as a target variable in this study makes it directly interpretable as the average absolute deviation between predicted and actual adoption rates across all country-year observations within a cluster and trial.

Properties

The *MAE* possesses several desirable properties that support its use as a primary evaluation metric [WM05]. It is non-negative, with a value of zero indicating perfect prediction. It is symmetric, meaning that it overestimates and underestimates of the same magnitude contribute equally to the metric. It is a convex function of the prediction errors, ensuring that it has a unique minimum. Moreover, it is also scale-dependent, meaning that its value is directly comparable across models evaluated on the same dataset but not across datasets with different scales or units.

Limitations

The primary limitation of *MAE* in heterogeneous panel settings is its sensitivity to extreme values [WM05]. Because *MAE* averages the absolute errors across all observations, a small number of country-year observations with extremely large prediction errors such as those arising from the rapid and unpredictable acceleration of BEV adoption in frontrunner markets like the Netherlands or Sweden in the early 2020s can substantially inflate the reported *MAE*, potentially misrepresenting the typical prediction accuracy of the model across the majority of observations. This limitation motivates the use of the Median Absolute Error as a complementary metric.

2.6.2 MedAE

Mathematical Formulation

MedAE replaces the arithmetic mean in the *MAE* formula with the statistical median, producing a measure of the typical prediction error that is robust to the influence of extreme outliers [WM05]. The MedAE is defined as:

$$MedAE = \text{median}\left(|y_1 - \hat{y}_1|, |y_2 - \hat{y}_2|, \ldots, |y_n - \hat{y}_n|\right) \tag{2.58}$$

The MedAE reports the value at the 50th percentile of the distribution of absolute errors, meaning that 50% of all country-year predictions have an absolute error below the reported MedAE and 50% have an error above it. Like *MAE*, it is expressed in the same units as the target variable and is directly interpretable as a percentage point deviation between predicted and actual BEV market share.

Properties

The key advantage of MedAE over *MAE* in the context of this study is its robustness to the influence of extreme prediction errors [WM05]. The median is a breakdown point estimator with a breakdown point of 0.5, meaning that up to 50% of the observations can take arbitrarily extreme values without affecting the median. In the context of BEV market forecasting across 27 heterogeneous EU member states, where a small number of frontrunner countries such as *Netherlands*, *Sweden*, and *Denmark* record adoption rates several times the EU average, MedAE provides a more representative measure of the prediction accuracy experienced by the typical country within each cluster than *MAE*, which would be inflated by the extreme errors associated with these outlier markets.

2.6.3 Complementarity of MAE and MedAE

The combined use of *MAE* and *MedAE* as evaluation metrics in this study reflects a deliberate strategy of characterizing the full distribution of prediction errors rather than relying on a single summary statistic [WM05]. *MAE* captures the overall average prediction accuracy across all country-year observations, including the influence of extreme errors in high-adoption markets. *MedAE* captures the typical prediction accuracy experienced by the median country, providing a measure that is representative of the majority of the panel rather than being dominated by outlier markets.

The divergence between *MAE* and *MedAE* for a given model provides diagnostic information about the distribution of its prediction errors. A large gap between *MAE* and *MedAE* indicates that the model produces a small number of very large errors that inflate the mean, while convergence between the two metrics suggests a more symmetric and uniformly distributed error profile.

2.6.4 Framework for Evaluating and Comparing Forecasting Performance

Drawing on the information-theoretic principles established by [Aka74] for model comparison and the forecasting evaluation framework of [WM05], this study adopts the following principled framework for evaluating and comparing model performance:

1. **Metric selection:** *MAE* and MedAE are computed on the absolute prediction errors in the original percentage scale of *BEV %*, following the back-transformation of log-scale predictions using the exponential function.This ensures that the metrics directly reflect the magnitude of prediction errors in the units of primary interest to policymakers and industry stakeholders.

2. **Cluster-level aggregation:** Metrics are aggregated separately for the *Leaders* and *Followers* clusters, reflecting the fundamental structural difference between high-adoption and low-adoption EU markets. Aggregating across all 27 member states would obscure the systematic differences in model performance between the two groups and reduce the diagnostic value of the evaluation.

3. **Trial-level comparison:** Metrics are reported for all three trials, enabling a direct assessment of how the temporal composition of the training data affects predictive accuracy for each model. This cross-trial comparison is the primary mechanism through which the four hypotheses of this study are evaluated.

4. **Model ranking:** Models are ranked in ascending order of Trial 3 performance within each cluster, reflecting the primary objective of achieving the most accurate near-term forecasts within the post-COVID market regime. This ranking prioritizes performance in the most recent and most policy-relevant forecasting scenario over average performance across all trials.

5. **Cross-metric consistency:** Model rankings are assessed for consistency across *MAE* and *MedAE*. A model that performs well on both metrics provides stronger evidence of genuine forecasting capability than one that excels on only one metric, as the latter may reflect sensitivity to the specific error distribution rather than robust predictive accuracy.

This framework ensures that the comparison of forecasting performance across models, clusters, and trials is conducted on a consistent and theoretically grounded basis, enabling unambiguous conclusions about the relative merits of each modelling approach and the impact of training data composition on predictive accuracy in the European BEV market context.

2.7 Research Gaps

The collective body of literature reviewed in this chapter advances from understanding the BEV adoption mechanisms to forecasting methodology substantially, yet exposes a consistent set of recurring gaps that motivate the design of the present study.

1. **Geographic concentration on single markets:** The overwhelming majority of BEV forecasting studies have been conducted on single national markets, with China representing the dominant focus of the machine learning and neural network literature [WGW19] and [Zha+17].

   While these studies produce valuable methodological insights, their findings are inherently market-specific and cannot be directly generalised to the European context, where the coexistence of 27 member states with fundamentally different economic, infrastructural, and policy conditions creates a level of heterogeneity that single-market models are not designed to capture. The clustering approach of [MSJ24] represents a notable exception, demonstrating that grouping markets by structural similarity can substantially improve the analytical efficiency and policy relevance of European BEV market research. This study builds directly on this insight by applying a cluster-based forecasting framework to all 27 EU member states simultaneously.

2. **Assumption of structural stability:** Most existing forecasting models including regression, ARIMA, and early neural network approaches were calibrated on historical data under the assumption that the structural relationships between BEV adoption and its drivers remain stable over time. This assumption has been increasingly challenged by the post-2020 acceleration of the European BEV market, driven by tightening EU CO_2 emissions regulations, large-scale government stimulus packages, and rapid improvements in vehicle technology and charging infrastructure [LH21]. The structural break introduced by the COVID-19 pandemic and its aftermath fundamentally altered the growth dynamics of BEV adoption across all EU member states, rendering models trained exclusively on pre-COVID data potentially unreliable for near-term forecasting. Remarkably, despite the significance of this structural break, no existing study has systematically investigated how the temporal composition of training data specifically the inclusion or exclusion of the post-COVID period from 2020 onwards affects the forecasting accuracy of different model classes across heterogeneous European markets.

3. **Short temporal horizons and data limitations:** Many existing studies use relatively short observation periods, often fewer than five years of data which constrains the long-term predictive reliability of the fitted models and limits the ability to detect structural changes in adoption dynamics [KSP24]. The limited availability of harmonized, multi-country, multi-variable panel data covering the full lifecycle of BEV market development from early diffusion through mainstream adoption has further constrained the scope and generalization of prior research. This study addresses this limitation by constructing a comprehensive country-year panel covering all 27 EU member states from 2011 to 2024, spanning both the early diffusion stage and the post-2020 acceleration phase within a single harmonized dataset.

4. **Methodological fragmentation:** The EV forecasting literature remains divided between interpretable causal and statistical models and flexible but opaque machine learning frameworks, with few studies systematically comparing model classes across a common experimental framework [DC23]. This fragmentation makes cross-study comparison of forecasting performance difficult and prevents the identification of robust conclusions about the relative merits of different modelling approaches. This study addresses this gap by evaluating nine model classes spanning Linear Regression, SVR, ARIMAX, LSTM, Random Forest, XGBoost and three variants of the proposed stacking ensemble within a unified experimental framework applied consistently across all models, clusters, and trials.

5. **Lack of ensemble approaches tailored to structural breaks:** Despite the theoretical advantages of ensemble and stacking methods for combining complementary model strengths, no existing EV forecasting study has proposed and evaluated a stacking architecture specifically designed to leverage the complementary properties of linear time series models and deep learning architectures in the context of a structural market break. The proposed ARIMAX-LSTM stacking framework in this study directly addresses this gap, combining the linear temporal structure of ARIMAX with the non-linear sequential learning of LSTM through a data-driven meta-learner, and evaluating its performance systematically across three temporally distinct training regimes.

6. **Weak validation protocols:** As discussed in Section 2.6, many existing studies rely on in-sample evaluation or single-metric assessment, limiting the credibility and reproducibility of their reported results [DC23] and [LH21]. This study addresses this limitation through a strict out-of-sample evaluation framework, evaluating all models on held-out test periods that the models have not been exposed to during training, and reporting two complementary metrics — *MAE* and MedAE, separately for two market clusters to provide a comprehensive and robust characterization of forecasting performance.

The following table gives us a quick summary to the research gaps mentioned in 2.7

Table 2.2: Comparison of studies across key aspects

Study	G1	G2	G3	G4	G5	G6
Sierzchula et al. (2014) [Sie+14]	●	○	○	○	○	○
Zhang et al. (2016) [Zha+16]	○	○	○	◐	○	○
Zhang et al. (2017) [Zha+17]	○	○	○	◐	○	◐
Wang et al. (2019) [WGW19]	○	○	○	○	◐	◐
Lieven & Hügler (2021) [LH21]	●	◐	◐	○	○	○
Domarchi & Cherchi (2023) [DC23]	●	◐	◐	●	○	◐
Möring-Martínez et al. (2024) [MSJ24]	●	○	●	○	○	◐
Kemala et al. (2024) [KSP24]	●	◐	◐	●	○	◐
This study (2026)	●	●	●	●	●	●

○ not fulfilled ◐ partially fulfilled ● fully fulfilled

G1 = Geographic scope of study (single-country to multiple countries), **G2** = Incorporation of structural breaks, **G3** = Temporal coverage (short to long-term history), **G4** = Method comparison (single model to multiple model comparison), **G5** = Use of hybrid or ensemble approaches, **G6** = Validation rigor (in-sample to robust out-of-sample evaluation)

3 Methodology

3.1 Data

This section describes the data used in this study, including the target variable, features, and their sources.

3.1.1 Data Variables (Features and Target) and Sources

The dataset consists of two types of variables - country level variables which carry distinct values for each member of the EU27 countries, and EU-level variables which are the annual averages across all the countries and therefore assigned uniformly for each country.

The following table provides an overview of the variables used in this study, along with their categories and sources.

Variable	Category	Source	Level
AF Market share of total registrations (M1)	Target	EAFO- [Eur25b]	Country-level
GDP (annual, constant prices)	Socioeconomic	Eurostat-[Off25b]	Country-level
Overall employment growth	Socioeconomic	Eurostat-[Off25a]	Country-level
Consumer Price Index	Socioeconomic	IMF-[Int25]	Country-level
Number of recharging points (–2019)	Infrastructure	EAFO-[Eur25b]	Country-level
Number of recharging points (2020–present)	Infrastructure	EAFO-[Eur25b]	Country-level
Average retail price of BEVs	Market / Cost Factor	EAFO-[Eur25a]	EU-level average
Average range of BEV models	Technological Performance	EAFO-[Eur25a]	EU-level average
Average battery capacity (kWh)	Engineering Feature	EAFO-[Eur25a]	EU-level average
Number of available BEV models in Europe	Market Structure	EAFO-[Eur25a]	EU-level average
Average AC and DC recharging speed (km/h)	Infrastructure	EAFO-[Eur25a]	EU-level average

Table 3.1: Overview of data variables, their categories, sources, and level of aggregation.

3.1.2 Timelines

The data set spans 2011-2024 with observations recorded on an annual basis, yielding 14 data points for each country. This period covers both the early diffusion stage of BEVs and the acceleration phase observed in the 2020s, ensuring an adequate representation of evolving market dynamics.

3.1.3 Countries Included

The analysis covers all 27 member states of the EU, ensuring geographic, socio-economic and infrastructural diversity across the data set.

3.2 Design and Implementation of the Experiments

3.2.1 Hypothesis

The COVID-19 pandemic marked a structural turning point for BEV adoption in Europe. Despite broad economic disruptions, EV sales did not stall but instead continued to rise, defying the overall decline in vehicle registrations [LH21]. This acceleration was driven by a combination of factors, including heightened public awareness of carbon emissions and increased consumer interest in cleaner mobility alternatives. These market dynamics, which became particularly pronounced from 2020 onwards, justify the selection of this year as a structural break point in the trial design of this study. With this context in mind, four hypotheses are proposed as follows.

Hypothesis	Description
H1	Models trained on 2011–2019 data will show a significant increase in error when applied to 2020–2024 test data, suggesting structural non-stationarity in market drivers.
H2	Extending the training data to 2022 will improve predictive performance; however, the "historical dominance" of the pre-COVID market in the 2010s will result in a systematic underestimation of the current market pace.
H3	Models trained exclusively on post-COVID data from 2020–2023 will yield the highest predictive performance, indicating that recent market dynamics are governed by a distinct set of drivers compared to the pre-COVID era.
H4	The 27 EU member states exhibits a stable two-cluster structure, namely *Leaders* and *Followers* based on their BEV adoption characteristics that persists across all three training configurations, and cluster-specific models. The country-level membership in the clusters may evolve over time.

Table 3.2: Research hypotheses for BEV market dynamics.

3.2.2 Data cleaning and Imputations

Data Merging and Integration

The Recharging points variables was constructed by combining two different datasets - Number of recharging points (– 2019) and Number of recharging points (2020 – present) into a single continuous series of data points.

Then some of the datasets exhibited inconsistencies in temporal coverage, with certain countries starting earlier from 2008–2011 than others. To ensure a harmonized and comparable dataset across all 27 EU countries, a multi-stage merging process was applied.

In the first step, all country-level features were merged into a single panel dataset using country and year as the common keys. In the second step, this consolidated feature dataset was merged with the target variable, representing the annual market share of alternative-fuel vehicle registrations for each country. This integration produced a complete country–year panel containing both the predictors and the target outcome. Finally, the merged dataset was combined with the EU-level average dataset. Since these EU-level variables represent averages across all countries for each year, the same values were assigned to all 27 countries for the corresponding year to maintain temporal consistency across the full panel.

Following this merge, only observations from the year 2011 to 2024 were retained to ensure uniformity across the panel and to align with the scope defined by the research hypotheses. The shape of the resultant data set was 378 rows and 13 columns.

Imputing Missing Values

Following the data merging process, the consolidated data set was examined for completeness. Table 3.3 summarizes the non-null counts, data types, and extent of missing values across all variables.

Variables	Non-Null Values	Missing Values	Total	Data Type
Country	378	0	378	object
Year	378	0	378	int64
BEV Percentage (Total Registrations)	359	19	378	float64
GDP	378	0	378	float64
CPI	378	0	378	float64
Employment Growth (EG)	378	0	378	float64
Recharging Points	328	50	378	float64
AC Recharging Speed (km/h)	324	54	378	float64
DC Recharging Speed (km/h)	324	54	378	float64
Available BEV Models	351	27	378	float64
Battery Capacity (kWh)	324	54	378	float64
Real Range (km)	324	54	378	float64
Purchase Price (EUR)	297	81	378	float64

Table 3.3: Summary of data completeness and variable data types after merging.

The key variables affected were: *BEV Percentage (Target), Recharging Points, AC Recharging Speeds DC Recharging Speeds, Available Models, Battery Capacity, Real Range*, and *Purchase Price*. In contrast, socioeconomic indicators such as *GDP, Consumer Price Index (CPI)*, and *Employment Growth (EG)* were complete and therefore did not require any imputation.

Following the examination of missing values, appropriate imputation methods were selected based on the nature of each variable and the pattern of missingness observed. The study uses three different methods to fill in the missing values - Log-Linear Interpolation, CAGR-Based Extrapolation and Hybrid method.

Given that the majority of missing values occurred at the boundaries of the study period, particularly in the early years when BEV markets were in the infant stage, standard linear interpolation was deemed unsuitable as it does not account for the multiplicative growth dynamics characteristic of emerging technologies. Two primary methods were therefore employed across the variables: log-linear interpolation and CAGR based extrapolation, applied individually or in combination depending on the structure of the missing data.

Log-Linear Interpolation

Log-linear interpolation was selected for variables exhibiting exponential growth patterns, where values grow multiplicatively rather than additively over time. By applying logarithm transformation to the series, the exponential trend is linearized, allowing standard linear interpolation to be applied in log space. The interpolated values are then converted back to the original scale through exponentiation. This approach preserves non-negativity, avoids unrealistic jumps between observed values, and adapts naturally to the growth dynamics observed in BEV-related variables [LAC17].

This method was applied to the target variable - *BEV Percentage (Total Number Of Registrations)*, where missing values were confined to specific smaller or less developed EU markets in the early years of the study period, reflecting delayed adoption rather than data error. Log-linear interpolation was also applied to *Purchase Price (EUR)* across all 27 countries for the years 2011, 2012, and 2024. CAGR extrapolation was not suitable in this case as missing values were present at both ends of the time series for every country, making it impossible to compute a reliable boundary-anchored growth rate. The observed linear trend in log-transformed prices between 2013 and 2023 confirmed the appropriateness of this method [LR02].

CAGR-Based Extrapolation

For terminal missing values, where observations were absent at the end of the time series rather than within it, CAGR-based extrapolation was applied. The CAGR was computed from the last three observed years for each variable, and the resulting growth rate was used to project values forward. This method is well suited for extrapolating near-term values when a stable historical trend is present and sufficient observations are available [LAC17]. CAGR extrapolation was applied to *Available BEV Models*, where only the year 2024 was missing, and to *Real Range (km)*, where both 2023 and 2024 were absent. In both cases, the preceding years provided a sufficient and stable basis for computing the growth rate.

Hybrid Approach

For *AC Recharging Speed (km/h)*, *DC Recharging Speed (km/h)*, and *Battery Capacity (kWh)*, a hybrid approach was employed, as missing values were present both within the time series and at the terminal year. Log-linear interpolation was first applied to fill internal gaps, specifically the year 2012, by capturing the multiplicative growth trend between two known observations. CAGR-based extrapolation was subsequently applied to project the missing 2024 values forward from the last three observed years. This sequential combination ensures that interpolated values reflect local growth trends, while extrapolated terminal values remain consistent with the most recent trajectory of each variable [LAC17].

For *Recharging Points*, missing values were concentrated in the early years across most countries, reflecting the limited charging infrastructure present at the outset of the study period. Backward log-linear extrapolation was applied, deriving the extrapolation slope from the first two observed values for each country. Where only a single observation was available, a conservative default growth rate was applied to avoid overfitting sparse data. Zero values were replaced with missing prior to imputation, as a country with recorded BEV activity is assumed to have had at least minimal charging infrastructure [LR02].

3.2.3 EDA

EDA was performed to understand the statistical properties, temporal evolution, and inter-dependencies of the variables prior to model training was performed.

Descriptive Analysis

Table 3.4 represents the descriptive statistics for *Country-level* variables. Table 3.5 represents the descriptive statistics for *EU-level average* variables over 378 observation from 27 EU countries.

Statistic	BEV %	GDP	CPI	EG	Recharging Points
Count	378	378	378	378	378
Mean	3.82	2,747,938	2.62	1.13	24,009
Std	7.44	9,282,203	3.28	1.96	80,950
Min	0.00	7,033	-2.10	-5.60	0
25%	0.11	57,203	0.62	0.13	155
50%	0.62	308,266	1.85	1.20	1,065
75%	3.56	1,861,603	3.29	2.20	7,423
Max	51.72	81,447,660	19.71	6.90	675,302

Table 3.4: Descriptive Statistics – Country Level Variables.

The target variable *BEV %* shows a mean of 3.82% and a median of 0.62%, indicating that it is right skewed; which means that a small a number of countries are in a forefront which pushes the mean up. The maximum value of 51.72% corresponds to the highest adoption rates recorded among leading EU markets in the most recent years of the study period. The standard deviation of 7.44 confirms considerable dispersion across countries and time.

GDP ranges from 7,033 to 81,447,660 with a mean of 2,747,938 and a standard deviation of 9,282,203, reflecting the substantial economic heterogeneity across EU member states, distinguishing large economies from smaller ones. In a similar manner *Recharging points* exhibit similarly extreme dispersion, with a minimum of 0 and a maximum of 675,302, highlighting the vast infrastructure inequality that exists across the EU, where a handful of countries account for the majority of charging infrastructure.

The *CPI* ranges from -2.10 to 19.71, capturing both deflationary periods and the sharp inflationary episode observed across Europe in 2022 and 2023. *EG* records a minimum of -5.60, reflecting the severe labour market contraction, assumed to be during the COVID-19 pandemic, and a maximum of 6.90, corresponding to the recovery period that followed.

Statistic	AC Recharge	DC Recharge	Available BEV	Battery Capacity	Real Range	Purchase Price
Mean	46.26	392.90	116.66	52.55	242.32	39779.79
Std. Dev.	16.75	163.02	166.99	20.69	91.45	5690.95
Min	17.50	175.00	4.00	16.38	95.00	30469.93
Max	84.22	711.43	580.20	85.05	378.92	46225.88

Table 3.5: Descriptive statistics of EU-average variables.

Table 3.5 presents the descriptive statistics for the *EU-level average* variables. *AC recharging speed (km/h)* and *DC recharging speed (km/h)* show considerable variation over the study period, with *AC recharging speed (km/h)* ranging from 17.50 to 84.22 km/h and *DC recharging speed (km/h)* from 175.00 to 711.43 km/h, reflecting the rapid technological advancement in charging infrastructure. The number of *Available BEV models* grew from as few as 4 to a maximum of 580, with a high standard deviation of 166.99, capturing the transition from a niche market with very limited consumer choice to a diverse and competitive product landscape.

Battery Capacity (kWh) and *Real Range (km)* both demonstrate steady technological progress, increasing from 16.38 to 85.05 kWh and from 95.00 to 378.92 km respectively. *Purchase Price (EUR)*, in contrast, shows comparatively modest variation with a standard deviation of only 5,690.95 relative to a mean of 39,779.79 EUR, suggesting a relatively stable pricing trend across the study period despite the significant improvements in vehicle performance.

Temporal Analysis

To identify trends from 2011-2024, temporal aggregations were performed. *Country-level* variable such as *Recharging Points* was aggregated by summation as this accumulate across countries to form meaningful EU-wide totals. *Country-level* variables such as *BEV%*, *GDP*, *CPI*, and *EG* were aggregated by mean because *BEV%* being a pre-calculated percentage unique to each and other economic indicators captures average EU-wide economic conditions. On the other hand, *EU-level average* variables were extracted as unique annual values, since they are consistent across countries within a given year.

Figure 3.1 illustrates the co-evolution of average *BEV%* and total *Recharging Points*) infrastructure from 2011 to 2024. Both variables remained near zero until 2018, after which they experienced sharp exponential growth. The average *BEV%* across EU countries rose from below 1% in 2018 to approximately 14% by 2024, while total recharging points grew from under 200,000 to over 3,000,000 over the same period. This parallel trajectory is consistent with the descriptive statistics presented in Table 3.4, where both variables exhibited extreme right skewness driven by rapid growth in recent years. The co-growth pattern suggests a mutually reinforcing relationship between charging infrastructure availability and BEV adoption, whereby greater infrastructure encourages adoption, and growing demand incentivises further infrastructure investment.

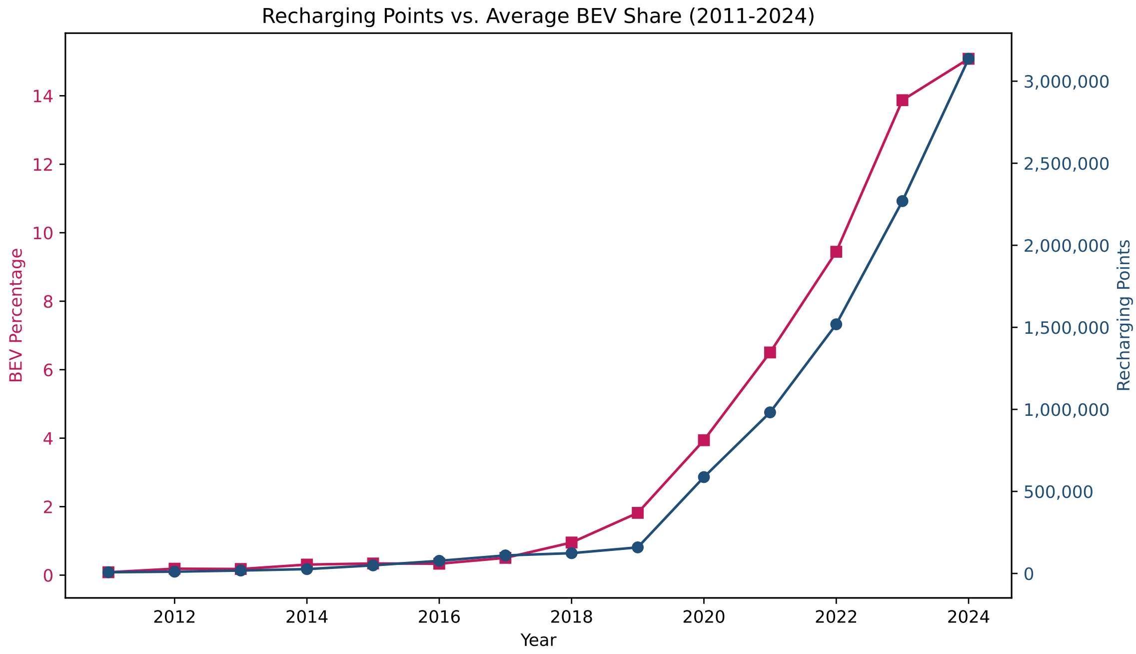

Figure 3.1: Total Recharging Points vs. Average BEV Market Share (2011–2024)

Figure 3.2 presents the Average *Real Range (km)* of BEV models over time. While the overall trajectory is upward, growing from approximately 95 km in 2011 to nearly 379 km by 2024, a notable dip is observed around 2017-2018. This is likely attributable to a shift in the composition of available models during that period, where a broader range of entry-level and shorter-range vehicles entered the market, rather than an actual technological regression. From 2019 onwards, real range increases steadily, reflecting continued improvements in battery technology and vehicle efficiency, consistent with the *Battery Capacity(kWh)* growth reported in Table 3.5.

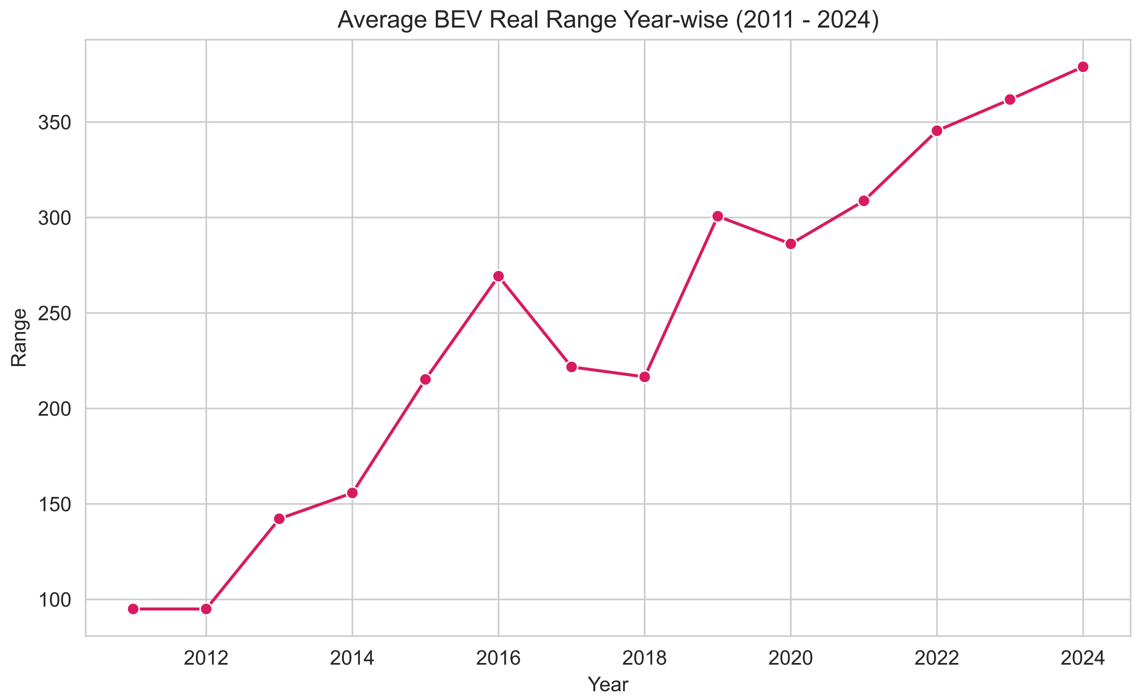

Figure 3.2: Average BEV Real Range Year-wise (2011–2024)

Figure 3.3 shows that the number of available BEV models grew modestly from just 4 models in 2011 to around 30 by 2019, after which it expanded rapidly, reaching more than 580 models by 2024. This sharp acceleration reflects the broader shift in the automotive industry toward electrification, with established manufacturers entering the BEV segment at a scale following the tightening of EU emissions regulations and improving battery economics. The high standard deviation of 166.99 reported in Table 3.5 is a direct consequence of this exponential expansion in the latter years of the study period.

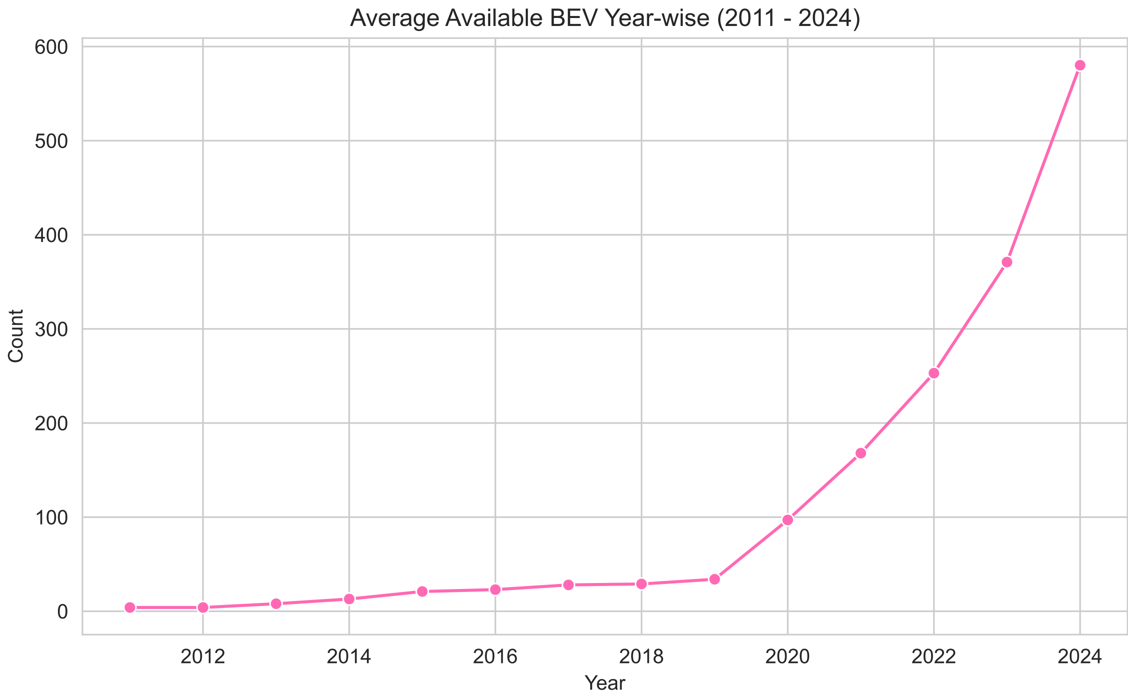

Figure 3.3: Available BEV Models Year-wise (2011–2024)

Overall, the temporal analysis reveals a clear structural shift around 2019-2020, where both market adoption and technology indicators accelerated significantly. This structural break aligns with the period following the announcement of the EU Green Deal [Eur19] in 2019 and the post-COVID economic recovery in 2020-2021 [Eur20]. The non-linear growth patterns observed across multiple variables, particularly the exponential increases in *BEV%*, *Recharging Points*, and *Available models*, highlight the presence of right skewness in several variables, which is addressed in the following section.

Distribution Assessment and Transformations

To prepare the data set for modelling, all numerical variables were examined for skewness. Variables with a skewness greater than +1 were considered highly skewed, following the classification by [MCD14].

Country-level variables exhibited right skewness due to the presence of early years with very low BEV counts and GDP disparities. Whereas, *EU-average level* variables had only one variable that was right skewed - *Available BEV models*. To reduce the skewness, a natural logarithmic transformation (log1p) was applied to all right-skewed variables.

EU-average variables were comparatively less skewed, except for the number of available BEV models (skew = +1.90), reflecting the rapid expansion in recent years.
On the other hand, *Average Purchase Price (EUR)* exhibited moderate left skewness (−0.67), indicating a longer tail toward lower price values. Logarithmic transformation is suitable for right-skewed distributions for its simplicity and interpretibility. Yeo–Johnson transformation was applied to the left-skewed variable as this approach adapts to both positive and negative skewness by optimally scaling data to approximate normality, as proposed by [Yeo00].

After applying Yeo-Johnson transformation, skewness of *Average Purchase Price (EUR)* reduced from −0.67 to -0.28, yielding a more symmetric distribution subsequent modelling [Yeo00].

Variable	Before Log Transform	After Log Transform
BEV Market Share	+3.06	+1.10
Recharging Points	+5.29	-0.21
GDP	+5.76	+0.29
CPI	+2.34	-0.58
Available	+1.90	+0.33
DC Recharging Speed (km/h)	+0.62	+0.04

Table 3.6: Skewness Before and After Log Transformation

The following visualisation shows the effect of log transformation on *BEV Market Share* and *Recharging Points* demonstrating the reduction in skewness and improved linearity of the relationship after transformation.

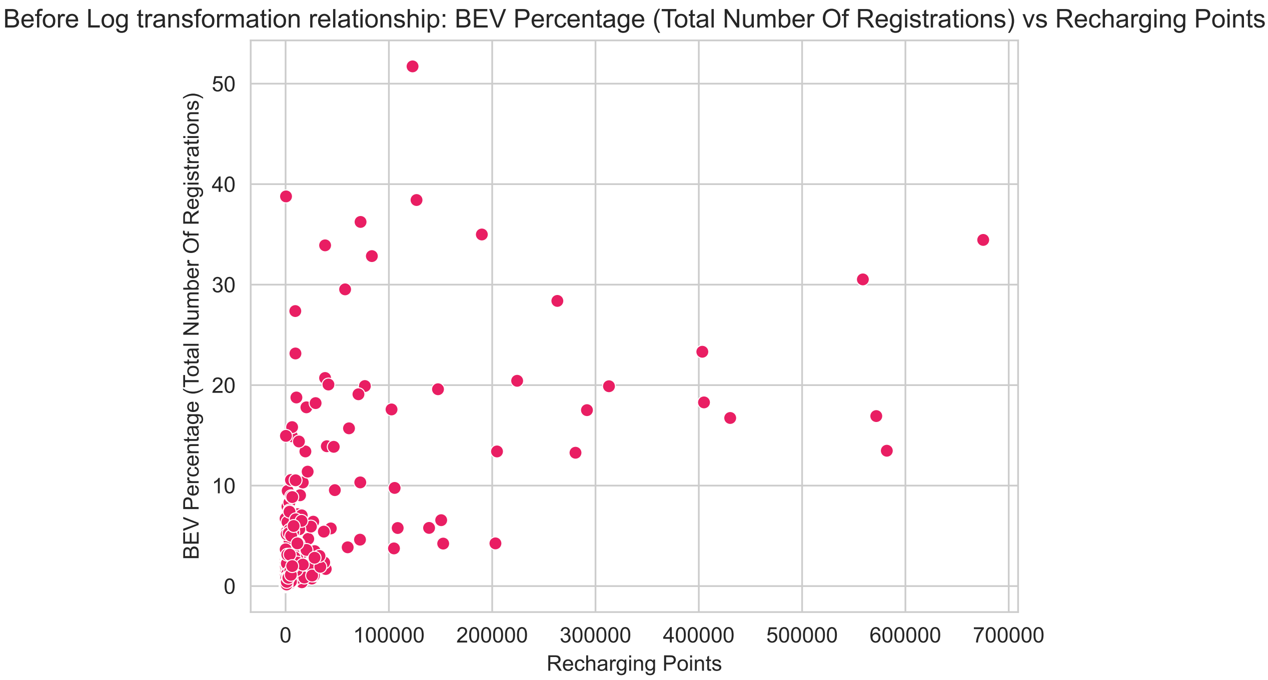

Figure 3.4: Before Log Transformation (BEV Market Share vs Recharging Points)

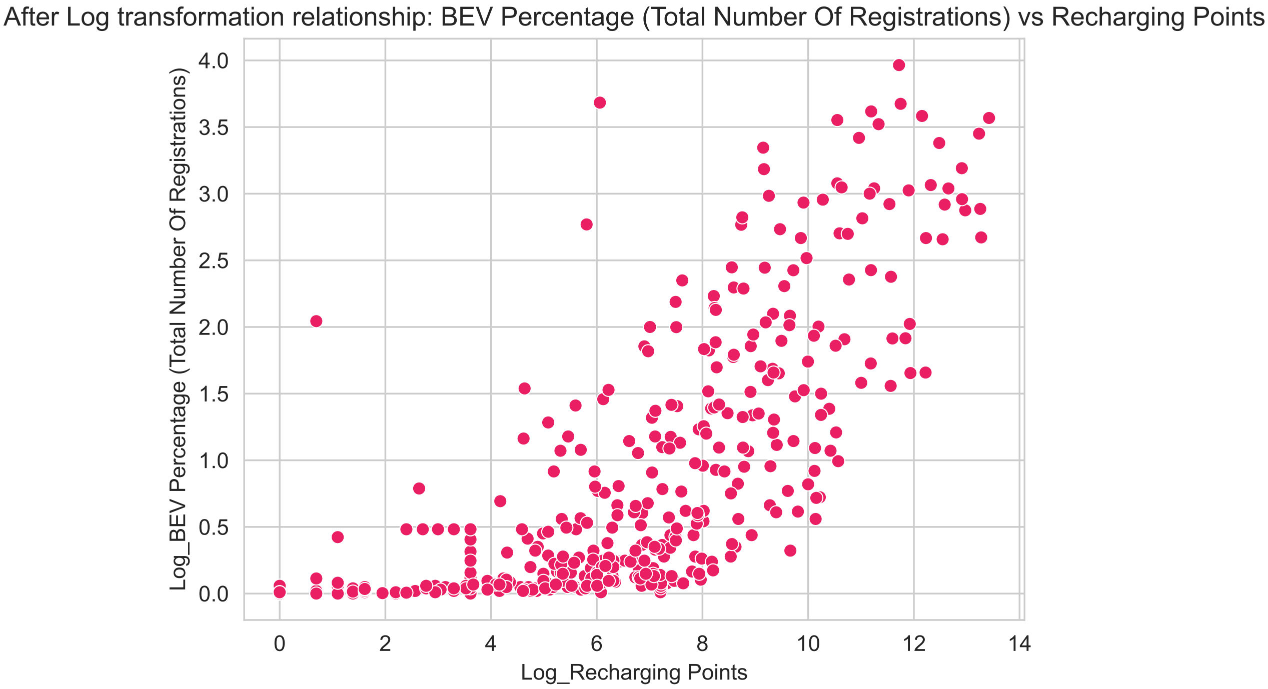

Figure 3.5: After Log Transformation (BEV Market Share vs Recharging Points)

Overall, the EDA revealed substantial heterogeneity across EU member states. It clearly shows structural acceleration in BEV adoption post-2019, and significant right skewness in several variables. The log and Yeo-Johnson transformations applied to address these distributional issues, along with the aggregated variables, form the basis for the preprocessing steps outlined in the following sub-section.

3.2.4 Feature Selection and Dimensionality Reduction

Prior to model development, feature selection and dimensionality reduction were performed exclusively on the training dataset for each trial to prevent data leakage. This ensured that no information from the test period influenced the feature transformation process, preserving the integrity of the out-of-sample evaluation. Table 3.7 sums up the training and test periods across the three trials.

Trial	Training Period	Test Period	Rationale
Trial 1	2011-2019	2020-2024	Pre-COVID training, post-COVID testing
Trial 2	2011-2022	2023-2024	Extended training including COVID period
Trial 3	2020-2023	2024	Post-COVID training, near-term testing

Table 3.7: Train and Test Split by Trial.

As the training data composition differs across trials, all subsequent steps including correlation analysis, VIF computation, PCA fitting, and cluster derivation were repeated independently for each trial. Results specific to each trial are reported separately in the results section.

Correlation Analysis

As a first step, the Pearson correlation between each feature (both transformed and untransformed) and the log-transformed target variable, *BEV%*, was computed and visualized using a heatmap. This provided an initial assessment of the linear relationships between the predictors and the target. However, the correlation matrix revealed substantial multicollinearity among several features, with multiple predictors exhibiting high inter-feature correlations. Since high multicollinearity can destabilize model coefficients and reduce interpretability, further investigation was warranted [Hai+19].

VIF

To quantify the extent of multicollinearity, VIF was computed for each feature. VIF measures how much the variance of a regression coefficient is inflated due to collinearity with other predictors, with values exceeding 10 generally considered indicative of severe multicollinearity [DGE12]. The VIF results confirmed the presence of high multicollinearity across several features. However, as a low VIF alone does not guarantee a meaningful relationship with the target variable, feature elimination based solely on VIF was not considered a sufficient solution. A more comprehensive dimensionality reduction approach was therefore adopted.

PCA

To simultaneously address multicollinearity and reduce the dimensionality of the feature space, PCA was applied [Jol02]. As discussed in the previous chapter, PCA transforms correlated features into a smaller set of uncorrelated principal components, resolving the conflict between high inter-feature correlation and the need to retain informative predictors.

Prior to applying PCA, all features were standardized using a standard scaler to ensure that variables measured on different scales contributed equally to the principal components [Bis06]. The scaler was fitted exclusively on the training data and subsequently applied to the test data to avoid data leakage.

As described in Section 2.3.3, PCA proceeds through the following steps: computation of the covariance matrix, eigenvalue decomposition, normalization of eigenvectors, and projection of the original data onto the principal component space.

The optimal number of principal components was determined using the elbow method applied to a scree plot of explained variance [Jol02]. Components above the elbow point were retained, as they capture the most meaningful variance in the data before the marginal contribution of additional components diminishes significantly. As the training data composition differs across the three trials, the number of retained components and the total explained variance varied accordingly and are reported separately for each trial in the results section.

PCA was fitted on the training dataset only, and the same transformation was subsequently applied to the test dataset by projecting it into the principal component space derived from the training data. This ensured that the dimensionality reduction process was fully blind to the test period. Across all trials, the ten original features were reduced to the number of components identified by the elbow method, with the corresponding explained variance values reported in the results section.

Loading Factors

To interpret the principal components, loading factors were examined for each trial. Loading factors represent the correlation between each original feature and the derived principal components, indicating the relative contribution of each variable to the component structure [Jol02]. As the loading factors differ across trials due to the varying temporal composition of the training data, they are discussed in detail in the results section alongside the corresponding scree plots and explained variance values.

3.2.5 Clustering

Following dimensionality reduction through PCA, K-means clustering was applied to group the 27 EU member states into distinct market segments based on their BEV adoption profiles. Clustering was performed separately on the training and test datasets for each trial, using the principal component scores as input features. As the PCA space differs across trials due to the varying temporal composition of the training data, the clustering results are reported independently for each trial in the results section. The methodological sequence of correlation analysis, multicollinearity assessment, dimensionality reduction, and clustering adopted in this study is consistent with the approach of [MSJ24] in their analysis of European EV markets, providing further validation of its suitability for this domain.

Determining the Optimal Number of Clusters

The optimal number of clusters was determined using two complementary methods. First, the elbow method was applied by plotting the WCSS against the number of clusters and identifying the point at which the marginal reduction in inertia diminishes significantly. Second, the silhouette score was computed as a validation step to confirm the cluster separation identified by the elbow method [Rou87]. The silhouette score measures how similar each observation is to its own cluster relative to other clusters, with higher values indicating better defined cluster boundaries. Both methods consistently identified two clusters as the optimal solution across all three trials, suggesting that the division of EU member states into two distinct BEV market groups is a robust and stable finding independent of the training period considered.

K-Means Clustering

K-means clustering was selected as the primary clustering algorithm due to its simplicity, computational efficiency, and suitability for continuous numerical data in a low-dimensional PCA space [Mac67]. The algorithm was initialized with a fixed random state for reproducibility and run with ten initializations to avoid convergence to local minima. With two clusters identified as optimal, the 27 EU member states were partitioned into two groups for both the training and test periods of each trial, enabling the analysis of cluster stability and country transitions across trials. The resulting clusters are referred to throughout this study as *Leaders* and *Followers*, reflecting the distinction between member states with high BEV adoption rates and those with comparatively lower adoption levels.

DBSCAN Consideration and Rejection

DBSCAN was also considered as an alternative clustering approach given its ability to identify clusters of arbitrary shape and handle outliers as noise points [Est+96]. However, DBSCAN was found to be unsuitable for this dataset, as the distribution of EU member states in PCA space did not exhibit the density-based structure required for meaningful partitioning. This finding is consistent with [MSJ24], who similarly found that density-based clustering was not well suited to European EV market data. K-means was therefore retained as the most appropriate clustering method for this analysis.

Cluster Analysis and Transitions

Following the application of K-means, the cluster memberships of each country were compared between the training and test periods to identify countries that transitioned between clusters and those that remained stable in the train and test periods. Cluster centroids were examined in PCA space to characterize the distinguishing features of each group, and loading factors derived in the previous section were used to interpret the economic and infrastructural meaning of the principal components driving the cluster separation. Countries that changed cluster membership between the training and test periods are of particular interest, as they reflect shifts in relative BEV adoption dynamics in response to the structural market changes observed post-2020. The detailed cluster compositions, country transition matrices, centroid characteristics, and visualisations for each trial are reported in the results section.

3.2.6 Model Development

The theoretical foundations, mathematical formulations, and underlying assumptions of each model are discussed in detail in Section 2.5. This section describes the implementation decisions, configurations, and adaptations applied to each model within the context of this study.

1. Baseline Models

Linear Regression

Linear Regression was implemented as the simplest baseline model to establish a reference level of predictive performance. The model was fitted separately for the *Leaders* and *Followers* clusters using the principal component scores as input features and the log-transformed *BEV%* as the target variable.

Predictions generated in log space were back-transformed using the exponential function to recover values in the original percentage scale, ensuring that evaluation metrics are directly interpretable in terms of actual BEV market share. The model was evaluated on both the training and test datasets using the *MAE*, computed in both log and real space [DGE12].

SVR

SVR was implemented using a RBF kernel with a regularization parameter $C = 100$ and an $\varepsilon = 0.1$, showing a preference for lower bias at the cost of potentially higher variance. The RBF kernel was selected for its ability to capture non-linear relationships between the principal components and the target variable [CV95].

Same as Linear Regression, the model was trained separately for the *Leaders* and *Followers* clusters, predictions were back-transformed from log to real scale using the exponential function, and performance was evaluated using *MAE* on both the training and test datasets.

2. Time Series Model

ARIMAX

The ARIMAX model was implemented at the individual country level, with a separate model fitted for each of the 27 EU member states. This approach was adopted to account for the country-specific temporal dynamics of BEV adoption, which vary considerably across member states in terms of both growth rate and market maturity.

The autoregressive (p) and moving average (q) orders were selected automatically for each country using the AIC [Aka74], with a maximum order of 2 considered for both parameters [Box+08]. A fixed integration order of d = 1 was applied across all countries to account for the non-stationarity commonly observed in BEV adoption time series. In cases where automatic order selection failed due to convergence issues, a fallback order of (1, 1, 0) was applied to ensure model completion across all countries. The principal component scores were included as exogenous variables to incorporate the influence of macroeconomic, infrastructural, and technological drivers alongside the autoregressive structure of the model. Forecasts were generated for the full test period using the fitted model parameters and the corresponding test period principal component scores. As with the other models, predictions were back-transformed from log to real scale using the exponential function prior to evaluation.

3. Tree Based Models

Tree-based models do not require feature scaling as a prerequisite for training, as they partition the feature space based on threshold values rather than distance metrics and are therefore invariant to the scale of input features [Bre01]. However, in this study the input features were already standardized prior to PCA, as described in Section 2.3.3, and the scaled principal component scores were used as inputs across all models for consistency. The log transformation applied to the target variable was retained for tree-based models to maintain consistency with the rest of the modelling pipeline and to account for the right-skewed distribution of BEV market share. Predictions were back-transformed from log to real scale using the exponential function prior to evaluation for all tree-based models. Both models were trained and evaluated separately for the *Leaders* and *Followers* clusters across all three trials.

Random Forest Regressor

Random Forest was implemented using 100 decision trees (`n_estimators=100`) with no restriction on maximum tree depth, allowing each tree to grow fully and capture complex non-linear patterns in the principal component space [Bre01]. A fixed random state of 42 was applied to ensure reproducibility across all trials.

Performance was assessed using *MAE* computed in both log and real space on the training and test datasets.

XGBoost Regressor

XGBoost was implemented with a maximum of 1,000 estimators (`n_estimators=1000`), a learning rate of 0.05, and a maximum tree depth of 5 [CG16] . To prevent overfitting, early stopping was applied with a patience of 10 rounds, whereby training was terminated if the validation score did not improve for 10 consecutive iterations. A validation set comprising 10% of the training data was held out internally to monitor performance during training and trigger early stopping where appropriate. This approach ensures that the model does not overfit the training data while making full use of the available observations.

4. Neural Network Model

LSTM

Unlike tree-based models, LSTM networks are sensitive to the scale of input features due to the activation functions used within the gating mechanisms. A separate MinMaxScaler was applied to both the input features and the target variable within the LSTM pipeline, independently of the StandardScaler applied during PCA [HS97]. The scaler was fitted on the training data and subsequently applied to the test data to prevent data leakage. The target variable was also scaled using a MinMaxScaler and predictions were inverse-transformed to log scale before applying the exponential back-transformation to recover values in the original percentage scale.

The input features were reshaped into a three-dimensional array of the form [samples, time steps, features], as required by the Keras LSTM Implementation [Cho21]. To assess the sensitivity of the standalone LSTM model, timesteps value of 2 was used, as it captures the short-term year-on-year dependencies without exceeding the complexity threshold imposed by the small training sample size. In stacking ensemble, single time step was retained for the LSTM base learner. Temporal dependencies were already modeled by the ARIMAX base learner, and increasing the timestep would cause LSTM to learn similar temporal patterns, reducing model diversity within the meta model which would increase the model variance and prediction error. Since stacking benefits from combining complementary models, timestep 1 allows LSTM to function primarily as a non-linear feature learner, while ARIMAX captures the temporal autocorrelation structure.

The LSTM architecture consisted of a single LSTM layer with 32 units and (`tanh`) activation function, followed by a Dropout layer with a rate of 0.1 to reduce overfitting, a Dense hidden layer with 16 units and a ReLU activation function, and a final Dense output layer with a single unit. The model was compiled using the Adam optimizer with a learning rate of 0.001 and mean squared error as the loss function. A lower learning rate was selected to ensure stable convergence given the relatively small size of the training dataset [KB17].

The model was trained for a maximum of 200 epochs with a batch size of 2, reflecting the small number of annual observations available per cluster. Early stopping was applied with a patience of 15 epochs, monitoring validation loss and restoring the best weights upon termination. A validation split of 20% of the training data was held out internally during training to monitor generalization performance. The model was trained and evaluated separately for the *Leaders* and *Followers* clusters across all three trials.

5. Proposed Hybrid Models

The proposed model is a two-stage stacking ensemble architecture that combines ARIMAX and LSTM as base learners, with a meta-learner trained on their combined outputs. The motivation for selecting ARIMAX and LSTM as base learners stems from their complementary modelling strengths — ARIMAX captures linear temporal autocorrelation and the influence of exogenous drivers, while LSTM captures non-linear sequential dependencies across time [HS97]. Both base learners were trained separately for each country using the same ARIMAX and LSTM configurations described in the previous subtopic.

Stage 1 — Base Learner Predictions

ARIMAX and LSTM were each trained on the country-level training data for every trial. Their predictions on both the training and test periods were collected and merged into a meta-dataset indexed by country and year. This meta-dataset contained the actual *BEV%* alongside the ARIMAX and LSTM predictions for each country-year observation, forming the input feature space for the second stage meta-learner. Rows corresponding to the test period were labelled accordingly to ensure a strict separation between training and test data at the meta-learning stage.

For the LSTM base learner within the ensemble, a single time step configuration was used, as the temporal dependencies in the data were already explicitly captured by the ARIMAX model. Increasing the number of time steps in this setting was therefore not necessary, as it would introduce redundant sequential structure and increase model complexity without adding additional informational gain.

Stage 2 — Meta-Learner

The meta-learner was trained on the training period predictions of the two base learners, using the actual *BEV%* as the target variable. Three variants of the meta-learner were implemented and evaluated to assess the sensitivity of the stacking architecture to the choice of combination method. All predictions were clipped to the range [0, 100] to prevent physically implausible values.

The 3 meta learners are as follows:

1. ElasticNet

ElasticNet was used as the meta-learner, combining L1 and L2 regularization to produce a sparse and stable weighting of the base learner predictions [ZH05]. The optimal regularization parameters were selected through cross-validated grid search over a range of alpha values spanning 10^{-3} to 10^{3} and l1_ratio values of [0.1, 0.5, 0.7, 0.9, 0.95, 0.99, 1.0], using 3-fold cross-validation. A positivity constraint was applied to ensure that the meta-learner assigned non-negative weights to both base learners, reflecting the assumption that both ARIMAX and LSTM contribute positively to the final prediction.

2. Gradient Boost

A Gradient Boosting Regressor was used as the meta-learner to capture non-linear interactions between the base learner outputs [Fri01]. Hyperparameters were selected through 3-fold cross-validated grid search over the following parameter space: number of estimators in {50, 100}, learning rate in {0.01, 0.05, 0.1}, maximum tree depth in {3, 4}, and squared error as the loss function. The best performing hyperparameter combination identified by the grid search was used for final prediction.

3. Weighted Average

A weighted average of the ARIMAX and LSTM predictions was used as the combination method, with the optimal weight determined through numerical optimization [BG69]. The initial weight for w was set to 0.5, reflecting an equal contribution from both base learners as the starting point for optimization. The optimal weight was determined independently for each cluster and each trial, reflecting the different relative strengths of the two base learners across different training periods and market segments.

3.2.7 Evaluation Metrics

Two evaluation metrics were selected to assess the predictive performance of all models across trials and clusters, *MAE* and *MedAE*. The theoretical foundations and mathematical formulations of both metrics are discussed in detail in Section 2.6 of the Literature Review. The application and results of these metrics across all models and trials are presented in Section 4.5.

MAE was selected for its interpretability as a measure of average prediction accuracy expressed in the same units as the target variable. MedAE was selected as a complementary metric due to its robustness to outliers, which is particularly relevant given the substantial heterogeneity in BEV adoption rates across EU member states. Both metrics were computed on the absolute prediction errors in the original percentage scale.

4 Results and Discussion

4.1 Overview

This chapter presents the results of the three experimental trials designed to investigate the impact of training data composition on the predictive accuracy of *BEV%* forecasting models across 27 EU member states. For each trial, the results are organised into two parts: first, the dimensionality reduction and clustering outcomes, which describe how the EU member states are partitioned into *Leaders* and *Followers* groups based on their BEV adoption profiles; and second, the model performance results, which compare the predictive accuracy of all models across both clusters.

The chapter is structured as follows. Sections 4.2, 4.3, and 4.4 presents the multicollinearity, PCA and clustering results for Trial 1, Trial 2, and Trial 3 respectively. Section 4.5 presents a consolidated cross-trial comparison of model performance using *MAE* and *MedAE*. Section 4.7 formally validates the four hypotheses proposed in Section 3.2.1. For the case study, the best model is subsequently tested on the year 2025 to evaluate its near-term forecasting capability beyond the original test horizon, providing a forward-looking validation of the proposed architecture under real-world conditions. Finally, the chapter concludes by interpreting the findings in the broader context of European BEV market dynamics and discusses the policy implications and limitations of the study.

4.2 Trial 1

The training period was from 2011-2019 and tested on 2020-2024.

4.2.1 Feature Selection and Dimensionality Reduction

Correlation Analysis

As a first step, the Pearson correlation between each log-transformed features and the target variable was computed on the Trial 1 training dataset covering the period 2011-2019. Table 4.1 presents the correlation coefficients in descending order.

Feature	Correlation with Log BEV %
Log Recharging Points	0.538
Real Range	0.460
Log Available BEV Models	0.460
YJ Purchase Price (EUR)	0.424
AC Recharging Speed (km/h)	0.402
Battery Capacity	0.401
Log DC Recharging Speed	0.384
Employment Growth	0.313
Log GDP	0.098
Log CPI	0.073

Table 4.1: Correlation of features with Log BEV % — Trial 1 Training Dataset (2011–2019).

The results indicate that infrastructure and technological performance variables exhibit the strongest correlations with BEV market share. *Log Recharging Points* recorded the highest correlation of 0.538, followed by *Real Range* (0.460) and *Log Available BEV Models* (0.460), suggesting that charging infrastructure availability and vehicle capability were the primary drivers of BEV adoption in the pre-COVID period. Socioeconomic variables showed comparatively weaker correlations, with *Log GDP* and *Log CPI* recording the lowest values of 0.098 and 0.073 respectively, indicating limited linear association with BEV market share in this period.

Variance Inflation Factor (VIF)

Despite the meaningful correlations observed between several features and the target variable, the VIF analysis revealed severe multicollinearity across the feature set, as shown in Table 4.2.

Feature	VIF
Real Range	537.36
Battery Capacity	271.27
Log DC Recharging Speed (km/h)	131.91
Log Available BEV Models	115.98
Log GDP	56.51
AC Recharging Speed (km/h)	39.43
Log Recharging Points	17.05
YJ Purchase Price (EUR)	4.67
Log CPI	3.29
Employment Growth	1.78

Table 4.2: Variance Inflation Factor (VIF) for all features — Trial 1 Training Dataset (2011–2019).

The VIF results confirm extreme multicollinearity among the technological performance variables. *Real Range* recorded the highest VIF of 537.36, followed by *Battery Capacity* (271.27) and *Log DC Recharging Speed* (131.91), reflecting the strong interdependence among vehicle capability metrics that tend to improve simultaneously over time. *Log Available BEV Models* (115.98) and *Log GDP* (56.51) also exceeded the threshold of 10 substantially. Only *YJ Purchase Price* (4.67), *Log CPI* (3.29), and *Employment Growth* (1.78) recorded acceptable VIF values. These results confirm that feature elimination alone would be insufficient to resolve the multicollinearity present in the dataset, justifying the application of PCA as described in Section 2.3.3.

Principal Component Analysis

Following standardization of the training data, PCA was applied to the full set of ten features. Figure 4.1 presents the scree plot derived from the elbow method.

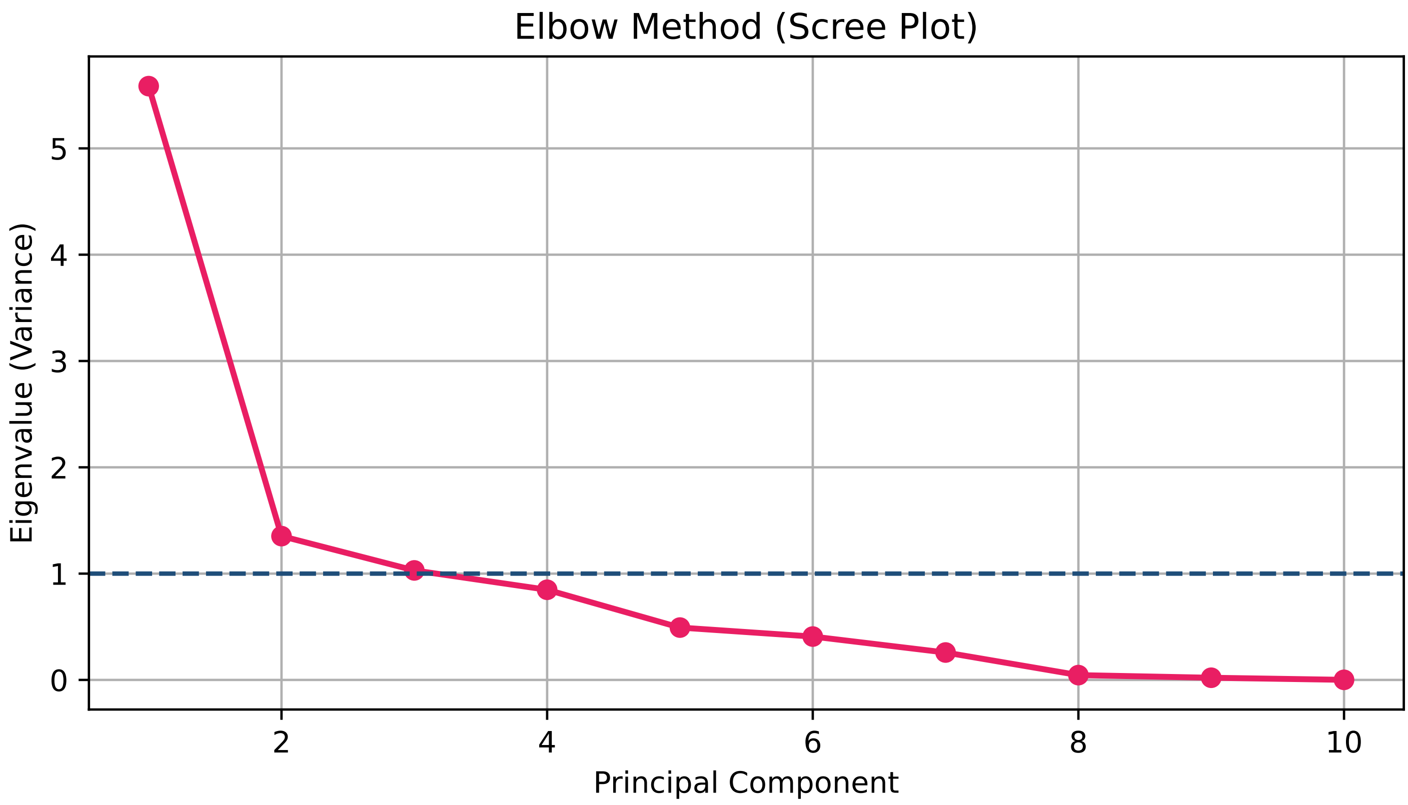

Figure 4.1: Scree Plot — Elbow Method for Trial 1 (2011–2019 Training Dataset).

The elbow in the curve is observed at Principal Component 3, above which the eigenvalues fall below 1.0 and the marginal contribution of additional components diminishes substantially. Three principal components were therefore retained, collectively explaining 79.36% of the total variance in the training data. The correlation of each principal component with the target variable confirmed that PC1 exhibited the strongest association with *Log BEV%* (0.486), followed by PC3 (0.264) and PC2 (0.206), as shown in Table 4.3.

Principal Component	**Correlation with Log BEV %**
PC1	0.486
PC3	0.264
PC2	0.206

Table 4.3: Correlation of Principal Components with Log BEV Market Share — Trial 1.

Following PCA, the VIF of all three principal components was exactly 1.0, verifying the complete elimination of multicollinearity as expected from orthogonal principal components.

Loading Factors

Table 4.4 presents the loading factors for each feature across the three retained principal components, indicating the contribution of each original variable to the component structure.

Feature	PC1	PC2	PC3
Log Recharging Points	0.229	0.555	0.085
Real Range	0.410	-0.038	-0.001
Log Available BEV Models	0.404	-0.009	0.148
YJ Purchase Price (EUR)	0.343	0.047	0.389
AC Recharging Speed (km/h)	0.362	-0.017	-0.022
Battery Capacity	0.396	-0.081	-0.210
Log DC Recharging Speed (km/h)	0.383	-0.090	-0.286
Employment Growth	0.186	-0.237	0.524
Log GDP	0.038	0.755	-0.160
Log CPI	-0.164	0.219	0.627

Table 4.4: PCA Loading Factors — Trial 1 (2011–2019 Training Dataset).

The loading factors reveal distinct and interpretable structures across the three components. PC1 is primarily associated with the technological performance and market availability variables, with *Real Range* (0.410), *Log Available BEV Models* (0.404), *Battery Capacity* (0.396), *Log DC Recharging Speed* (0.383), *AC Recharging Speed* (0.362), and *YJ Purchase Price* (0.343) all loading strongly and positively. PC1 can therefore be interpreted as a *BEV Technology and Market Maturity* component, capturing the overall advancement of the BEV product offering available to European consumers in the pre-COVID period.

PC2 is characterised by strong positive loadings from *Log GDP* (0.755) and *Log Recharging Points* (0.555), with a moderate negative loading from *Employment Growth* (-0.237). This component can be interpreted as an *Economic and Infrastructure* dimension, reflecting the distinction between wealthier member states with more developed charging networks and those with limited infrastructure investment.

PC3 is primarily driven by *Log CPI* (0.627), *Employment Growth* (0.524), and YJ Purchase Price (0.389), with negative loadings from *Log DC Recharging Speed* (-0.286) and *Battery Capacity* (-0.210). PC3 can be interpreted as a *Macroeconomic and Cost* component, capturing the influence of inflation, labour market conditions, and vehicle purchase cost on BEV adoption dynamics in the pre-COVID period.

It is important to note that the appearance of certain variables across multiple principal components does not introduce multicollinearity into the analysis. In PCA, every original variable carries a loading on every principal component by construction, and a variable contributing to more than one component simply reflects the fact that it captures variation along multiple dimensions simultaneously. For instance, *YJ Purchase Price* loads meaningfully on both *PC1* (0.343) and *PC3* (0.389), which is conceptually consistent — purchase price is simultaneously a reflection of the technological sophistication of the vehicle and a determinant of consumer affordability under macroeconomic conditions. Similarly, *Employment Growth* contributes to both *PC2* (-0.237) and *PC3* (0.524), reflecting its dual role as both an indicator of broader economic conditions and a measure of macroeconomic volatility driven by events such as the COVID-19 pandemic.

The orthogonality of the principal components ensures that despite these shared variable contributions, the components themselves remain completely uncorrelated with one another. This is confirmed by the VIF of exactly 1.0 recorded for all three principal components, verifying the complete elimination of multicollinearity from the feature space prior to model training. The three components therefore capture genuinely distinct and non-overlapping dimensions of BEV market dynamics are *Technology and Market Maturity*, *Economic and Infrastructure Development*, and *Macroeconomic and Cost Conditions*. These components provide a clean and interpretable input space for the subsequent clustering and modelling procedures.

4.2.2 Clustering Results

Optimal Number of Clusters

Following PCA, K-Means clustering was applied to the principal component scores of the training dataset (2011-2019) and the test dataset (2020-2024) separately. The optimal number of clusters was determined using the elbow method and validated using the silhouette score, as described in Section 3.2.5.

Figure 4.2 present the elbow plots for the pre-COVID and post-COVID periods respectively. In both cases, the WCSS shows a sharp decline from $k = 1$ to $k = 2$, after which the rate of reduction diminishes substantially, indicating that two clusters capture the most meaningful partition of the data.

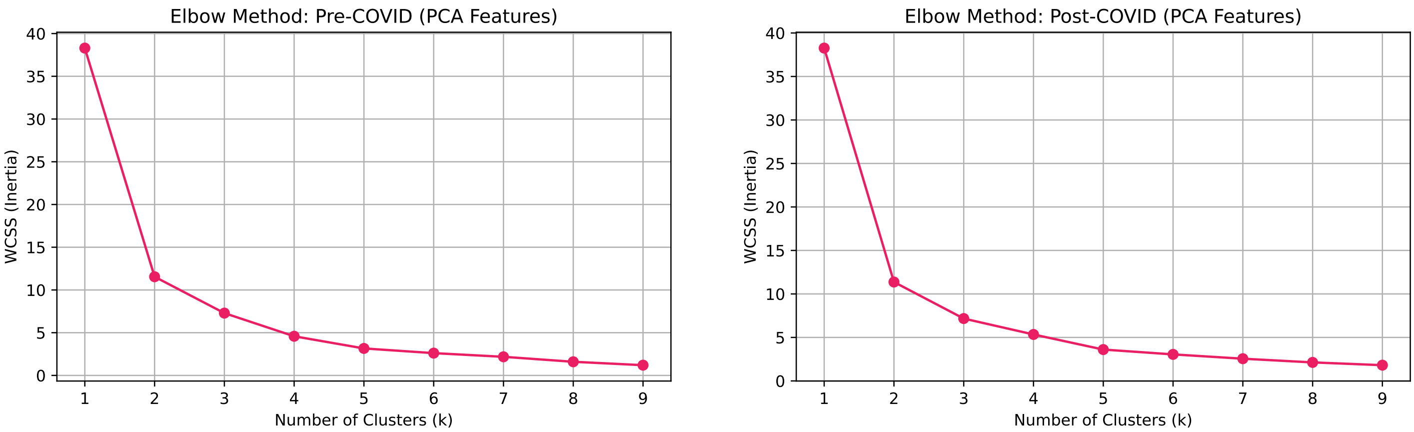

Figure 4.2: Elbow Method - Pre-COVID (left) and Post-COVID (right) PCA Features (Trial 1).

This finding was validated using the silhouette score, presented in Figure 4.3. The silhouette score is highest at $k = 2$ for both the pre-COVID (0.575) and post-COVID (0.55) periods, confirming that two clusters provide the most well-defined and internally cohesive partition of the EU member states in the principal component space. The silhouette score declines monotonically for higher values of k, further supporting the selection of two clusters.

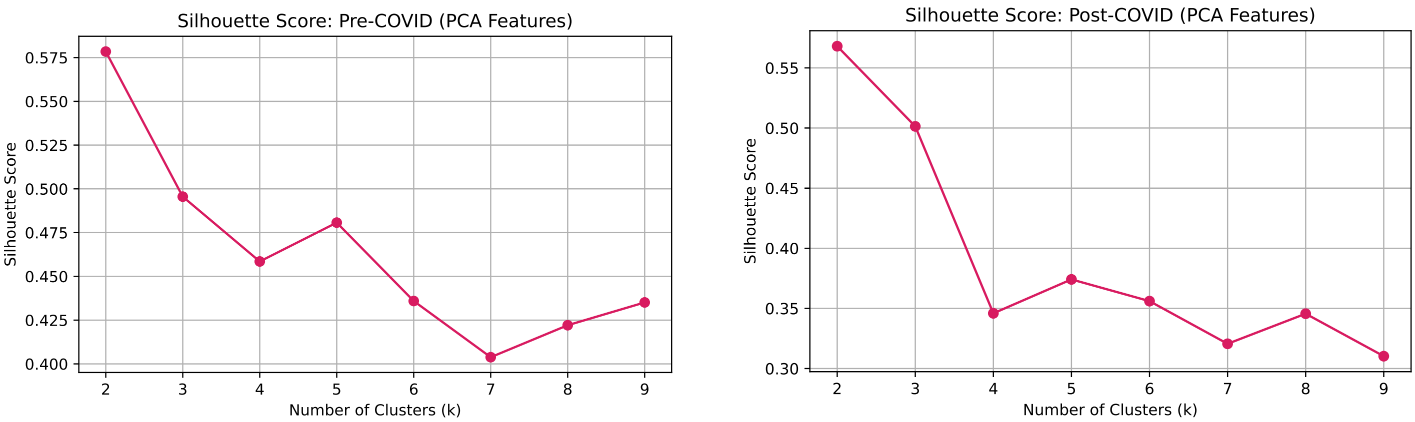

Figure 4.3: Silhoutte Method - Pre-COVID (left) and Post-COVID (right) PCA Features (Trial 1).

Cluster Composition

Based on the K-Means clustering results, the 27 EU member states were partitioned into two groups, referred to throughout this study as *Leaders* and *Followers*, reflecting their relative BEV adoption profiles. The *Leaders* cluster comprises 14 member states: *Austria*, *Belgium*, *Czech Republic*, *Denmark*, *Finland*, *France*, *Germany*, *Hungary*, *Italy*, *Netherlands*, *Poland*, *Romania*, *Spain*, and *Sweden*. The *Followers* cluster comprises the remaining 13 member states: *Bulgaria*, *Croatia*, *Cyprus*, *Estonia*, *Greece*, *Ireland*, *Latvia*, *Lithuania*, *Luxembourg*, *Malta*, *Portugal*, *Slovakia*, and *Slovenia*.

Cluster visualisation and Interpretation

Figure 4.4 presents the cluster visualisation in the *PC1 - PC2* space for both the pre-COVID and post-COVID periods.

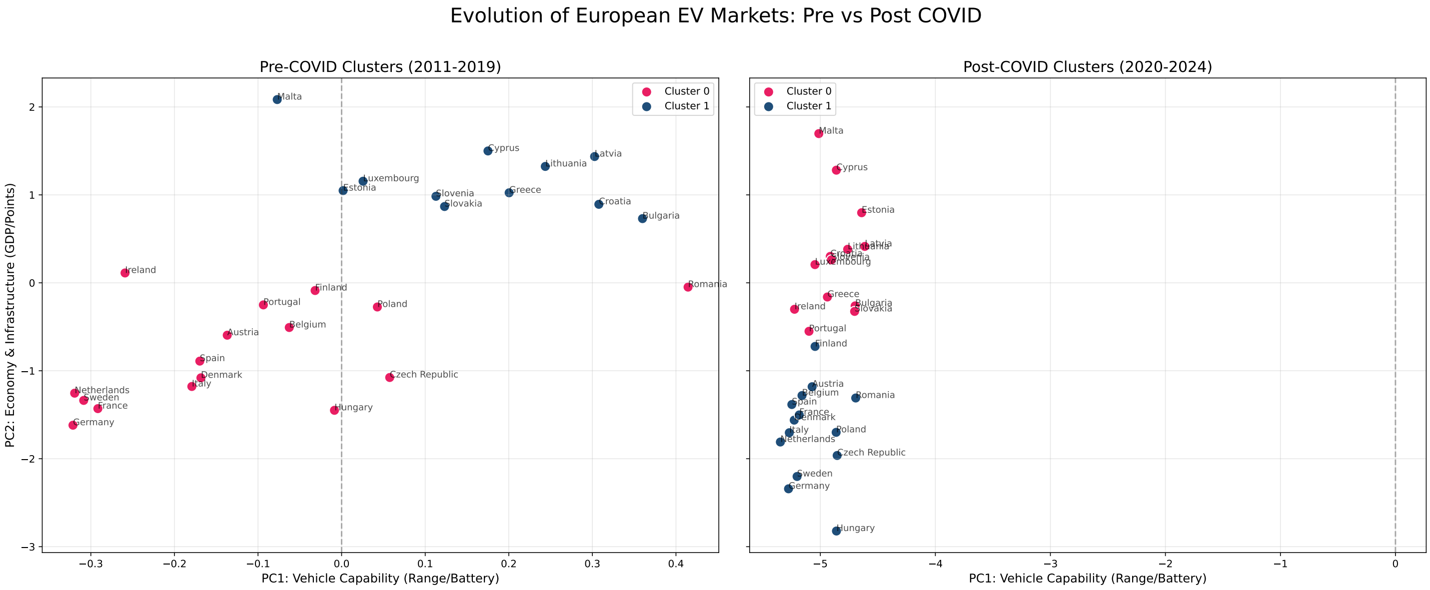

Figure 4.4: Evolution of European EV Markets: Pre-COVID (2011-2019) vs Post-COVID (2020-2024) Cluster Visualisation - Trial 1.

In the pre-COVID period, the two clusters are separated primarily along the *PC2* axis, which as established in the loading factor analysis captures *Economic and Infrastructure* conditions, driven by *Log GDP* and *Log Recharging Points*. Countries such as *Germany*, *France*, *Netherlands*, and *Sweden* occupy the upper right quadrant, reflecting both higher vehicle capability scores on *PC1* and stronger economic and infrastructure foundations on *PC2*. In contrast, countries such as *Bulgaria*, *Latvia*, *Croatia*, and *Malta* are positioned in the lower left quadrant, indicating lower scores on both dimensions during the pre-COVID period.

In the post-COVID period, the entire distribution shifts dramatically to the right along the *PC1* axis, with all 27 member states recording substantially higher *PC1* scores compared to the pre-COVID period. This shift reflects the universal advancement of BEV technology across Europe post-2020, captured by the EU-level variables that load strongly on *PC1*, namely *Real Range*, *Log Available BEV Models*, *Battery Capacity*, and *Recharging Speed*. The post-COVID separation between clusters becomes primarily driven by *PC2*, distinguishing member states by their economic size and infrastructure development, with *Hungary*, *Germany*, *Sweden*, and *Czech Republic* occupying the highest positions on the *PC2* axis in the *Leaders* cluster.

Cluster Centroids

Table 4.5 presents the cluster centroids in the principal component space for both the pre-COVID and post-COVID periods.

Period	Cluster	PC1	PC2	PC3
Pre-COVID	Cluster 0 (Leaders)	0.114	0.811	-0.090
Pre-COVID	Cluster 1 (Followers)	-0.166	-1.180	0.131
Post-COVID	Cluster 0 (Followers)	4.882	-0.293	0.741
Post-COVID	Cluster 1 (Leaders)	5.093	1.677	0.470

Table 4.5: Cluster Centroids in PCA Space - Pre-COVID and Post-COVID Periods (Trial 1).

The centroid values reveal several important findings. In the pre-COVID period, the *Leaders* cluster centroid records a positive *PC2* score of 0.811, reflecting the stronger economic capacity and charging infrastructure of this group relative to the *Followers* cluster, which records a negative *PC2* centroid of -1.180. The difference in *PC1* between the two clusters is modest in the pre-COVID period (0.114 vs -0.166), indicating that the technological dimension was relatively homogeneous across all EU member states during this period, as BEV technology was still at an early stage of development and uniformly limited across markets.

In the post-COVID period, both cluster centroids record dramatically elevated *PC1* values of 4.882 and 5.093 respectively, reflecting the universal technological advancement of BEV products across all EU markets. The separation between clusters in the post-COVID period is again driven by *PC2*, with the *Leaders* centroid recording a positive score of 1.677 compared to -0.293 for the *Followers* cluster, confirming that economic capacity and infrastructure development remain the primary differentiating factors between the two groups.

Cluster Transitions

A notable feature of the Trial 1 clustering results is the apparent transition of 25 out of 27 member states between cluster labels when comparing the pre-COVID and post-COVID periods. Two interpretations of this finding are worth considering. The first is a technical explanation is K-Means cluster labels are arbitrary and not guaranteed to be consistent across separate model runs, meaning that what appears as a mass transition may simply reflect a label swap between Cluster 0 and Cluster 1 rather than a genuine reordering of country groupings [Mac67]. This interpretation is supported by the cluster visualisation in Figure 4.4, which shows that historically strong BEV markets such as *Germany*, *Sweden*, *France*, and *Netherlands* continue to occupy the upper positions on the *PC2* axis in the post-COVID period, consistent with their pre-COVID positioning.

However, the second and more substantively interesting interpretation is that the apparent transitions reflect a genuine structural shift in the European BEV market following 2020. As established in Section 2.1, the post-COVID period was characterized by a fundamental acceleration in BEV adoption driven by tightening EU emissions regulations, government stimulus packages, and rapidly improving vehicle technology [LH21]. This structural break may have altered the relative positioning of member states within the principal component space, as countries that had previously lagged in BEV adoption began to accelerate at different rates in response to these new market conditions. The dramatic rightward shift of all countries along the *PC1* axis in the post-COVID period; reflecting the universal advancement of *BEV Technology and Market Maturity*, supports this view, as it indicates that the technological landscape changed so fundamentally after 2020 that pre-COVID cluster structures may no longer be directly comparable to post-COVID ones [MSJ24].

Taken together, both interpretations are consistent with the central premise of this study that the pre and post-COVID BEV market regimes are structurally distinct, and that models trained on pre-2020 data cannot be expected to generalise effectively to the post-2020 period. Whether the cluster transitions reflect a label swap or a genuine redistribution of market positions, the finding reinforces the justification for the trial design adopted in this study, particularly the motivation for Trial 3, which isolates the post-COVID regime entirely to capture the new market dynamics.

The two member states that remained nominally stable across periods were *Ireland* and *Portugal*, both of which stayed in *Followers* across both the pre-COVID and post-COVID periods, suggesting that their BEV adoption profile was consistently aligned with the lower-adoption group throughout this window. Both countries occupy mid-range positions on *PC1* and *PC2* in the visualisation, consistent with their moderate and gradually evolving BEV adoption trajectories relative to the broader EU market. Their stability may reflect a more gradual and steady adoption pattern that was less susceptible to the abrupt structural changes observed across the majority of EU member states following 2020.

4.3 Trial 2

For this trial, The training period was from 2011-2022 and tested on 2023-2024.

4.3.1 Feature Selection and Dimensionality Reduction

Correlation Analysis

The Pearson correlation between each feature and the log-transformed target variable was computed on the Trial 2 training dataset covering the extended period 2011-2022. Table 4.6 presents the correlation coefficients in descending order.

Feature	Correlation with Log BEV %
Log Available BEV Models	0.780
Real Range	0.680
Log Recharging Points	0.680
Log DC Recharging Speed (km/h)	0.678
Battery Capacity	0.603
Log CPI	0.358
YJ Purchase Price (EUR)	0.353
AC Recharging Speed (km/h)	0.277
Employment Growth	0.211
Log GDP	0.159

Table 4.6: Correlation of features with Log BEV Market Share — Trial 2 Training Dataset (2011–2022).

Compared to Trial 1, the correlation structure in Trial 2 shows notable differences reflecting the inclusion of post-2020 observations in the training data. *Log Available BEV Models* emerges as the strongest predictor (0.780), overtaking *Log Recharging Points* which led in Trial 1 (0.538). *Real Range* (0.680), *Log Recharging Points* (0.680), and *Log DC Recharging Speed* (0.678) follow closely, all recording substantially higher correlations than in Trial 1. This increase in correlation strength across technological variables reflects the more pronounced relationship between BEV product improvements and market adoption that became apparent during the post-2020 acceleration phase. Notably, *Log CPI* increased considerably from 0.073 in Trial 1 to 0.358 in Trial 2, suggesting that the inflationary pressures of the post-COVID period became an increasingly relevant factor in BEV adoption dynamics. *Log GDP* continued to show the weakest association with the target variable (0.159), consistent with the Trial 1 findings.

Variance Inflation Factor (VIF)

The VIF analysis for Trial 2 produced identical results to Trial 1 (Table 4.2), as the same feature set was used across both trials and the VIF is determined by the inter-feature relationships rather than the target variable. The severe multicollinearity identified in Trial 1 therefore persists in Trial 2, with *Real Range* (537.36), *Battery Capacity* (271.27), and *Log DC Recharging Speed* (131.91) continuing to record extremely high VIF values. This confirms that PCA remains necessary for Trial 2 to resolve multicollinearity prior to model training.

Principal Component Analysis

Following standardisation of the Trial 2 training data, PCA was applied to the full set of ten features. Figure 4.5 presents the scree plot derived from the elbow method.

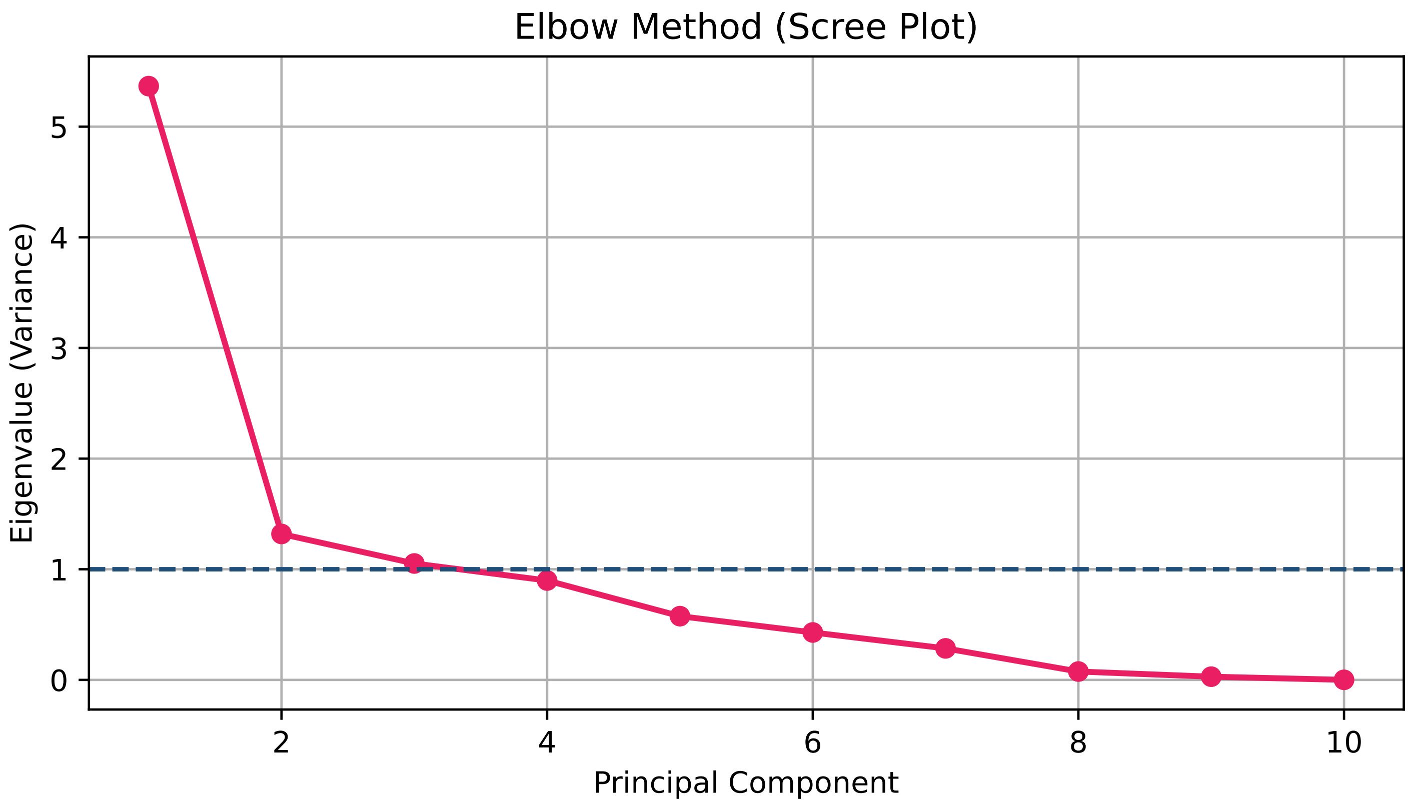

Figure 4.5: Scree Plot — Elbow Method for Trial 2 (2011–2022 Training Dataset).

Consistent with Trial 1, the elbow is observed at Principal Component 3, with eigenvalues falling below 1.0 from the fourth component onwards. Three principal components were therefore retained, collectively explaining 77.14% of the total variance in the Trial 2 training data. This represents a slight improvement in explained variance compared to Trial 1, reflecting the stronger and more consistent relationships between features observed when the post-2020 acceleration phase is included in the training period. Following PCA, the VIF of all three principal components was confirmed to be exactly 1.0, verifying the complete elimination of multicollinearity.

Loading Factors

Table 4.7 presents the loading factors for each feature across the three retained principal components for Trial 2.

Feature	PC1	PC2	PC3
Log Recharging Points	0.294	0.446	-0.031
Real Range	0.420	-0.021	-0.030
Log Available BEV Models	0.396	0.064	0.107
YJ Purchase Price (EUR)	0.331	-0.161	0.058
AC Recharging Speed (km/h)	0.325	-0.193	-0.217
Battery Capacity	0.409	-0.061	-0.138
Log DC Recharging Speed (km/h)	0.404	-0.010	-0.072
Employment Growth	0.164	-0.385	0.391
Log GDP	0.059	0.720	-0.143
Log CPI	0.070	0.254	0.859

Table 4.7: PCA Loading Factors — Trial 2 (2011–2022 Training Dataset).

The loading structure in Trial 2 remains broadly consistent with Trial 1, confirming the stability of the underlying feature relationships across the extended training period. *PC1* continues to be dominated by technological performance and market availability variables, with *Real Range* (0.420), *Battery Capacity* (0.409), *Log DC Recharging Speed* (0.404), *Log Available BEV Models* (0.396), *AC Recharging Speed* (0.325), and *YJ Purchase Price* (0.331) all loading strongly and positively. *PC1* can again be interpreted as a *BEV Technology and Market Maturity* component.

PC2 retains its *Economic and Infrastructure* character, driven by *Log GDP* (0.720) and *Log Recharging Points* (0.446), with a negative loading from *Employment Growth* (-0.385). The loading of *Log Recharging Points* on *PC2* decreased from 0.555 in Trial 1 to 0.446, while *Log GDP* remained the dominant contributor at 0.720, consistent with Trial 1.

PC3 shows the most notable change relative to Trial 1, now dominated almost entirely by *Log CPI* (0.859), with a secondary contribution from *Employment Growth* (0.391). The substantially higher loading of *Log CPI* on *PC3* in Trial 2 compared to Trial 1 (0.859 vs 0.627) reflects the importance of inflationary conditions as a differentiating factor across EU member states when the high-inflation period 2020-2022 is included in the training data. *PC3* can therefore be more specifically interpreted as an *Inflationary Conditions* component in Trial 2, capturing the cross-country variation in consumer price dynamics that became particularly pronounced in the aftermath of the COVID-19 pandemic and the energy crisis of 2022.

4.3.2 Clustering Results

Optimal Number of Clusters

Following PCA, K-Means clustering was applied separately to the training dataset (2011–2022) and the test dataset (2023–2024) to identify the optimal partition of EU member states. The optimal number of clusters was determined using the elbow method and validated using the silhouette score.

Figure 4.6 present the elbow plots for the training and test periods respectively. In both cases, the WCSS shows a sharp decline from $k = 1$ to $k = 2$, after which the rate of reduction diminishes substantially, indicating that two clusters provide the most meaningful partition of the data.

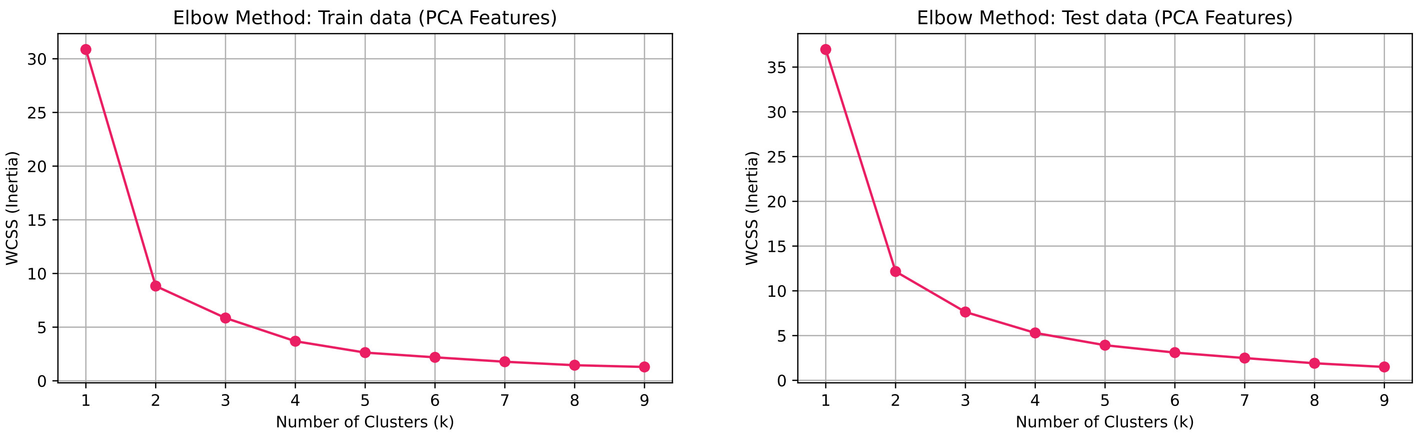

Figure 4.6: Elbow Method - Train Data (left) and Test Data (right) PCA Features (Trial 2).

The silhouette score validation is presented in Figure 4.7. The highest silhouette score is recorded at $k = 2$ for both the training period (0.56) and the test period (0.55), confirming that two clusters provide the most well-defined and internally cohesive partition of the EU member states in the principal component space. The silhouette score declines consistently for higher values of k across both periods, further supporting the selection of two clusters as the optimal solution.

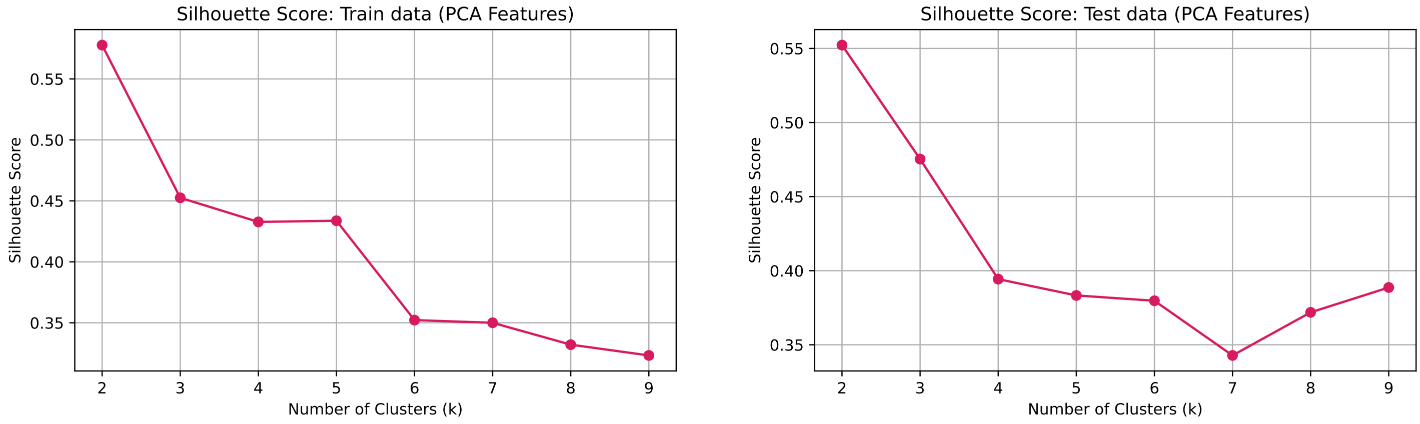

Figure 4.7: Silhoutte Method - Train Data (left) and Test Data (right) PCA Features (Trial 2).

Cluster Composition

Based on the K-Means clustering results, the 27 EU member states were again partitioned into two groups. The *Leaders* cluster comprises 14 member states: *Austria*, *Belgium*, *Czech Republic*, *Denmark*, *Finland*, *France*, *Germany*, *Hungary*, *Italy*, *Netherlands*, *Poland*, *Romania*, *Spain*, and *Sweden*. The *Followers* cluster comprises the remaining 13 member states: *Bulgaria*, *Croatia*, *Cyprus*, *Estonia*, *Greece*, *Ireland*, *Latvia*, *Lithuania*, *Luxembourg*, *Malta*, *Portugal*, *Slovakia*, and *Slovenia*. The cluster composition in Trial 2 is identical to Trial 1, suggesting that the fundamental division of EU member states into high and low adoption groups is a robust and stable finding that persists regardless of whether the training period includes post-2020 observations.

Cluster visualisation and Interpretation

Figure 4.8 presents the cluster visualisation in the *PC1 - PC2* space for both the training and test periods.

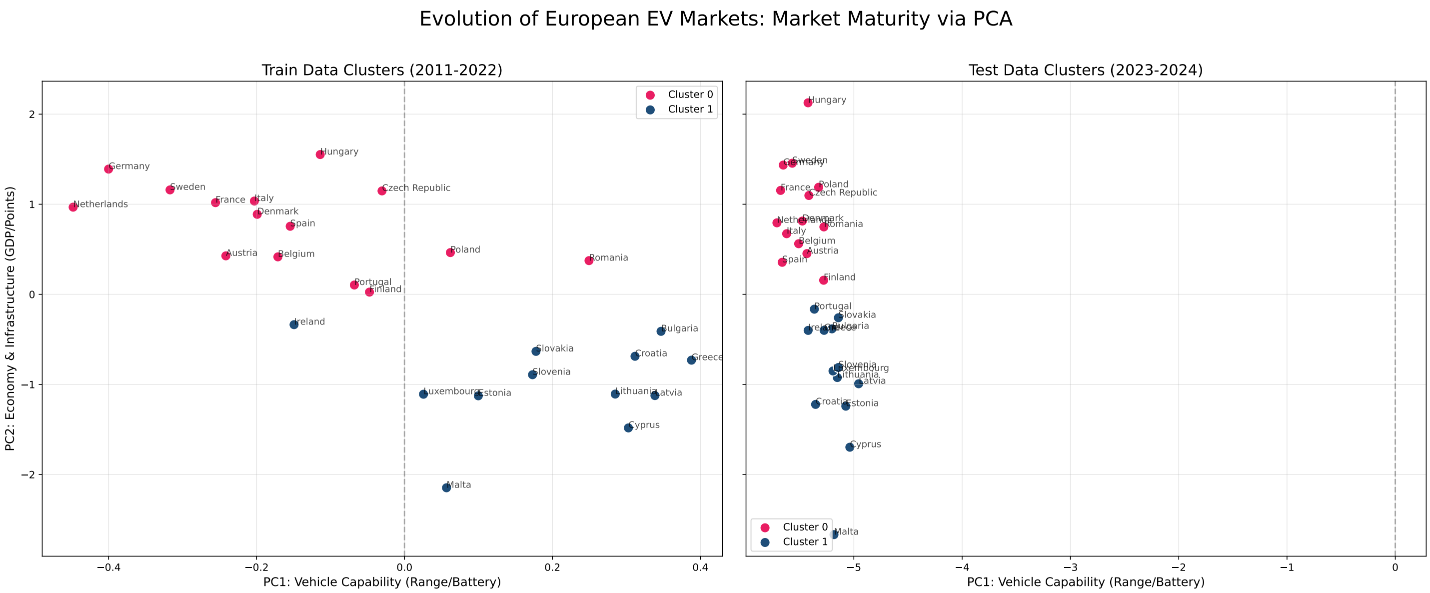

Figure 4.8: Evolution of European EV Markets: Train Data (2011–2022) vs Test Data (2023–2024) Cluster Visualisation – Trial 2.

In the training period, the two clusters are separated along both the *PC1* and *PC2* axes, consistent with Trial 1. The *Leaders* cluster occupies the left side of the plot with predominantly positive *PC2* scores, reflecting stronger economic capacity and infrastructure development. Countries such as *Germany*, *Netherlands*, *Sweden*, and *France* occupy the upper left quadrant, while *Romania* and *Poland* are positioned on the **right side** of the plot, reflecting their membership in the *Leaders* cluster by virtue of economic scale despite lower vehicle capability scores on *PC1*. The *Followers* cluster is distributed across the lower portion of the plot, with *Malta* recording the most negative *PC2* position and *Cyprus* and *Latvia* occupying the lower right area.

In the test period (2023–2024), the entire distribution shifts dramatically to the left along *PC1*, with all 27 member states recording substantially more negative *PC1* scores compared to the training period, consistent with the universal technological advancement of BEV products across all EU markets observed in Trial 1. The separation between clusters in the test period is primarily driven by *PC2*, with the *Leaders* cluster maintaining higher scores on the *Economic and Infrastructure* dimension. *Hungary* occupies the highest position on *PC2* in the test period, followed by *Germany*, *Sweden*, and *Denmark*, reflecting their continued leadership in economic capacity and charging infrastructure development.

Cluster Centroids

Table 4.8 presents the cluster centroids in the principal component space for both the training and test periods.

Period	Cluster	PC1	PC2	PC3
Train (2011-2022)	Cluster 0 (Leaders)	0.156	0.781	-0.126
Train (2011-2022)	Cluster 1 (Followers)	-0.196	-0.982	0.138
Test (2023-2024)	Cluster 0 (Leaders)	5.502	0.929	-0.269
Test (2023-2024)	Cluster 1 (Followers)	5.192	-0.924	0.115

Table 4.8: Cluster Centroids in PCA Space - Training and Test Periods (Trial 2).

The centroid values reveal patterns consistent with Trial 1 but with notable differences. In the training period, the *Leaders* cluster centroid records a positive *PC2* score of 0.781, while the *Followers* centroid records -0.982, confirming that economic capacity and infrastructure development remain the primary differentiating factors between the two groups. The *PC1* difference between clusters in the training period remains modest (0.156 vs -0.196), consistent with Trial 1, as the EU-level technological variables that dominate *PC1* evolve uniformly across all member states.

In the test period, both centroids record substantially elevated *PC1* values of 5.502 and 5.192 respectively, reflecting the universal technological advancement of the post-2020 period. The *PC2* separation between clusters is maintained in the test period, with the *Leaders* centroid recording 0.929 compared to -0.924 for the *Followers* cluster. Notably, the *PC1* gap between clusters in the test period is wider in Trial 2 (0.310) compared to Trial 1 (0.211), indicating that the inclusion of post-COVID period does not eliminate the differences of technological adoption across the two groups when the training data includes post-2020 observations.

Cluster Transitions

A striking contrast with Trial 1 is the stability of cluster memberships in Trial 2. All 27 EU member states remained in their assigned cluster when comparing the training and test periods, with no genuine transitions observed. This stands in sharp contrast to the apparent mass transition in Trial 1, which was attributed to a K-Means label swap. The full consistency of cluster assignments in Trial 2 confirms that the label swap interpretation of the Trial 1 transitions was correct, when the training data includes post-2020 observations, the algorithm assigns cluster labels consistently across both periods, revealing the true stability of the EU member state groupings.

It is worth noting that *Portugal*'s position in the PCA space shifted noticeably between the training and test periods, placing it increasingly close to the cluster boundary by 2023–2024. While this movement did not constitute a cluster reassignment, *Portugal* remained in the *Followers* cluster throughout both the training and test periods of Trial 2, the progressive shift along *PC2* signals a gradual convergence toward the *Leaders* group in terms of economic readiness and charging infrastructure. This boundary proximity foreshadows the repositioning observed in Trial 3, where *Portugal*'s PCA scores cross into *Leaders* territory under the exclusively post-COVID training window.

The complete stability of cluster compositions across the training and test periods confirms the robustness of the fundamental division between *Leaders* and *Followers* in the European BEV market. The persistence of this partition across an extended training window that includes both pre and post-COVID observations suggests that the structural differences between the two groups that is driven primarily by economic capacity and infrastructure development as captured by *PC2*, and are deeply embedded characteristics of these markets rather than transient features of any particular time period.

4.4 Trial 3

The training period for the last trial was from 2020-2023 and tested on 2024.

4.4.1 Feature Selection and Dimensionality Reduction

Correlation Analysis

The Pearson correlation between each feature and the log-transformed target variable was computed on the Trial 3 training dataset covering the post-COVID period 2020-2023. Table 4.9 presents the correlation coefficients in descending order.

Feature	Correlation with Log BEV %
Log Recharging Points	0.617
Log Available BEV Models	0.484
YJ Purchase Price (EUR)	0.482
Battery Capacity	0.480
Log DC Recharging Speed (km/h)	0.479
Real Range	0.478
AC Recharging Speed (km/h)	0.444
Employment Growth	0.306
Log GDP	0.273
Log CPI	0.248

Table 4.9: Correlation of features with Log BEV Market Share — Trial 3 Training Dataset (2020-2023).

The correlation structure in Trial 3 differs notably from both Trial 1 and Trial 2, reflecting the distinct dynamics of the exclusively post-COVID training period. *Log Recharging Points* re-emerges as the strongest predictor (0.617), reclaiming the position it held in Trial 1 and surpassing *Log Available BEV Models* which led in Trial 2. This shift suggests that within the post-COVID period, the availability of charging infrastructure became the most critical driver of cross-country variation in BEV adoption, as the rapid expansion of charging networks during 2020-2023 created meaningful differences in infrastructure readiness across EU member states.

Notably, the correlation values across the remaining technological variables — *Log Available BEV Models* (0.484), *YJ Purchase Price* (0.482), *Battery Capacity* (0.480), *Log DC Recharging Speed* (0.479), and *Real Range* (0.478) are remarkably similar to one another, forming a tight cluster of correlations between 0.478 and 0.484. This convergence suggests that within the post-COVID period, all EU-level technological variables contribute approximately equally to BEV adoption variation, having all advanced sufficiently to become relevant rather than acting as binding constraints as they did in the earlier diffusion period. *Log GDP* (0.273) and *Log CPI* (0.248) recorded higher correlations in Trial 3 compared to Trial 1, reflecting the increased relevance of macroeconomic conditions in the post-COVID period characterised by inflationary pressures and economic recovery dynamics.

Variance Inflation Factor (VIF)

As in Trials 1 and 2 (Table 4.2), the VIF analysis confirmed severe multicollinearity across the feature set, with identical VIF values recorded across all three trials due to the shared feature set and the inter-feature relationships being independent of the target variable. *Real Range* (537.36), *Battery Capacity* (271.27), and *Log DC Recharging Speed* (131.91) again recorded the highest VIF values, confirming that PCA remains necessary to resolve multicollinearity prior to model training in Trial 3.

Principal Component Analysis

Following standardization of the Trial 3 training data, PCA was applied to the full set of ten features. Figure 4.9 presents the scree plot derived from the elbow method.

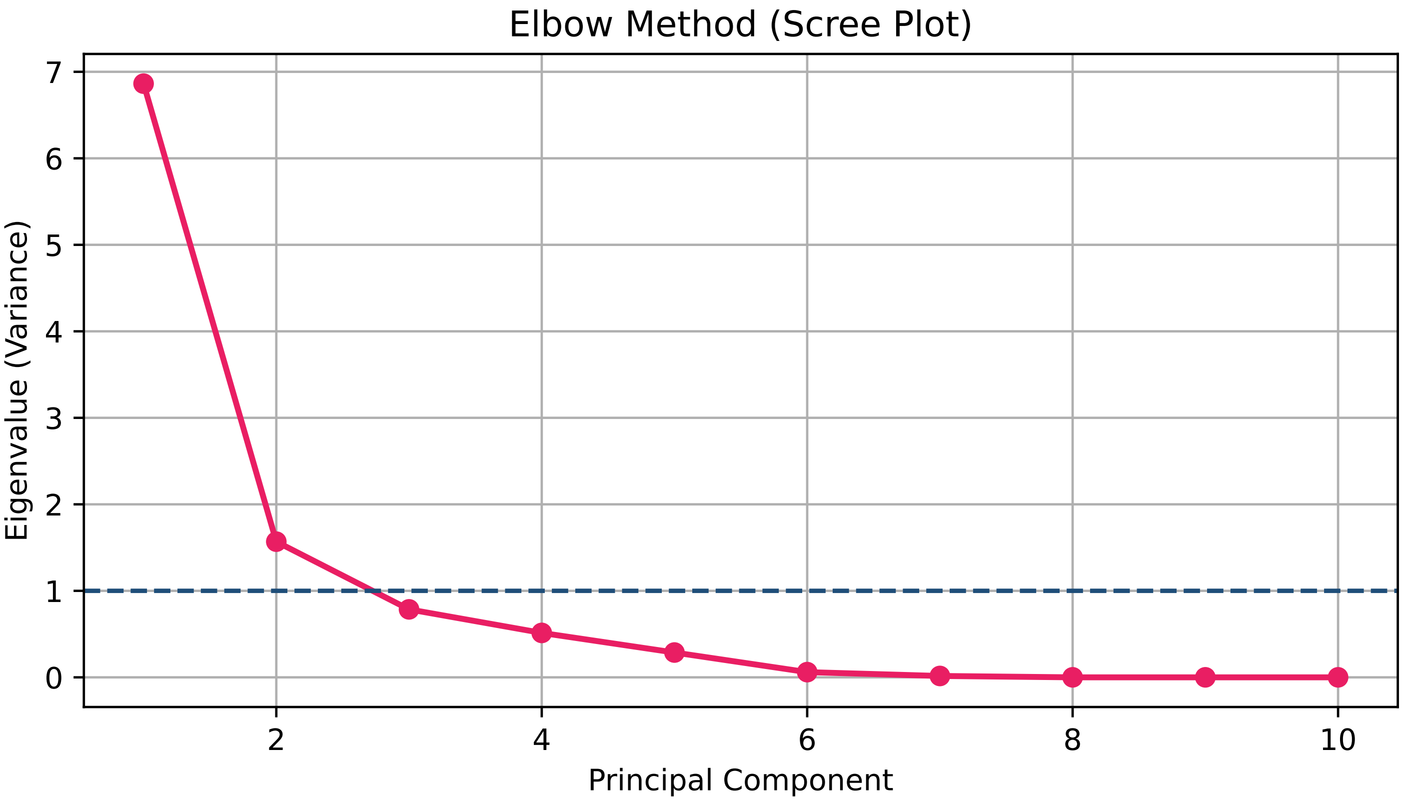

Figure 4.9: Scree Plot — Elbow Method for Trial 3 (2020-2023 Training Dataset).

In contrast to Trials 1 and 2, where three principal components were retained, the elbow method applied to the Trial 3 training data identifies the elbow at principal component = 2, with the eigenvalue falling below 1.0 from the third component onwards. Only two principal components were therefore retained for Trial 3. This reduction in the number of retained components reflects the more compact and concentrated variance structure of the post-COVID dataset, where the shorter and more homogeneous training period (2020-2023) results in a stronger dominant component that captures a larger proportion of total variance of 83.53%. The correlation of each principal component with the target variable confirmed that both *PC1* (-0.498) and *PC2* (-0.358) are meaningfully associated with *Log BEV%*. The negative signs of these correlations reflect the sign convention of the PCA solution for this training period rather than an inverse relationship, as the direction of principal components is mathematically arbitrary. Following PCA, the VIF of both principal components was confirmed to be exactly 1.0, verifying the complete elimination of multicollinearity.

Loading Factors

Table 4.10 presents the loading factors for each feature across the two retained principal components for Trial 3.

Feature	PC1	PC2
Log Recharging Points	-0.132	-0.644
Real Range	-0.381	0.029
Log Available BEV Models	-0.381	0.028
YJ Purchase Price (EUR)	-0.380	0.021
AC Recharging Speed (km/h)	-0.336	-0.009
Battery Capacity	-0.381	0.026
Log DC Recharging Speed (km/h)	-0.381	0.034
Employment Growth	-0.218	0.264
Log GDP	-0.038	-0.715
Log CPI	-0.310	0.016

Table 4.10: PCA Loading Factors — Trial 3 (2020–2023 Training Dataset).

The loading structure in Trial 3 presents the most distinctive pattern across all three trials, reflecting the concentrated nature of the post-COVID training period. All features load negatively on *PC1*, which as noted above reflects the sign convention of the PCA solution rather than genuine inverse relationships. *PC1* is dominated by technological performance variables, with *Real Range* (-0.381), *Battery Capacity* (-0.381), *Log Available BEV Models* (-0.381), *Log DC Recharging Speed* (-0.381), *YJ Purchase Price* (-0.380), and *AC Recharging Speed* (-0.336) all loading at near-identical magnitudes. This remarkable convergence in loading magnitudes across all technological variables at EU-level is consistent with the correlation analysis, which showed that these variables clustered tightly between 0.478 and 0.484. It confirms that within the post-COVID period, all BEV technological dimensions evolved in a highly synchronised manner across all EU member states, causing the variables to contribute in a very similar patterns of variations in the post-COVID period. *PC1* can therefore be interpreted as a *Unified BEV Technology* component capturing the collective advancement of the European BEV product landscape during 2020-2023.

PC2 presents a markedly different structure, dominated by *Log GDP* (-0.715) and *Log Recharging Points* (-0.644), with a secondary contribution from *Employment Growth* (0.264). This component retains the *Economic and Infrastructure* interpretation established in Trials 1 and 2, capturing the cross-country variation in economic capacity and charging network development that remained the primary differentiating factor across EU member states even within the exclusively post-COVID training window. The consistency of this *PC2* interpretation across all three trials despite the varying temporal composition of the training data reinforces the conclusion that economic and infrastructural heterogeneity is a structural and persistent feature of the European BEV market rather than a period-specific phenomenon.

4.4.2 Clustering Results

Optimal Number of Clusters

Following PCA, K-Means clustering was applied separately to the training dataset (2020-2023) and the test dataset (2024) to identify the optimal partition of EU member states. In the same way as the previous, optimal number of clusters was determined using the elbow method and validated using the silhouette score.

Figure 4.10 present the elbow plots for the training and test periods respectively. In both cases, the WCSS shows a sharp decline from $k = 1$ to $k = 2$, after which the rate of reduction diminishes substantially, confirming that two clusters provide the most meaningful partition of the data. This finding is consistent across all three trials, reinforcing the robustness of the two-cluster solution for the European BEV market.

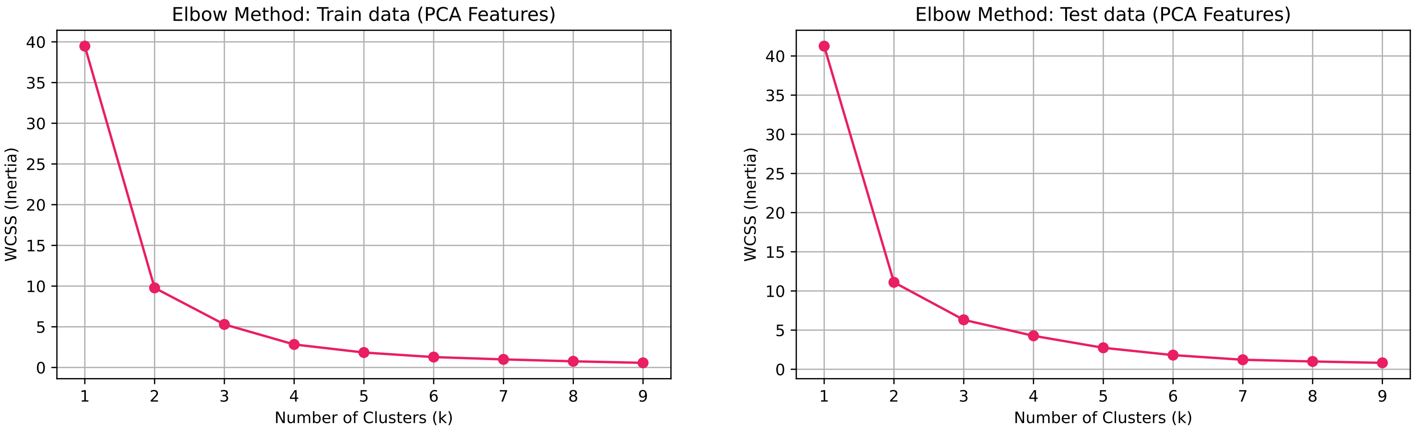

Figure 4.10: Elbow Method - Train Data (left) and Test Data (right) PCA Features (Trial 3).

The silhouette score validation is presented in Figure 4.11. The highest silhouette score is recorded at $k = 2$ for both the training period (0.620) and the test period (0.600), representing the strongest cluster separation observed across all three trials. The consistently high silhouette scores at $k = 2$ in Trial 3 compared to Trials 1 and 2 suggest that the post-COVID training period produces more clearly defined and internally cohesive clusters, reflecting the sharper differentiation between high and low adoption member states that emerged during the acceleration phase 2020-2023.

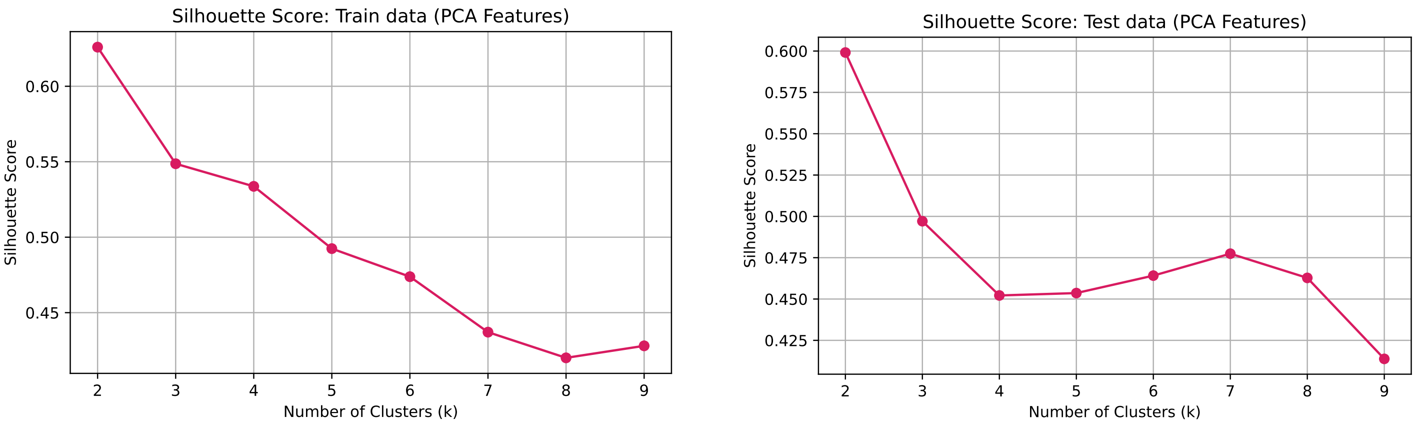

Figure 4.11: Silhoutte Method - Train Data (left) and Test Data (right) PCA Features (Trial 3).

Cluster Composition

Based on the K-Means clustering results, the 27 EU member states were again partitioned into two groups. The *Leaders* cluster comprises the 15 member states: *Austria*, *Belgium*, *Czech Republic*, *Denmark*, *Finland*, *France*, *Germany*, *Hungary*, *Italy*, *Netherlands*, *Poland*, *Portugal*, *Romania*, *Spain*, and *Sweden*. The *Followers* cluster consists of: *Bulgaria*, *Croatia*, *Cyprus*, *Estonia*, *Greece*, *Ireland*, *Latvia*, *Lithuania*, *Luxembourg*, *Malta*, *Slovakia*, and *Slovenia*. The broadly stable two-cluster structure across all three trials, with only *Portugal* shifting into the *Leaders* group from the *Followers* group in Trial 3, further confirms that the fundamental division of EU member states is a robust feature of the European BEV market.

Cluster visualisation and Interpretation

Figure 4.12 presents the cluster visualisation in the *PC1 – PC2* space for both the training and test periods.

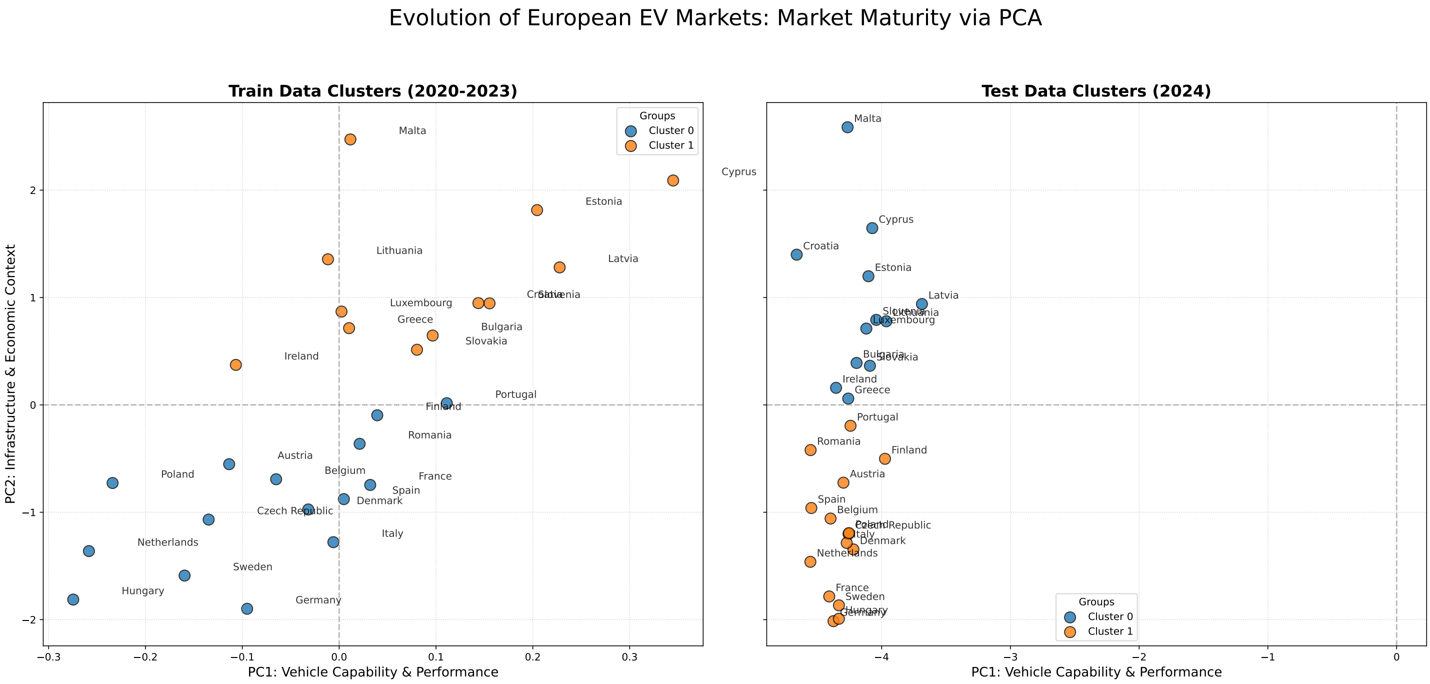

Figure 4.12: Evolution of European EV Markets: Train Data (2020-2023) vs Test Data (2024) Cluster Visualisation - Trial 3.

The Trial 3 visualisation presents the most distinctive cluster structure across all three trials, driven by two important differences. First, as established in the loading factor analysis, all features load negatively on *PC1* in Trial 3 due to the sign convention of the PCA solution, meaning that higher BEV adoption is associated with more negative *PC1* values. The most advanced *Leaders* such as *Hungary*, *Germany*, and *Sweden* now occupy the far left of the plot rather than the far right.

In the training period (2020–2023), the two clusters are separated primarily along the *PC2* axis, which retains its *Economic and Infrastructure* interpretation from previous trials. *Cluster 1* corresponds to *Followers* in the training period that occupies the upper portion of the plot with positive *PC2* scores, while *Cluster 0* corresponding to *Leaders* occupies the lower portion with negative *PC2* scores. This inversion of the *PC2* axis relative to Trials 1 and 2 again reflects the sign convention of the Trial 3 PCA solution rather than a genuine reversal of economic and infrastructure relationships. Notably, *Malta* appears in the upper left quadrant during the training period despite being classified as a *Follower*, suggesting an unusual combination of infrastructure investment and lower vehicle capability scores in the post-COVID period.

In the test period (2024), the entire distribution shifts further right along *PC1*, consistent with the continued technological advancement of BEV products in 2024. The separation between clusters remains driven by *PC2*, with the *Leaders* cluster maintaining lower *PC2* scores in the inverted coordinate system of Trial 3. *Malta* and *Cyprus* occupy notably high *PC2* positions in the test period, consistent with their *Followers* classification despite relatively high infrastructure scores on this inverted axis.

Cluster Centroids

Table 4.11 presents the cluster centroids in the principal component space for both the training and test periods.

Period	Cluster	PC1	PC2
Train (2020–2023)	Cluster 0 (Leaders)	-0.078	-0.935
Train (2020–2023)	Cluster 1 (Followers)	0.097	1.169
Test (2024)	Cluster 0 (Leaders)	-4.151	0.919
Test (2024)	Cluster 1 (Followers)	-4.334	-1.200

Table 4.11: Cluster Centroids in PCA Space – Training and Test Periods (Trial 3).

The centroid values in Trial 3 reflect the inverted coordinate system of the PCA solution. In the training period, the *Leaders* cluster centroid records a negative *PC2* score of -0.935, while the *Followers* centroid records a positive *PC2* of 1.169, consistent with the sign inversion noted in the loading factor analysis. In the test period, both centroids shift substantially to more negative *PC1* values (-4.151 and -4.334 respectively), reflecting the continued advancement of BEV technology in 2024. The *PC2* separation between clusters in the test period is maintained, with the *Leaders* centroid recording 0.919 and the *Followers* centroid recording -1.200, again consistent with the inverted sign convention.

It is important to note that despite the sign inversions in Trial 3, the substantive interpretation of the cluster centroids is consistent with Trials 1 and 2. In all three trials, the *Leaders* cluster is characterised by stronger economic capacity and infrastructure development as captured by *PC2*, and by higher overall BEV technological performance as captured by *PC1*. The sign differences across trials are a mathematical artefact of the PCA solution and do not alter the underlying market interpretation.

Cluster Transitions

All 27 EU member states transitioned between cluster labels when comparing the training period (2020-2023) and the test period (2024), with no country remaining in the same cluster. As established in the Trial 1 discussion, this pattern is consistent with a K-Means label swap rather than a genuine mass reordering of country groupings [Mac67]. This interpretation is strongly supported by two observations. First, the cluster visualisation in Figure 4.12 confirms that the relative positioning of countries within the principal component space remains broadly consistent between the training and test periods, with historically strong BEV markets such as *Germany*, *Sweden*, *France*, and *Hungary* continuing to occupy the lower portion of the plot corresponding to the *Leaders* cluster in both periods. Second, the cluster composition used for modelling remains broadly stable across all three trials, with the only exception of *Portugal* transitioning from *Followers* to *Leaders* group, confirming that the fundamental grouping of member states is stable regardless of the label assigned by the algorithm in any given run.

The complete label swap observed in Trial 3 in contrast to the partial swap in Trial 1 and the near-complete stability of Trial 2 can be attributed to the shorter and more homogeneous training window of 2020-2023. With only four years of post-COVID observations, the within-cluster variance is more evenly distributed across the two groups, making the algorithm more susceptible to arbitrary label assignment. This finding further highlights the sensitivity of unsupervised clustering to the temporal composition of the training data, consistent with the broader argument of this study that the pre and post-COVID periods represent fundamentally distinct market regimes.

These findings collectively confirm the robustness of the *Leaders* and *Followers* partition across all three trials and provide the clustering foundation upon which the model performance results presented in Section 4.5 are based.

4.5 Model Performance

Following is a summary of the clustering results presented in Sections 4.2, 4.3, and 4.4.

Aspect	**Trial 1 (2011–2019)**	**Trial 2 (2011–2022)**	**Trial 3 (2020–2023)**
Notable shift	Infrastructure leads	Technology surpasses infrastructure	All tech variables cluster tightly
VIF (highest)	Real Range (537.36)	Identical to Trial 1	Identical to Trial 1
VIF after PCA	1.0	1.0	1.0
Components retained	3	3	2
Variance explained	79.36%	77.14%	83.53%
PC1 interpretation	BEV Technology & Market Maturity	BEV Technology & Market Maturity	Unified BEV Technology
PC2 interpretation	Economic & Infrastructure	Economic & Infrastructure	Economic & Infrastructure
PC3 interpretation	Macroeconomic & Cost	Inflationary Variables (CPI dominant)	(dropped)

Table 4.12: Cross-trial comparison of correlation structure, multicollinearity, and PCA outcomes.

4.5.1 Evaluation Metrics

As described in Section 3.2.7 and discussed in detail in Section 2.6, two complementary metrics were used to evaluate model performance across all trials and clusters — *MAE* and *MedAE*, both expressed in percentage points of BEV market share.

MAE captures the average prediction error across all country-year observations within each cluster and trial, giving equal weight to every error regardless of its magnitude. MedAE captures the typical prediction error by taking the median rather than the mean, making it robust to the influence of extreme outliers. Given the substantial heterogeneity in BEV adoption rates across EU member states where a small number of frontrunner countries such as *Germany*, *Sweden*, and *Netherlands* record adoption rates several times higher than the EU average; MedAE is particularly relevant for this study as it reflects the error experienced by the typical country rather than being inflated by extreme values. The two metrics are therefore complementary; MAE provides a complete picture of overall prediction accuracy while MedAE provides a more robust measure of typical performance across the heterogeneous EU market landscape.

All results are reported separately for the *Leaders* and *Followers* clusters and sorted in ascending order of Trial 3 performance, reflecting the primary objective of achieving the most accurate near-term forecasts within the post-COVID market regime.

4.5.2 Model Performance — MAE

Table 4.13 presents the MAE across all models, clusters, and trials, sorted in ascending order of Trial 3 performance.

Table 4.13: MAE % by Model, Cluster and Trial.

Cluster	Model	Trial 1	Trial 2	Trial 3
Followers	Proposed (ElasticNet)	5.539	5.048	**3.683**
	ARIMAX	4.896	5.723	4.122
	Proposed (Weighted)	5.247	5.445	4.442
	Proposed (GradientBoost)	5.708	5.106	4.508
	Linear Regression	5.918	7.269	7.407
	RF	4.877	6.708	8.172
	XGB	5.868	6.873	8.750
	LSTM	6.328	11.330	11.760
	SVR	17.139	162.848	12.994
Leaders	Proposed (ElasticNet)	9.281	5.071	**2.971**
	Proposed (GradientBoost)	9.938	5.633	3.146
	ARIMAX	8.644	5.298	3.542
	Proposed (Weighted)	9.101	5.467	4.130
	Linear Regression	10.559	12.750	11.649
	RF	8.803	11.830	12.789
	XGB	11.260	12.866	12.821
	LSTM	11.846	11.793	14.620
	SVR	150.710	19.156	19.487

Followers Cluster

The results for the *Followers* cluster are most meaningfully interpreted through the performance achieved in Trial 3, where the training data fully reflects the post-COVID market regime. In this setting, the proposed models clearly outperform all baseline approaches. *Proposed (ElasticNet)* achieves the lowest MAE of 3.683%, followed by *Proposed (GradientBoost)* at 4.508% and *Proposed (Weighted)* at 4.442%.

A key observation is the consistent improvement exhibited by the proposed models across all three trials. *Proposed (ElasticNet)* reduces its error from 5.539% in Trial 1 to 3.683% in Trial 3, while *Proposed (GradientBoost)* improves from 5.708% to 4.508%. This steady reduction indicates that the stacking framework becomes increasingly effective as the training data better captures the structural characteristics of the post-2020 period.

Among the baseline models, *ARIMAX* remains the strongest comparator with a Trial 3 MAE of 4.122%, although its performance fluctuates across trials, suggesting limited robustness to changes in the training window. In contrast, machine learning models such as *RF* and *XGB* exhibit clear deterioration in Trial 3, indicating an inability to generalise to the evolving dynamics of the *Followers* markets. *SVR* consistently records the highest errors, confirming its lack of suitability for this task.

These findings highlight that, within the *Followers* cluster, the advantage of the proposed approach is not limited to marginal gains in accuracy but extends to improved stability and adaptability across training regimes.

Leaders Cluster

The superiority of the proposed models becomes even more pronounced in the *Leaders* cluster, particularly in Trial 3. *Proposed (ElasticNet)* achieves an MAE of 2.971%, the lowest error observed across all models, clusters, and trials. *Proposed (GradientBoost)* follows closely with 3.146%, reinforcing the strength of the stacking framework in more complex market environments.

The magnitude of improvement from Trial 1 to Trial 3 is substantial. *Proposed (ElasticNet)* reduces its error by approximately 68%, while *Proposed (GradientBoost)* shows a similarly strong decline. This pattern indicates that the proposed architecture is particularly effective when sufficient post-COVID information is incorporated into the training data.

ARIMAX again emerges as the most competitive baseline, achieving a Trial 3 MAE of 3.542%. Other approaches, including *LSTM*, *RF*, *XGB*, and *Linear Regression* perform substantially worse, with errors remaining above 10% in Trial 3. This suggests that both deep learning and tree-based methods struggle to capture the rapid and heterogeneous adoption dynamics characteristic of the *Leaders* markets.

The close alignment between *Proposed (ElasticNet)* and *Proposed (GradientBoost)* in Trial 3 further indicates that the performance gains are driven by the underlying ensemble structure rather than reliance on a single model specification.

Overall Assessment

Across both clusters, Trial 3 provides the most reliable indication of model capability, as it reflects forecasting performance within a consistent post-COVID regime. In this setting, the proposed models clearly establish a new performance benchmark.

Proposed (ElasticNet) consistently achieves the lowest MAE, while *Proposed (GradientBoost)* provides closely comparable results. Their joint performance demonstrates that combining complementary modelling approaches within a structured stacking framework leads to both higher accuracy and greater robustness.

While *ARIMAX* remains a strong individual baseline, no standalone model matches the consistency and level of performance achieved by the proposed methods. The results therefore support the conclusion that the stacking architecture offers a more reliable and effective approach for forecasting BEV market share under current market conditions.

A detailed visualisation of how these models performed for each country can be found in Section 6.1 and in the GitHub Repository 6.3 of Appendix 6. These findings are further examined using the Median Absolute Error in the following section.

4.5.3 Model Performance — MedAE

Table 4.14 presents the MedAE across all models, clusters, and trials, sorted in ascending order of Trial 3 performance.

Table 4.14: MedAE % by Model, Cluster and Trial.

Cluster	Model	Trial 1	Trial 2	Trial 3
Followers	Proposed (ElasticNet)	2.9437	3.3987	**2.2079**
	Proposed (Weighted)	2.9485	3.1991	2.1896
	Proposed (GradientBoost)	3.2251	3.1560	2.2462
	ARIMAX	2.7177	3.2305	2.9195
	Linear Regression	3.5502	3.3647	4.2537
	LSTM	4.170	7.453	4.253
	RF	2.4503	4.9279	5.6356
	XGB	3.4803	4.3348	6.6140
	SVR	5.0478	152.8129	10.1787
Leaders	Proposed (ElasticNet)	5.3177	2.5746	**1.2821**
	Proposed (GradientBoost)	6.0523	3.2924	1.7208
	Proposed (Weighted)	4.4244	3.6201	1.9152
	ARIMAX	4.6018	3.5760	1.9570
	LSTM	7.328	5.314	6.165
	RF	5.7276	12.1185	8.8497
	XGB	6.6631	11.1940	9.0384
	Linear Regression	6.6732	7.6936	10.6022
	SVR	21.8152	15.2423	17.8809

Followers Cluster

The MedAE results for the *Followers* cluster reinforce the patterns observed in the MAE analysis, with the most meaningful comparison arising in Trial 3. In this setting, the proposed models again achieve the strongest performance. *Proposed (ElasticNet)* records the lowest median error at 2.2079%, closely followed by *Proposed (Weighted)* at 2.1896% and *Proposed (GradientBoost)* at 2.2462%. The narrow range among these models indicates a consistently strong performance of the stacking framework.

A notable feature is the stability of the proposed models across trials. While some fluctuation is present, their Trial 3 performance remains clearly superior to all baseline approaches. *Proposed (GradientBoost)* shows a steady reduction from 3.2251% in Trial 1 to 2.2462% in Trial 3, while *Proposed (ElasticNet)* achieves the lowest overall error despite a temporary increase in Trial 2.

Among the baselines, *ARIMAX* again performs relatively well with a Trial 3 MedAE of 2.9195%, but it remains notably less accurate than the proposed models. Other approaches such as *RF* and *XGB* exhibit substantial deterioration in Trial 3, while *SVR* continues to produce extreme errors, confirming its lack of robustness.

Overall, the MedAE results highlight that the proposed models not only reduce average errors but also improve the distributional consistency of predictions, limiting the impact of large deviations.

Leaders Cluster

The advantages of the proposed approach are even more pronounced in the *Leaders* cluster. In Trial 3, *Proposed (ElasticNet)* achieves a MedAE of 1.2821%, representing the lowest median error across all models and clusters. *Proposed (GradientBoost)* follows with 1.7208%, further confirming the effectiveness of the stacking framework in more complex market conditions.

The reduction in median error from Trial 1 to Trial 3 is substantial. *Proposed (ElasticNet)* improves from 5.3177% to 1.2821%, while *Proposed (GradientBoost)* decreases from 6.0523% to 1.7208%. This consistent improvement indicates that the proposed models are able to adapt effectively as the training data increasingly reflects the post-COVID environment.

ARIMAX again serves as the strongest baseline, achieving a Trial 3 MedAE of 1.9570%, but it does not match the accuracy of the proposed models. All remaining baselines record significantly higher errors, with *LSTM*, *RF*, and *XGB* exceeding 7% in Trial 3, indicating poor robustness to structural changes in the *Leaders* markets.

The close performance of *Proposed (ElasticNet)* and *Proposed (GradientBoost)* further supports the conclusion that the gains arise from the ensemble structure rather than any single modelling component.

Overall Assessment

The MedAE results provide strong complementary evidence to the MAE analysis. When evaluated under the most representative setting of Trial 3, the proposed models consistently achieve the lowest errors across both clusters.

Proposed (ElasticNet) emerges as the most accurate model, while *Proposed (GradientBoost)* delivers closely comparable performance. Their superiority in terms of median error indicates not only improved accuracy but also greater robustness to outliers and extreme deviations.

Although *ARIMAX* remains a competitive baseline, it does not reach the same level of performance or consistency. The findings therefore confirm that the proposed stacking architecture provides a more reliable and stable forecasting framework for BEV market share in the current European context.

4.6 Post-Assessment of Forecasting Models: Theoretical Expectations vs. Empirical Results

This section evaluates the empirical performance of each model against the theoretical expectations established in the pre-assessment in Section 2.5. For each model, the observed MAE and MedAE results across both clusters and all three trials are compared against the a-priori predictions, deviations are identified and explained, and the implications for the broader conclusions of this study are discussed.

4.6.1 Linear Regression

Expected: Linear Regression was expected to serve as the simplest possible baseline model against which all other models would be benchmarked. Given its strong linearity assumption and inability to capture non-linear dynamics or structural breaks, it was predicted to record high errors across all trials, particularly in Trial 1 where the structural disconnect between the pre-COVID training period and the post-COVID test period is most pronounced.

Observed: The empirical results partially confirm these expectations but reveal an important nuance. In the *Followers* cluster, Linear Regression recorded a Trial 3 MAE of 7.41% and MedAE of 4.25%, placing it in the lower half of the performance ranking but not at the bottom. In the *Leaders* cluster, it recorded a Trial 3 MAE of 11.65% and MedAE of 10.60%, again in the lower half of the ranking. Notably, Linear Regression performed comparably to or better than several more complex models including *RF* and *XGB*, particularly in the *Followers* cluster across Trials 1 and 2.

Explanation of deviation: The relatively competitive performance of Linear Regression in the *Followers* cluster can be attributed to the nature of BEV adoption in lower-adoption markets, where growth trajectories are more gradual and linear than in the *Leaders* cluster. In markets where adoption follows a steady upward trend without sharp acceleration, the linearity assumption of the model is less violated, and the simplicity of the model provides a degree of robustness against overfitting to pre-COVID patterns. This finding is consistent with the theoretical prediction that linear models perform adequately when the true relationship is approximately linear, and confirms that the *Followers* cluster exhibits more linearly tractable adoption dynamics than the *Leaders* cluster.

4.6.2 SVR

Expected: SVR was expected to outperform Linear Regression by capturing non-linear relationships between BEV adoption and its drivers through the RBF kernel. It was predicted to be competitive in Trial 3 where training and test data are drawn from the same post-COVID regime but it underperformed significantly in Trial 1 due to its reliance on distributional similarity between training and test data.

Observed: SVR recorded the worst performance across all models in both clusters and all trials, with a Trial 1 MAE of 17.14% in the *Followers* cluster and an extreme 150.71% in the *Leaders* cluster. Even in Trial 3, where conditions were most favorable, SVR recorded a MAE of 12.99% and 19.49% in the *Followers* and *Leaders* clusters respectively, having the highest errors of any model.

Explanation of deviation: The failure of SVR in the *Leaders* cluster in Trial 1 significantly exceeded the theoretical expectation. This extreme result is likely attributable to the sensitivity of the RBF kernel to the scale and distribution of the training data combined with the dramatic distributional shift between the pre-COVID training period and the post-COVID test period. The RBF kernel learned a similarity structure based on the slow and uniform growth dynamics of the pre-COVID *Leaders* markets. It was characterized by relatively low and gradually increasing adoption rates that became entirely inapplicable when tested on the rapid and heterogeneous post-2020 acceleration phase. The failure of SVR to recover competitive performance even in Trial 3 suggests that the hyperparameters selected $C = 100$ and $\varepsilon = 0.1$ were not well suited to the scale and dynamics of the post-COVID BEV market, and that SVR is fundamentally ill-suited to this forecasting task regardless of the training window.

4.6.3 ARIMAX

Expected: ARIMAX was expected to perform well in trials where the training data is drawn from a structurally stable period, capturing both temporal autocorrelation and the linear influence of exogenous drivers. It was predicted to degrade in Trial 1 due to the stationarity violation introduced by the structural break, and to recover competitive performance in Trial 3 where the training data is drawn exclusively from the post-COVID regime.

Observed: The empirical results strongly confirm these theoretical expectations. ARIMAX recorded a Trial 3 MAE of 4.12% in the *Followers* cluster and 3.54% in the *Leaders* cluster, making it the best performing individual model in Trial 3 across both clusters. Its MedAE in Trial 3 was 2.92% and 1.96% in the *Followers* and *Leaders* clusters, respectively. In Trial 1, ARIMAX recorded a MAE of 4.90% in *Followers* and 8.64% in *Leaders*, reflecting the impact of the structural break on its stationarity assumption. In Trial 2, performance showed mixed behavior by improving in the *Leaders* cluster but deteriorating slightly in *Followers*, which is consistent with the theoretical prediction that the conflicting pre-COVID and post-COVID signals in the extended training window create instability in the auto regressive coefficient estimates.

Explanation of Deviation: The strong Trial 3 performance of ARIMAX confirms the theoretical prediction that its linear temporal structure is well suited to capturing the more homogeneous and stationary dynamics of the post-COVID training period. The country-level implementation that fits a separate ARIMAX model for each of the 27 EU member states, allowed the model to adapt its auto regressive order to the specific temporal dynamics of each country, providing flexibility that a single pooled model would not have achieved. This finding further validates the selection of ARIMAX as a base learner in the proposed stacking architecture, as its strong individual performance in Trial 3 confirms that it captures meaningful temporal signal that can be leveraged by the meta-learner.

4.6.4 LSTM

Expected: LSTM was expected to capture non-linear sequential dependencies and acceleration phases in BEV adoption that linear models cannot represent. It was predicted to be sensitive to the small training sample size and to exhibit high variance, potentially struggling in Trial 2 where conflicting pre-COVID and post-COVID temporal patterns are present simultaneously in the training data.

Observed: The empirical results reveal a clear divergence in LSTM performance across clusters and trials. In the *Followers* cluster, LSTM performance degraded over the trials, recording a Trial 3 MAE of 11.76% and MedAE of 4.25%. In the *Leaders* cluster, LSTM shows even stronger instability. MAE decreases from 11.846% in Trial 1 to 11.793% in Trial 2 before increasing sharply to 14.620% in Trial 3, making it one of the weakest-performing models in this cluster, especially in Trial 3. This suggests that the model struggles to generalise across heterogeneous adoption regimes, particularly in high-acceleration markets where structural breaks are more pronounced.

A sensitivity analysis was conducted using timestep values of 2 and 3 in the standalone LSTM configuration. The results indicate that timestep selection has a non-linear effect on predictive performance. A timestep of 2 produced lower prediction error, indicating that a short lag structure is sufficient to capture meaningful year-on-year dependencies in annual BEV adoption data without introducing excessive model complexity. In contrast, timestep 3 consistently increased prediction error across clusters, reflecting overfitting and instability due to the extremely limited number of annual observations per country. This confirms that LSTM performance in this dataset is maximised at a minimal temporal window, where short-range dynamics are preserved without over-parameterising the sequence structure.

The strong deterioration of LSTM in the *Leaders* cluster during Trial 2 exceeds initial expectations and reflects the impact of structural breaks in BEV adoption dynamics. Countries such as *Germany*, *Sweden*, and the *Netherlands* exhibit sharp post-COVID acceleration, creating conflicting pre- and post-2020 patterns that the LSTM's gating mechanism cannot effectively adapt within a limited annual dataset. This leads to unstable learning and higher error in mixed-regime training. *MAE* and *MedAE* scores worsens in Trial 3 when only post-COVID data is used because the model is constrained by a small sample size of 4 years which limits the model's ability to learn patterns increases sensitivity to temporal window specification.

Explanation of deviation: The Trial 2 deterioration of LSTM in the *Leaders* cluster substantially exceeded the theoretical expectation mentioned in the pre-assessment 2.5. This result confirms the theoretical prediction regarding sensitivity to conflicting temporal signals but shows that the magnitude of this conflict is considerably stronger in the *Leaders* cluster than in the *Followers* cluster. The *Leaders* cluster contains EU member states such as *Germany*, *Sweden*, and the *Netherlands*, which experienced pronounced post-COVID acceleration in BEV adoption, creating a sharp discontinuity between pre-2020 and post-2020 dynamics in the extended training window for Trial 2. The LSTM's gating mechanism, designed to regulate the flow of temporal information, is unable to effectively reconcile these conflicting regimes when both are present in the training data, leading to unstable learning and degraded performance. The absence of improvement in Trial 3 further indicates that restricting the training window does not fully resolve this issue, as performance remains constrained by the limited sample size and the model's sensitivity to structural changes in temporal patterns.

4.6.5 Random Forest

Expected: Random Forest was expected to handle non-linearities and variable interactions robustly through its ensemble averaging mechanism, with competitive performance across trials. However it was predicted to be limited by its inability to extrapolate beyond the range of its training distribution, particularly in Trial 1.

Observed: Random Forest performed competitively in Trial 1 recording a MAE of 4.88% in *Followers* and 8.80% in *Leaders* but deteriorated substantially in subsequent trials. By Trial 3, its MAE had increased to 8.17% in *Followers* and 12.79% in *Leaders*, placing it among the weaker performing models. Its MedAE in Trial 3 was 5.64% and 8.85% in the two clusters respectively.

Explanation of deviation: The deterioration of Random Forest across trials directly confirms the theoretical prediction about extrapolation limitations. The relatively competitive Trial 1 performance which initially appeared promising is best interpreted as overfitting to the pre-COVID patterns of the training data rather than genuine generalization capability.

Random Forest cannot extrapolate beyond the range of values observed during training, meaning that as the post-COVID acceleration pushed BEV adoption rates well beyond the levels seen in the pre-COVID training period, the model's predictions became increasingly unreliable. The progressive deterioration across trials confirms this interpretation as the test period shifts closer to the present and adoption rates continue to accelerate, the out-of-distribution nature of the test data becomes more pronounced.

4.6.6 XGBoost

Expected: XGBoost was expected to be a strong competitor due to its demonstrated empirical performance on structured regression tasks and its regularization mechanisms. Like Random Forest, it was predicted to struggle with extrapolation beyond the training distribution, particularly in Trial 1.

Observed: XGBoost followed a similar pattern to Random Forest, recording moderate Trial 1 performance with MAE of 5.87% in *Followers* and 11.26% in *Leaders*, however, deteriorated substantially in Trials 2 and 3. Its Trial 3 MAE of 8.75% in *Followers* and 12.82% in *Leaders* places it among the weakest performers, with MedAE values of 6.61% and 9.04% respectively.

Explanation of deviation: The failure of XGBoost to maintain competitive performance in Trial 3 is surprising given its theoretical advantages in terms of regularization and sequential residual fitting. The most likely explanation is that the early stopping mechanism using an internal 10% validation split from the training data was insufficient to prevent overfitting to the pre-COVID patterns in Trials 1 and 2, and that the small post-COVID training window of Trial 3 (2020–2023) provided insufficient data for the gradient boosting algorithm to converge to a stable and generalizable solution. This finding suggests that XGBoost's theoretical advantages over Random Forest for structured data are not realized in this particular application, where the combination of small sample sizes, panel data structure, and structural breaks creates conditions that favor simpler and more parsimonious models.

4.6.7 Proposed Stacking Model

Expected: The proposed stacking architecture was theoretically expected to outperform all individual baseline models in Trial 3 by combining the complementary strengths of ARIMAX and LSTM through a data-driven meta-learner. The ensemble bias-variance analysis predicted that the combination of a high-bias low-variance model (ARIMAX) and a low-bias high-variance model (LSTM) would produce an intermediate bias-variance profile superior to either individual model. All three meta-learner variants were expected to be competitive, with ElasticNet providing the most stable combination under small sample sizes.

Observed: The empirical results strongly confirm and in several respects exceed the theoretical expectations. In Trial 3, *Proposed (ElasticNet)* achieved the lowest MAE of 3.68% across both clusters and all models in *Followers* and 2.97% in *Leaders* representing reductions of 33.5% and 68.0% respectively from its Trial 1 values. *Proposed (GradientBoost)* achieved Trial 3 MAE of 4.51% in *Followers* and 3.15% in *Leaders*, and *Proposed (Weighted)* recorded 4.44% and 4.13% respectively.

All three proposed model variants demonstrated consistent and monotonic improvement across all three trials in both clusters, a pattern not observed for any individual baseline model.

In terms of MedAE, *Proposed (ElasticNet)* achieved 2.21% in *Followers* and 1.28% in *Leaders* in Trial 3, confirming its superiority on both metrics.

Explanation: The strong performance of the proposed models in Trial 3 confirms the theoretical prediction about complementary base learner combination. The dominance of *Proposed (ElasticNet)* over the other two meta-learner variants is consistent with the theoretical expectation that ElasticNet provides the most stable combination under small sample sizes, as its L1 and L2 regularization prevents either base learner from dominating the ensemble through unstable coefficients [ZH05]. The under-performance of *Proposed (GradientBoost)* relative to *Proposed (ElasticNet)* in the *Followers* cluster is consistent with the theoretical concern about overfitting in small-sample meta-learning contexts, where the non-linear Gradient Boosting meta-learner has insufficient data to reliably learn the optimal non-linear combination function. The monotonic improvement of all three proposed variants across trials represents the most important finding of this post-assessment. This confirms that the stacking architecture learns progressively more effectively as the training data becomes more representative of the current market regime, demonstrating a robustness to structural breaks that no individual baseline model achieves.

4.6.8 Theoretical contribution:

The empirical results validate the core theoretical claim of this study, that a stacking ensemble combining ARIMAX and LSTM captures complementary aspects of BEV adoption dynamics that neither model addresses alone, and that this complementarity becomes most pronounced when the training data is drawn exclusively from the structurally distinct post-COVID market regime. The low error correlation between ARIMAX and LSTM predicted theoretically by [BG69] and confirmed empirically by the superiority of their combination over either individual model provides direct validation of the base learner selection rationale presented in Section 2.7.

4.7 Validation of Hypothesis

This section formally evaluates the four hypotheses proposed in Section 3.2.1 against the empirical results obtained across all three trials, two clusters, and two evaluation metrics.

4.7.1 Hypothesis 1: Confirmed

Hypothesis 1: Models trained on 2011–2019 data will show a significant increase in error when applied to 2020–2024 test data, suggesting structural non-stationarity in market drivers.

Hypothesis 1 is confirmed. The Trial 1 results provide unambiguous evidence of a structural break between the pre-COVID training period and the post-COVID test period. Across both clusters and all models, prediction errors in Trial 1 were substantially higher than in subsequent trials, confirming that models trained exclusively on pre-COVID data cannot reliably forecast the post-COVID acceleration phase of the European BEV market.

In the *Followers* cluster, Trial 1 MAE values ranged from 4.87% (*Random Forest*) to 17.14% (*SVR*), with no model achieving a MAE below 4%. In the *Leaders* cluster, the situation was even more pronounced, with Trial 1 MAE ranging from 8.60% (*ARIMAX*) to 150.71% (*SVR*), reflecting the greater difficulty of forecasting the rapid and heterogeneous adoption trajectories of the more advanced EU markets when trained on pre-COVID data alone.

The non-stationarity of BEV market drivers is further evidenced by the progressive improvement of all proposed models across trials if the pre-COVID dynamics were representative of the post-COVID period, Trial 1 performance would be comparable to Trial 3. The systematic degradation of models that rely heavily on historical patterns — including *RF* and *XGB* further confirms that the structural relationships between BEV adoption and its drivers changed fundamentally after 2020, consistent with the broader literature on the structural impact of the COVID-19 pandemic on European BEV markets [LH21].

4.7.2 Hypothesis 2: Partially Confirmed

Hypothesis 2: Extending the training data to 2022 will improve predictive performance; however, the "historical dominance" of the pre-COVID market in the 2010s will result in a systematic underestimation of the current market pace.

Hypothesis 2 is partially confirmed. Trial 2 produced meaningful improvements in forecasting accuracy relative to Trial 1 for the majority of models, confirming the first part of the hypothesis. In the *Leaders* cluster, *Proposed (ElasticNet)* reduced its MAE from 9.28% in Trial 1 to 5.07% in Trial 2, and *Proposed (GradientBoost)* from 9.94% to 5.63%, demonstrating that the inclusion of post-2020 observations in the training data meaningfully enhanced the predictive capability of the proposed models.

However, the latter part of H2 states that the dominance of pre-2020 observations would result in systematic underestimation of current adoption rates is only partially supported. While performance remained suboptimal across most models relative to Trial 3, the pattern of degradation was not uniform. Several models including *XGB* in the *Leaders* cluster, which deteriorated from 11.26% in Trial 1 to 12.86% in Trial 2 exhibited deterioration rather than improvement, suggesting that the conflicting temporal signals introduced by combining pre-COVID and post-COVID observations in a single training window created instability rather than simply biasing predictions downward. This finding extends the theoretical prediction of H2 by revealing that the impact of historical data dominance is model-specific; models that are sensitive to conflicting temporal dynamics, such as LSTM, are more severely affected than models with explicit temporal regularization such as ARIMAX.

4.7.3 Hypothesis 3: Strongly Confirmed

Hypothesis 3: Models trained exclusively on post-COVID data from 2020–2023 will yield the highest predictive performance, indicating that recent market dynamics are governed by a distinct set of drivers compared to the pre-COVID era.

Hypothesis 3 is strongly confirmed. Trial 3 produced the strongest results across all proposed models and both clusters, with *Proposed (ElasticNet)* achieving the lowest MAE and MedAE of any model across the entire study. 3.68% MAE and 2.21% MedAE in the *Followers* cluster, and 2.97% MAE and 1.28% MedAE in the *Leaders* cluster. These results represent reductions of 33.5% and 68.0% respectively from the Trial 1 values of *Proposed (ElasticNet)*, confirming the substantial benefit of isolating the post-COVID market regime in the training data.

The consistent and monotonic improvement of all three proposed model variants across trials shows a pattern not observed for any individual baseline model; which provides the strongest empirical support for H3. This trajectory confirms that the post-2020 BEV market operates under a distinct and more internally consistent set of dynamics than the earlier diffusion period, and that models trained within this regime generalise more effectively to near-term forecasting. Furthermore, the convergence of the *Leaders* and *Followers* clusters in Trial 3 performance where the gap between the best and worst performing models narrows substantially compared to Trials 1 and 2 suggests that the post-COVID market dynamics are more uniformly tractable across both market segments than the pre-COVID dynamics.

4.7.4 Hypothesis 4: Confirmed

Hypothesis 4: The 27 EU member states exhibits a stable two-cluster structure, namely Leaders and Followers based on their BEV adoption characteristics that persists across all three training configurations, and cluster-specific models. The country-level membership in the clusters may evolve over time.

Hypothesis 4 is confirmed. The elbow method consistently identified two optimal clusters across all three trials, which was further validated using silhouette score analysis. This result remained robust despite differences in the temporal composition of training data and dimensionality reduction, with Trials 1 and 2 retaining three principal components and Trial 3 retaining two principal components. Nevertheless, the optimal number of clusters remained unchanged, confirming the presence of a stable structural segmentation of the EU BEV market into *Leaders* and *Followers*. The principal component loading factors further support this interpretation, with the first component consistently dominated by technological indicators such as *Battery Capacity (kWh)*, while the second component captured infrastructure-related factors, like *GDP*.

While the two-cluster structure remained stable, country-level membership exhibited meaningful variation across trials, reflecting the evolving nature of BEV adoption. Trial 1 showed widespread label reassignment between pre-COVID and post-COVID periods, attributable to a K-Means label swap rather than genuine transitions. Trial 2 demonstrated complete stability across training and test periods, with all 27 member states remaining in their assigned cluster, though *Portugal*'s PCA position progressively approached the cluster boundary during 2023-2024. In Trial 3, this gradual convergence appears as a genuine transition, where *Portugal* crossed into the *Leaders* cluster under the post-COVID training window, reducing the *Followers* group from 13 to 12 member states. These findings confirm that although the EU BEV market exhibits a stable two-cluster structure, the relative positioning of countries within these clusters evolves over time — supporting the use of cluster-specific modelling and confirming Hypothesis 4.

4.8 Case Study To Forecast for 2025

As a forward-looking validation of the proposed architecture, all nine models were retrained on the 2020-2024 configuration to forecast BEV market share for the year 2025 across all 27 EU member states. This case study extends the evaluation beyond the original test horizon, providing an additional out-of-sample assessment of near-term forecasting capability under real-world conditions. The training configuration mirrors the Trial 3 in all respects.

In this case study, the cluster assignments were re-derived from the 2020-2024 training data. *Portugal* remained in the *Leaders* cluster, consistent with its Trial 3 assignment. Additionally, *Greece* transitioned from *Followers* to *Leaders* under this extended training window, reflecting improvements in its relative infrastructure and economic readiness over 2020-2024. Both countries were positioned near the cluster boundary, and their reassignment reflects gradual PCA score shifts rather than abrupt market changes. This is illustrated in Section 6.2 of Appendix 6.

Tables 4.15 and 4.16 present the MAE and MedAE results respectively.

Table 4.15: Case Study — MAE (%) for 2025 Forecast by Model and Cluster.

Cluster	**Model**	**MAE (%)**
Followers	Proposed (GradientBoost)	4.582
	Proposed (ElasticNet)	5.221
	Random Forest	5.720
	XGB	7.166
	Proposed (Weighted)	7.996
	Linear Regression	8.129
	SVR	10.709
	ARIMAX	13.160
	LSTM	16.114
Leaders	Proposed (ElasticNet)	5.167
	Proposed (Weighted)	5.939
	Proposed (GradientBoost)	6.115
	ARIMAX	6.419
	Random Forest	11.251
	SVR	13.325
	Linear Regression	13.419
	XGB	16.259
	LSTM	19.184

Table 4.16: Case Study — MedAE (%) for 2025 Forecast by Model and Cluster.

Cluster	Model	MedAE (%)
Followers	Proposed (Weighted)	1.289
	Proposed (ElasticNet)	1.864
	XGB	2.619
	Random Forest	2.722
	ARIMAX	3.021
	Proposed (GradientBoost)	3.228
	Linear Regression	5.615
	SVR	6.795
	LSTM	10.027
Leaders	ARIMAX	2.775
	Proposed (GradientBoost)	3.773
	Proposed (Weighted)	4.247
	Proposed (ElasticNet)	4.349
	LSTM	8.989
	Random Forest	9.597
	Linear Regression	11.736
	SVR	11.750
	XGB	14.072

Followers Cluster

In the *Followers* cluster, the proposed stacking models dominated performance across both metrics. *Proposed (GradientBoost)* achieved the lowest MAE of 4.582%, followed by *Proposed (ElasticNet)* at 5.221% and *Random Forest* at 5.720%. For MedAE, *Proposed (Weighted)* recorded the strongest result at 1.289%, closely followed by *Proposed (ElasticNet)* at 1.864%, confirming that the ensemble framework produces not only lower average errors but also a more consistent distribution of predictions across *Followers* markets. A notable feature of the MedAE ranking is that *XGB* and *Random Forest* performed competitively at 2.619% and 2.722% respectively, suggesting that for the typical *Followers* country their predictions are reasonably well calibrated, even though their MAE values of 7.166% and 5.720% indicate a small number of larger deviations in specific markets. *ARIMAX* recorded a competitive MedAE of 3.021% despite a substantially higher MAE of 13.160%, suggesting that while its typical prediction error remains reasonable, it produces a small number of large errors in specific Followers markets. *SVR* and *LSTM* recorded the worst MedAE values of 6.795% and 10.027% respectively, confirming their unsuitability for near-term forecasting in this cluster.

Leaders Cluster

In the *Leaders* cluster, all three proposed model variants achieved the lowest MAE values, with *Proposed (ElasticNet)* at 5.167%, *Proposed (Weighted)* at 5.939%, and *Proposed (GradientBoost)* at 6.115%. *ARIMAX* followed closely at 6.419%, with all remaining models recording MAE values above 11%. This confirms the consistent superiority of the stacking architecture in the most advanced EU markets. An interesting finding is that *ARIMAX* achieved the lowest MedAE of any model in the *Leaders* cluster at 2.775%, suggesting that for the majority of *Leaders* countries the linear temporal structure of *ARIMAX* captures 2025 adoption dynamics accurately at the typical country level. However, its MAE of 6.419% is notably higher than its MedAE, indicating that *ARIMAX* produces a small number of large errors in specific *Leaders* markets most likely reflecting policy discontinuities such as the withdrawal of German purchase subsidies, which created sharp deviations from the post-COVID trend that the autoregressive structure cannot anticipate. The three proposed variants follow closely on MedAE at 3.773%, 4.247%, and 4.349% for *Proposed (GradientBoost)*, *Proposed (Weighted)*, and *Proposed (ElasticNet)* respectively, confirming the robustness of the stacking framework even in the most complex and heterogeneous EU markets. *LSTM* recorded the worst MAE of 19.184% and a MedAE of 8.989%, reinforcing its unsuitability for this forecasting context given the small annual training sample.

Overall Assessment

The 2025 case study results confirm that the proposed stacking architecture retains the strongest overall performance beyond the original test horizon across both clusters and both metrics, with *Proposed (GradientBoost)* leading in *Followers*'s MAE and *Proposed (ElasticNet)* leading in *Leaders*'s MAE. The divergence between MAE and MedAE observed for several models like *ARIMAX* in the *Leaders* cluster and *XGB* and *Random Forest* in the *Followers* cluster suggests that 2025 market dynamics, including policy reversals and varying rates of market maturation across member states, introduced country-specific deviations that are not captured by models trained on 2020–2024 data alone. The case study therefore provides qualified support for the near-term applicability of the proposed model as a practical forecasting tool for European BEV market planning, while highlighting the need for periodic retraining as new observations become available to maintain alignment with evolving market conditions. A detailed visualisation of how these models performed for each country can be found in Section 6.2 of Appendix 6.

5 Final Outlook & Future Directions

5.1 Summary and Critical Appraisal

This study is set out to investigate a question that has received surprisingly little systematic attention in the BEV forecasting literature: how does the temporal composition of training data affect the predictive accuracy of forecasting models across heterogeneous European markets in the presence of a structural market break? To address this question, a comprehensive country-year panel covering all 27 EU member states from 2011 to 2024 was constructed, integrating macroeconomic, infrastructural, and technological variables from multiple official sources. A three-trial experimental design was developed to systematically vary the temporal composition of the training data, and nine model classes, namely Linear Regression, SVR, ARIMAX, LSTM, Random Forest, XGBoost, and three variants of a novel ARIMAX-LSTM stacking ensemble were evaluated within a unified framework across all trials, clusters, and metrics.

The central research objective was substantially fulfilled. The four hypotheses were confirmed empirically — H1 and H3 strongly and H2 partially and H4 confirmed, provided a clear and consistent evidence that the structural break introduced by the COVID-19 pandemic and the subsequent post-2020 acceleration of the European BEV market fundamentally altered the relationships between BEV adoption and its drivers, rendering models trained exclusively on pre-COVID data unreliable for near-term forecasting. The *Proposed (ElasticNet)* model achieved the best overall performance in Trial 3 across both clusters and both evaluation metrics, validating the theoretical motivation for the stacking architecture and the base learner selection rationale.

Several challenges were encountered during the course of this study that are worth acknowledging. The construction of a harmonized multi-country panel from heterogeneous data sources required substantial preprocessing effort, including the development of a hybrid imputation strategy to address the systematic early-period missingness in several key variables. The annual frequency of the data constrained the temporal resolution of the models and limited the training sample size in Trial 3 to only four observations per country, which is a particularly significant limitation for data-intensive architectures such as LSTM. The absence of country-specific values for EU-level average variables like the *Purchase Price (EUR), Real Range(km)* and *Battery Capacity (kWh)* prevented the models from capturing country-level variation in vehicle market composition, a limitation that is likely to become less binding as EAFO data [Eur25a] coverage improves in coming years.

The exclusion of explicit policy variables represents perhaps the most significant methodological limitation of this study. The literature consistently identifies financial incentives including purchase subsidies, tax exemptions, and registration fee waivers as among the strongest drivers of cross-national variation in BEV adoption rates [Sie+14]. The absence of these variables from the feature set means that the models cannot anticipate the impact of sudden policy changes such as the abrupt withdrawal of German purchase subsidies in December 2023 on near-term adoption trajectories, contributing to the performance degradation observed in the 2025 case study. Future iterations of this framework should prioritise the integration of harmonised, time-varying policy indicators as additional exogenous variables in the ARIMAX and stacking model pipelines.

Despite these limitations, the study makes several meaningful contributions to the literature. It is the first study to systematically investigate the impact of training data temporal composition on BEV forecasting accuracy across all 27 EU member states simultaneously, using a consistent experimental framework and a diverse set of model classes. It introduces a novel ARIMAX-LSTM stacking architecture specifically designed for structural break contexts, and demonstrates its superiority over all individual baseline models in the most policy-relevant forecasting scenario. And it provides a replicable methodological template for adaptive BEV market forecasting that can be updated annually as new data become available.

5.2 Further Improvements

5.2.1 Addressing Data Limitations

The most direct path to improving the performance of the proposed forecasting framework lies in expanding the availability and granularity of the input data. The current annual frequency of the dataset represents a fundamental constraint on the temporal resolution of all models, particularly LSTM which requires sufficiently long sequential training windows to exploit its architectural capacity for learning long-range temporal dependencies [HS97]. As EAFO [Eur25a] and Eurostat [Off25c] continue to expand their data collection infrastructure, quarterly or monthly data are likely to become available for an increasing number of variables and member states, and migrating the proposed framework to high frequency data should be a priority for future research.

The absence of country-specific values for vehicle technology variables that are currently available only as EU-level averages represents a second important data limitation. In reality, the mix of BEV models available and their associated prices, ranges, and battery capacities vary considerably across member states due to differences in import regulations, local manufacturer presence, and consumer preferences. The incorporation of country-specific vehicle market data if available through expanding EAFO coverage would enable the models to capture this cross-country variation in the technological dimension of BEV adoption, potentially improving forecasting accuracy particularly in the *Followers* cluster where technology availability constraints may play a larger role than in the more mature *Leaders* markets.

5.2.2 The Post-COVID Structural Shift and Its Implications for Forecasting

The findings of this study confirm that the COVID-19 pandemic marked a genuine structural turning point in the European BEV market, not merely a temporary disruption [LH21]. The post-2020 period is characterized by a qualitatively different set of market dynamics simultaneously driven by regulatory pressure from the EU Green Deal [Eur19], rapid technological maturation, and a fundamental shift in consumer attitudes toward sustainable mobility that renders pre-2020 historical data increasingly irrelevant as a basis for near-term forecasting.

This structural shift has two important implications for the design of future forecasting frameworks. First, the training window for operational BEV market forecasting models should be anchored in the post-2020 period and updated annually, rather than seeking to exploit the full available history of BEV market data. The Trial 3 results demonstrate conclusively that a four-year post-COVID training window outperforms an eleven-year window that includes both pre and post-COVID dynamics, confirming that recency of training data is more important than its volume in this context. Second, forecasting frameworks must be designed with explicit structural break detection mechanisms that can identify when a new regime change has occurred post-2020 which triggered automatic retraining of the model on the most recent data. Such mechanisms would enable the proposed framework to remain accurate across future market transitions, including potential disruptions from new battery technologies, changes in EU emissions policy post-2035, or geopolitical events affecting critical mineral supply chains for battery production.

5.2.3 Financial Incentives and Consumer Perspectives

A critical dimension of BEV adoption dynamics that is not captured in the current modelling framework is the role of financial incentives and evolving consumer attitudes. [Sie+14] demonstrate that financial incentives including purchase subsidies, tax exemptions, and charging infrastructure investments are among the strongest predictors of cross-national BEV adoption rates, and that their removal can rapidly destabilize previously stable adoption trajectories. The abrupt termination of German purchase subsidies in December 2023, which contributed to a sharp decline in German BEV registrations in early 2024 illustrates precisely this vulnerability, and is a likely contributor to the performance degradation of several models in the 2025 case study.

Future versions of the proposed framework should incorporate explicit financial incentive variables including the value and duration of purchase subsidies, registration fee exemptions, and annual ownership tax benefits as additional exogenous inputs to the ARIMAX and stacking model pipelines. The construction of a harmonized, time-varying policy database covering all 27 EU member states over the full study period represents a substantial but high-value data collection effort that would substantially enhance the policy sensitivity of the forecasting framework [Sie+14].

Beyond financial incentives, consumer attitudes and psychological factors represent a second dimension of BEV adoption dynamics that the current feature set cannot capture. [LH21] demonstrate that the COVID-19 pandemic heightened environmental awareness among potential car buyers globally, accelerating a pre-existing shift in consumer preferences toward sustainable mobility options that transcended the direct financial incentive effects. This attitudinal shift is not reflected in any of the macroeconomic or technological variables in the current dataset which contributed to the structural acceleration of BEV adoption post-2020 and represents an important source of variation that future models should attempt to quantify. Potential approaches include the incorporation of consumer sentiment indices derived from survey data, social media analysis, or search engine trend data as proxy measures of evolving attitudes toward BEV adoption across member states [KSP24].

5.2.4 Methodological Improvements

Several methodological enhancements could further improve the performance and interpretability of the proposed forecasting framework.

- Probabilistic forecasting. The current framework produces point estimates without quantifying the uncertainty associated with each prediction. In practice, infrastructure planners and policymakers require not only a central forecast but also a range of plausible outcomes to support robust decision-making under uncertainty. The extension of the proposed stacking architecture to produce prediction intervals through approaches such as conformal prediction, Bayesian neural networks, or quantile regression meta-learners that would substantially increase the practical utility of the framework for policy applications [KSP24].

- Dynamic cluster assignment. The current framework applies fixed cluster assignments derived from each trial's training period to the corresponding test period, without allowing for gradual evolution of cluster membership over time. The gradual boundary convergence of *Portugal* during Trial 2 and its eventual transition into the *Leaders* cluster in Trial 3 illustrates that market positioning can shift meaningfully over multi-year periods. A dynamic clustering approach such as online K-Means or time-varying mixture models that updates cluster assignments annually as new observations become available could improve the alignment between cluster structure and current market conditions, potentially improving forecasting accuracy in markets undergoing rapid structural change [Mac67].

- Transfer learning. The small training sample sizes encountered in Trial 3, particularly for LSTM suggests that transfer learning approaches could be valuable in this context. By pre-training the LSTM on a larger related dataset such as monthly EV sales data from a subset of member states or from non-EU markets with longer post-2020 histories and fine-tuning on the EU annual panel, the model could potentially exploit its full architectural capacity despite the limited post-COVID training window [HS97].

- Explainability. The proposed stacking architecture, while more interpretable than a single black-box deep learning model, does not provide direct insight into which features or time periods most strongly influence the final predictions. The incorporation of explainability methods such as SHAP values applied to the meta-learner outputs would enhance the interpretability of the framework and enable policymakers to understand which drivers are most influential in the current forecasting regime, supporting more targeted and evidence-based policy design.

5.2.5 Broader Applicability

While this study focuses exclusively on the 27 EU member states, the methodological framework developed here has broader applicability to other multi-market BEV contexts. The trial design, clustering approach, and stacking architecture are directly transferable to other regional groupings such as the G20 economies or the broader European Economic Area including Norway, Iceland, and the United Kingdom , where heterogeneous national markets are undergoing simultaneous but uneven transitions toward electrification. Applying the framework across these contexts would provide a more comprehensive test of its generalisability and enable cross-regional comparisons of the structural drivers of BEV adoption dynamics, contributing to the development of a more universal forecasting methodology for emerging technology markets [DC23].

5.3 Closing Remarks

The European BEV market stands at an inflection point. The structural transformation of the past five years that is driven by regulatory ambition, technological progress, and shifting consumer values. It created a market environment that is fundamentally different from the one that existed before 2020, and that will continue to evolve rapidly as the EU's 2035 zero-emission vehicle target approaches. Accurate near-term forecasting of BEV market dynamics is not merely an academic exercise but it is a practical necessity for the infrastructure planners, energy system operators, policymakers, and industry stakeholders who must make long-term investment decisions in an environment of substantial uncertainty.

This study demonstrates that the choice of training data is as important as the choice of model in determining forecasting accuracy in structural break contexts, and that a principled ensemble architecture combining complementary model strengths can consistently outperform individual baseline models when trained within the most recent market regime. These findings invite a reconsideration of the conventional practice of calibrating forecasting models on the longest available historical series, suggesting instead that recency, representativeness, and structural coherence of training data should take precedence over data volume in rapidly evolving technology markets.

The road to full electrification of European road transport is neither straight nor smooth. Policy reversals, infrastructure bottlenecks, consumer affordability constraints, and geopolitical uncertainties will continue to introduce structural shifts that challenge the stability of any forecasting framework. The contribution of this study lies not in producing a definitive forecasting model, but in establishing a methodological foundation that is adaptive, cluster-aware, and structurally sensitive, upon which more robust and practically relevant BEV market forecasting tools can be built.

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6 AI Declaration

The following AI tools were used to complete this thesis:

#	AI Tool	Purpose
1	Gemini 3	Assistance with complex code generation and debugging
2	Claude Sonnet 4.6	Assisting in drafting and refining parts of the text

The above mentioned AI tools were occasionally used for hyperparameter tuning, complex code generation, assist in drafting the outline of the document and refining parts of the text. The content was subsequently reviewed and verified by the author, and all ideas and conclusions are the author's own.

Appendix

6.1 Visualisation

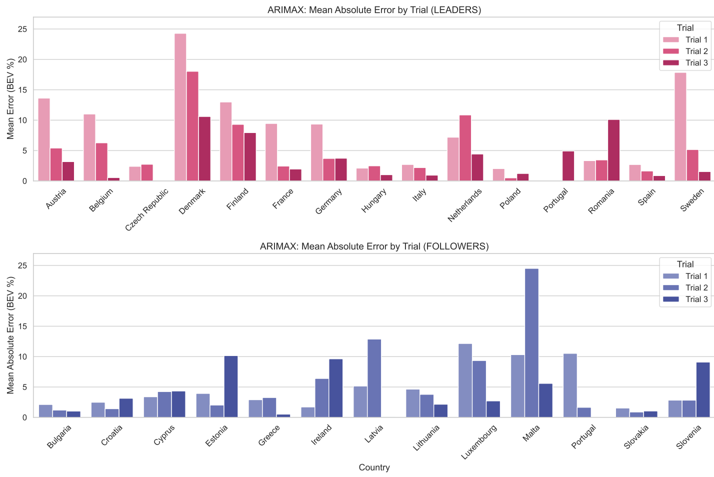

Figure 6.1: ARIMAX: Mean Absolute Error by Trial (Leaders and Followers)

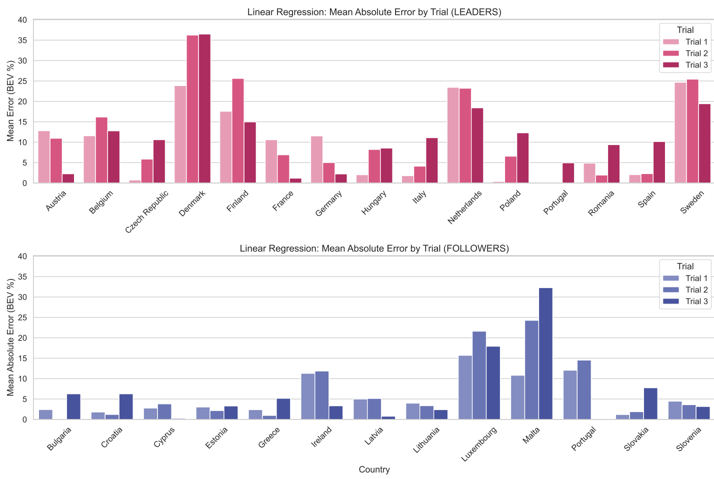

Figure 6.2: Linear Regression: Mean Absolute Error by Trial (Leaders and Followers)

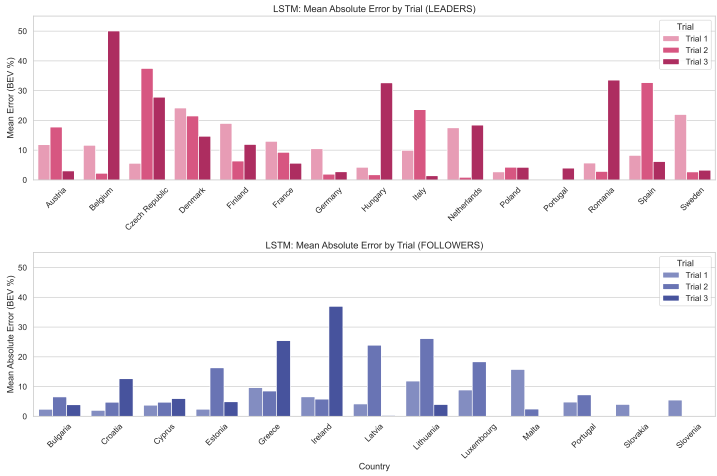

Figure 6.3: LSTM: Mean Absolute Error by Trial (Leaders and Followers)

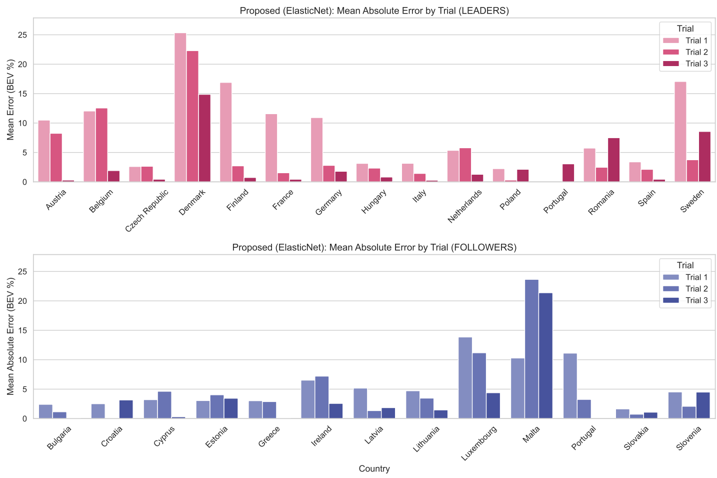

Figure 6.4: Proposed (ElasticNet): Mean Absolute Error by Trial (Leaders and Followers)

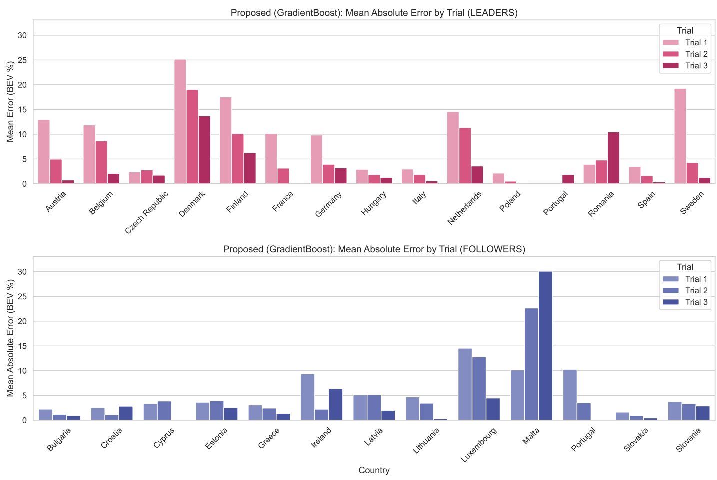

Figure 6.5: Proposed (GradientBoost): Mean Absolute Error by Trial (Leaders and Followers)

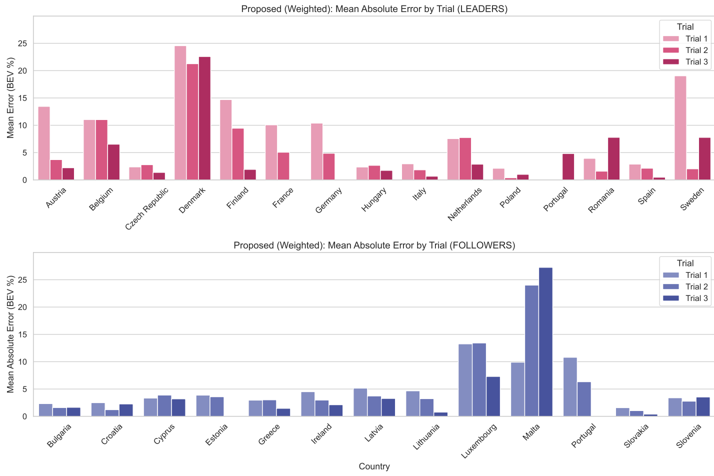

Figure 6.6: Proposed (Weighted): Mean Absolute Error by Trial (Leaders and Followers)

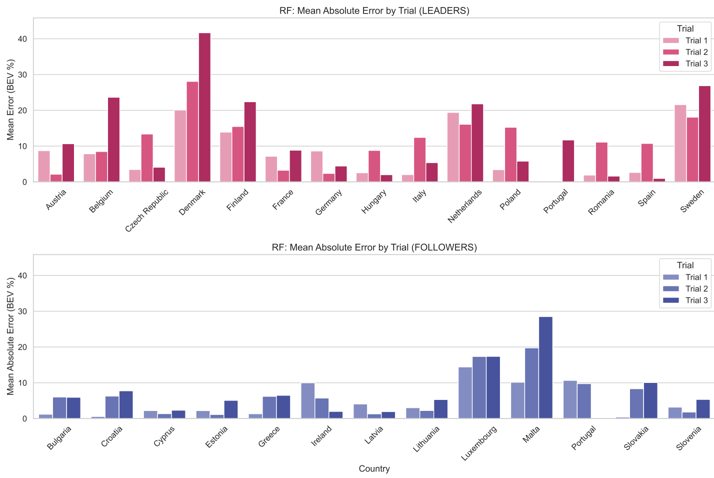

Figure 6.7: Random Forest: Mean Absolute Error by Trial (Leaders and Followers)

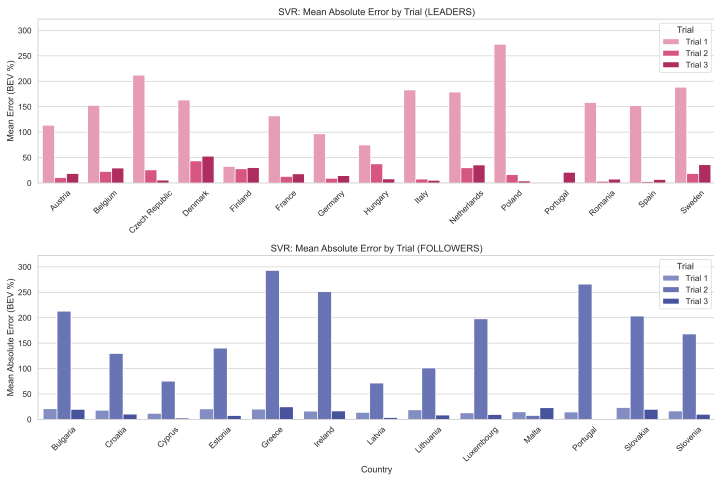

Figure 6.8: SVR: Mean Absolute Error by Trial (Leaders and Followers)

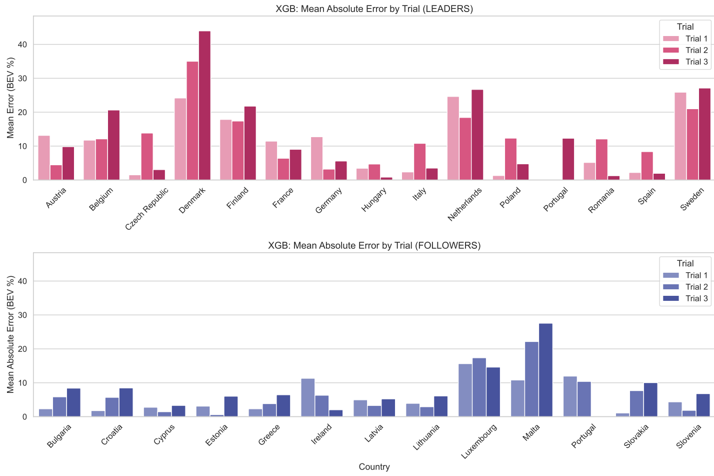

Figure 6.9: XGBoost: Mean Absolute Error by Trial (Leaders and Followers)

6.2 Visualisations for Case Study

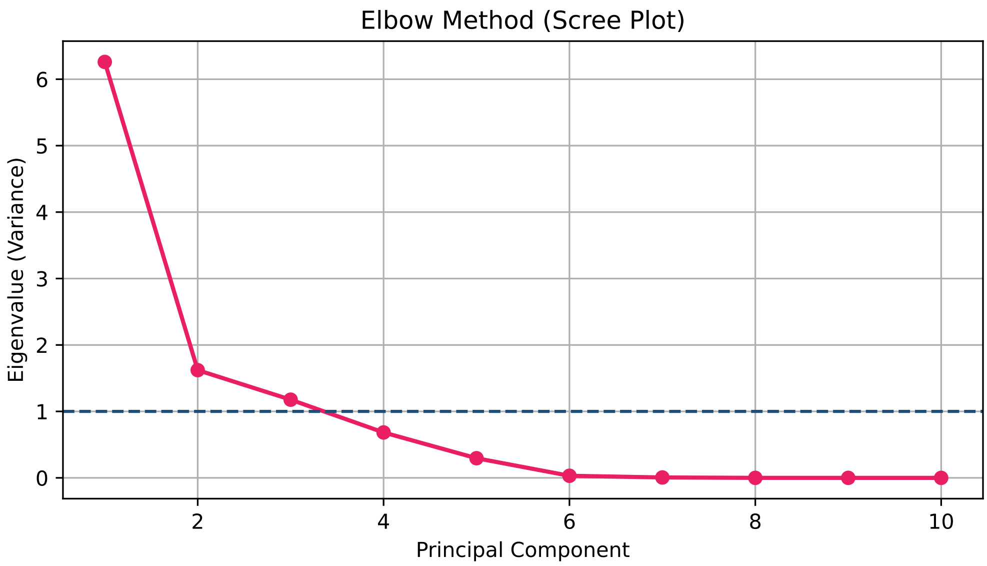

Figure 6.10: Elbow Method for Principal Component Detection

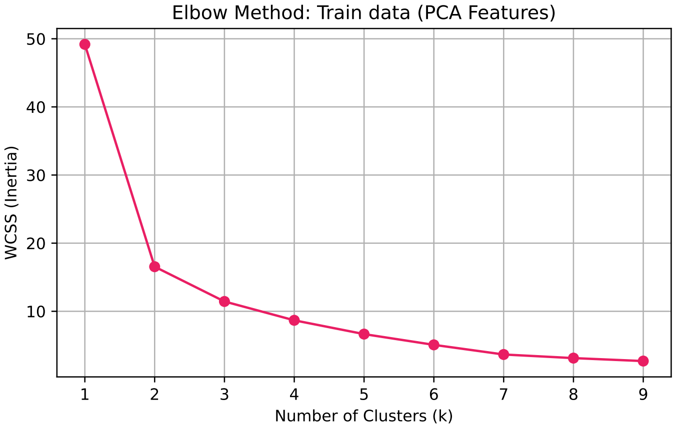

Figure 6.11: Number of clusters for Train set using Elbow Method

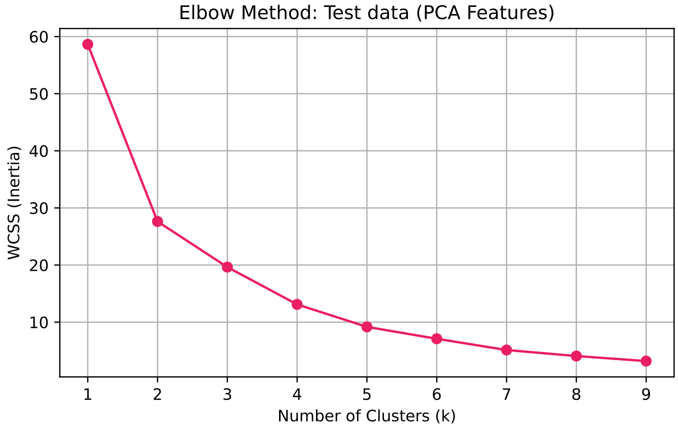

Figure 6.12: Number of clusters for Test set using Elbow Method

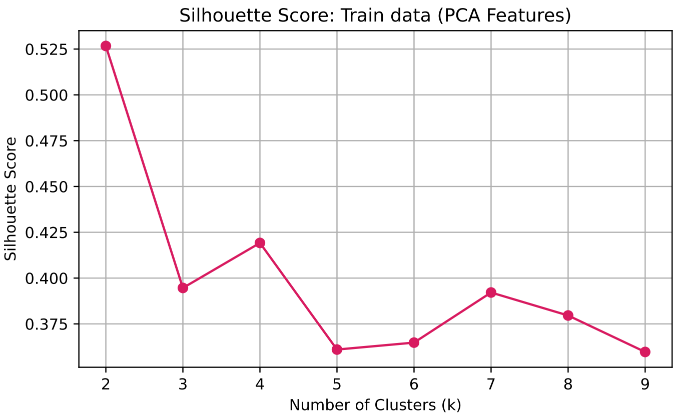

Figure 6.13: Number of clusters for Train set using Silhouette Method

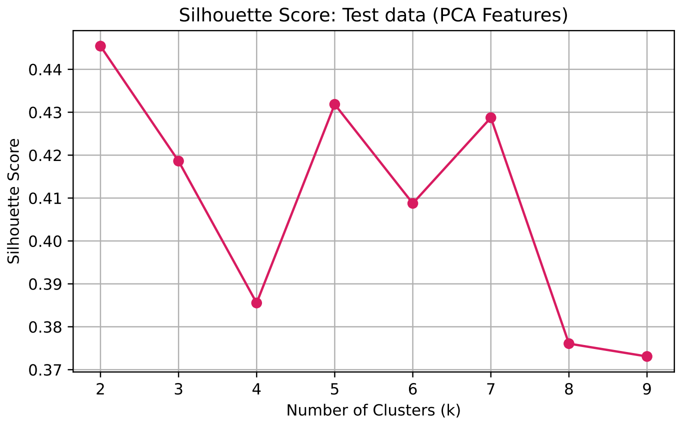

Figure 6.14: Number of clusters for Test set using Silhouette Method

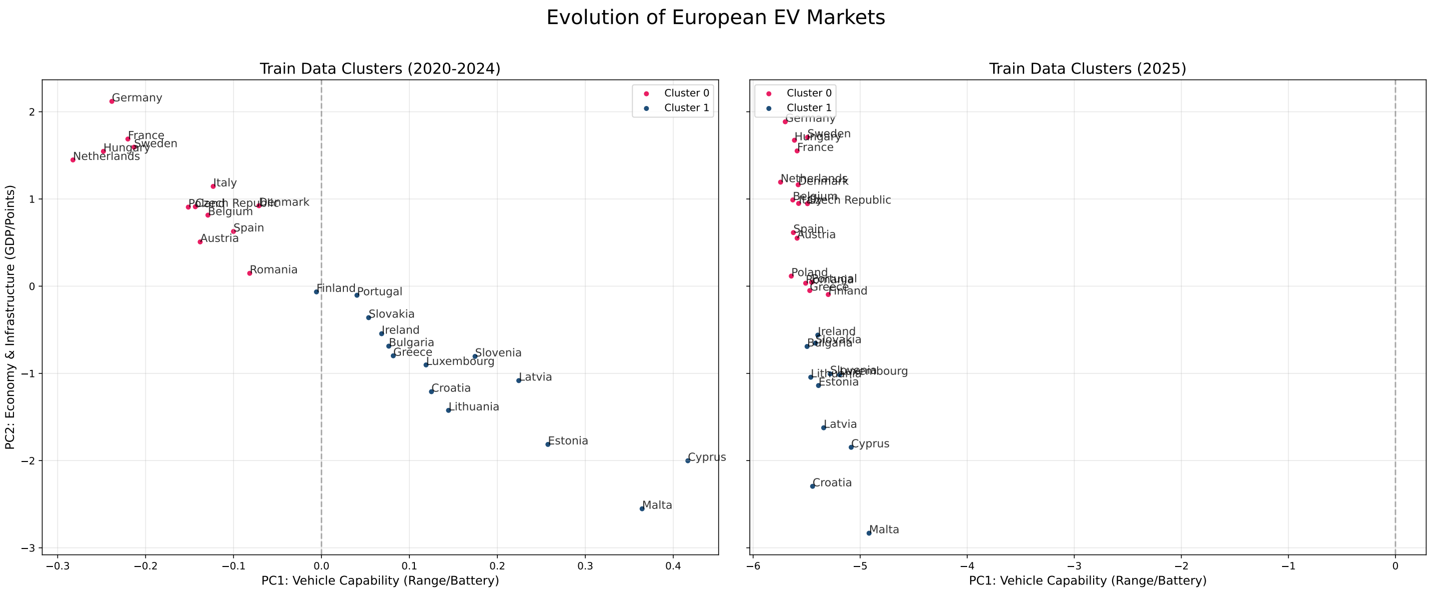

Figure 6.15: Evolution of the markets in the Train set (2020-2024) and Test set (2025)

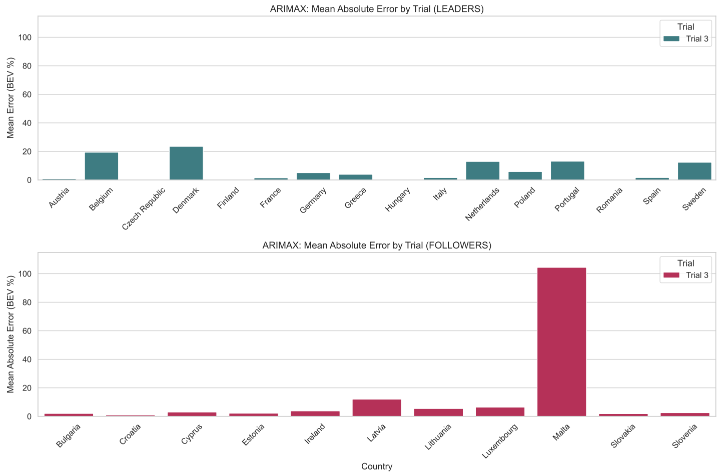

Figure 6.16: ARIMAX: Mean Absolute Error by Trial (Leaders and Followers)

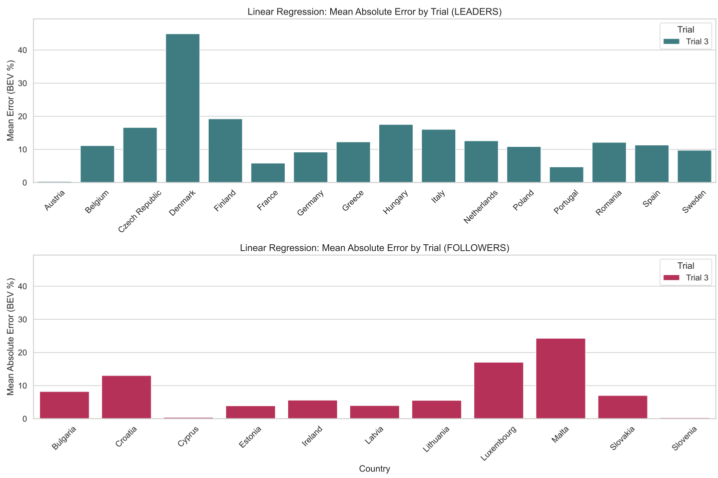

Figure 6.17: Linear Regression: Mean Absolute Error by Trial (Leaders and Followers)

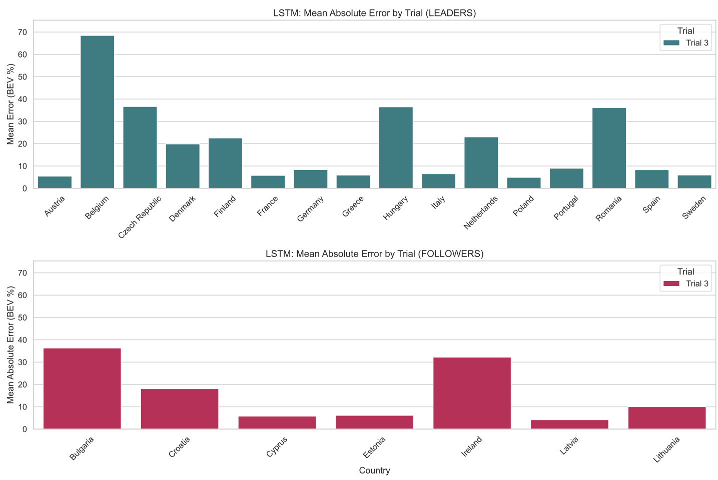

Figure 6.18: LSTM: Mean Absolute Error by Trial (Leaders and Followers)

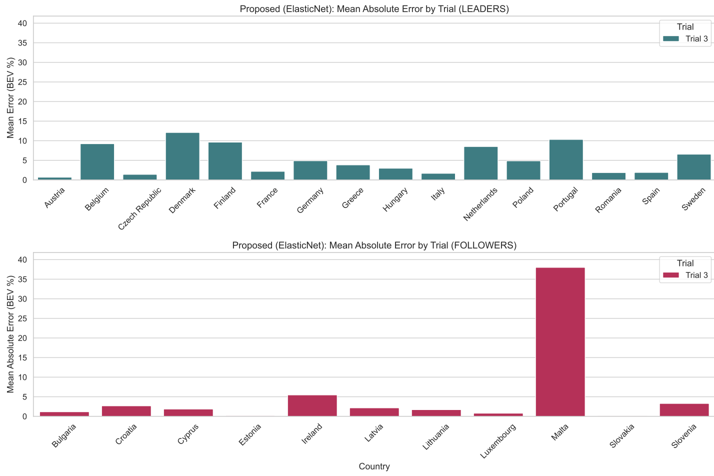

Figure 6.19: Proposed (ElasticNet): Mean Absolute Error by Trial (Leaders and Followers)

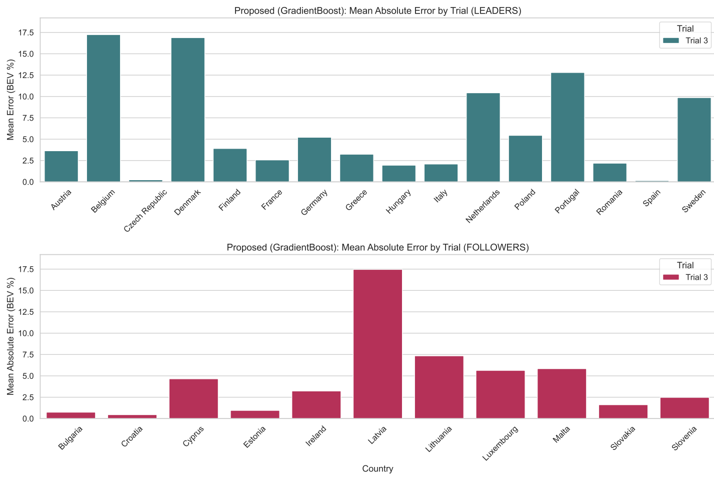

Figure 6.20: Proposed (GradientBoost): Mean Absolute Error by Trial (Leaders and Followers)

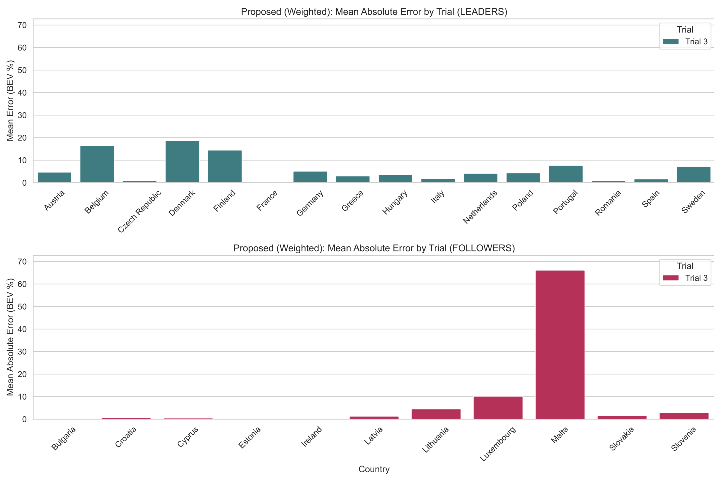

Figure 6.21: Proposed (Weighted): Mean Absolute Error by Trial (Leaders and Followers)

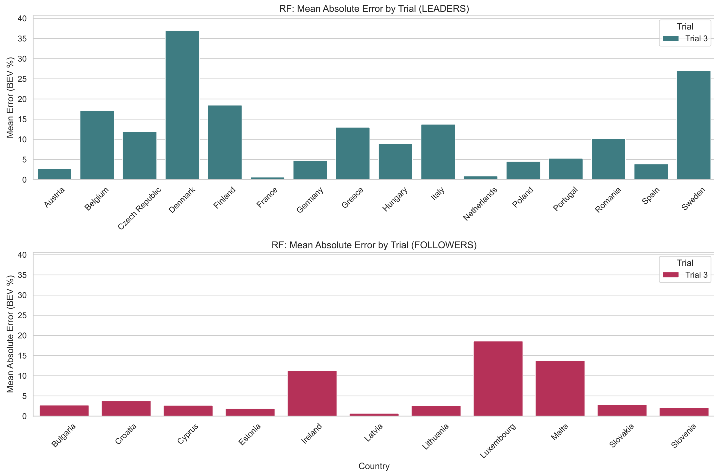

Figure 6.22: Random Forest: Mean Absolute Error by Trial (Leaders and Followers)

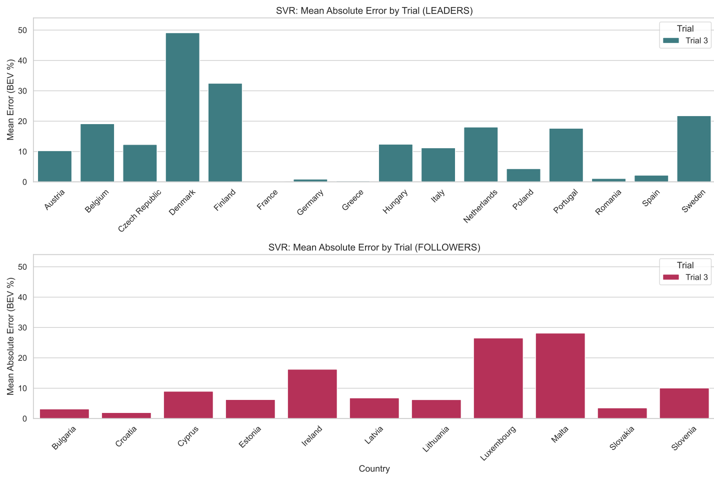

Figure 6.23: SVR: Mean Absolute Error by Trial (Leaders and Followers)

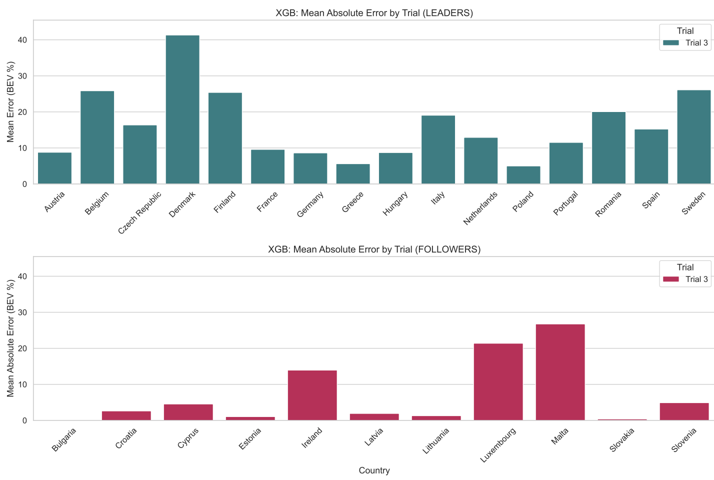

Figure 6.24: XGBoost: Mean Absolute Error by Trial (Leaders and Followers)

6.3 Code Availability

The code for this thesis is publicly available at the following GitHub repository: Battery Electric Vehicles Sales Prediction.

Repository Structure

```
Codes/          # Main experiment code - Trials 1-3
    Trial1/
    Trial2/
    Trial3/
Codes_2025/     # 2025 Case Study code
Data/           # Datasets for Trials 1-3
Data_2025/      # Datasets for the 2025 Case Study
README.md
```

Codes/ - for Trial 1 to 3

Each trial subfolder contains a consistent set of scripts following the full experimental pipeline:

Script	**Description**
Merging of Datasets	Combines and pre-processes raw datasets into analysis-ready form
Exploratory Data Analysis	Distributions, trends, correlations and outlier detection
KPI of the Results	Computes key performance indicators (MAE and MedAE) across models
Visualisation of the Results	Generates bar plots and comparative charts of forecasting outcomes

Codes_2025 - for Case Study to Forecast for 2025

Contains the source code for the 2025 real-world BEV market case study.

The folder structure mirrors `Codes/` with the exact same pipeline stages (merging, EDA, KPIs, and visualisation), and identical result file naming conventions apply.

Datasets

- `Data/` – All datasets used by the scripts in Codes/ for Trials 1–3.
- `Data_2025/` – All datasets used by the scripts in Codes_2025/ for the case study.

Note

The thesis document is also uploaded in the repository for reference.